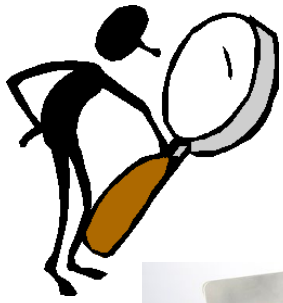


CHEM 1043 General Chemistry II

Topic 3 Solutions and their Properties

Prof. Seungkyu Lee

Examples of Solutions:



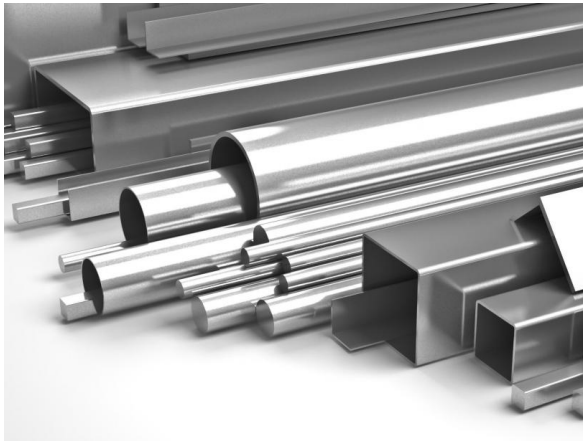
FOCUS in this Topic

Liquid





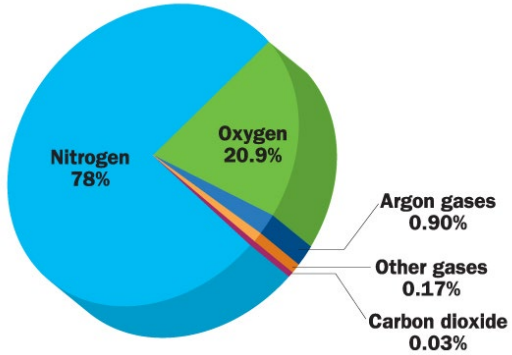
Solid





Gas





What to learn in this topic?



Solutions and their Properties

- ✓ Types of solutions
- ✓ Solution concentration
- ✓ Intermolecular forces and the solution process

  (enthalpy of solution, intermolecular forces in mixtures, formation of ionic solutions)
- ✓ Solution formation and equilibrium

  (solubility as a function of temperature, recrystallization)
- ✓ Solubilities of gases
- ✓ Vapor pressures of solutions
- ✓ Osmotic pressure
- ✓ Freezing-point depression and boiling-point elevation of non-electrolyte solutions
- ✓ Solutions of electrolyte solutions
- ✓ Colloidal mixtures

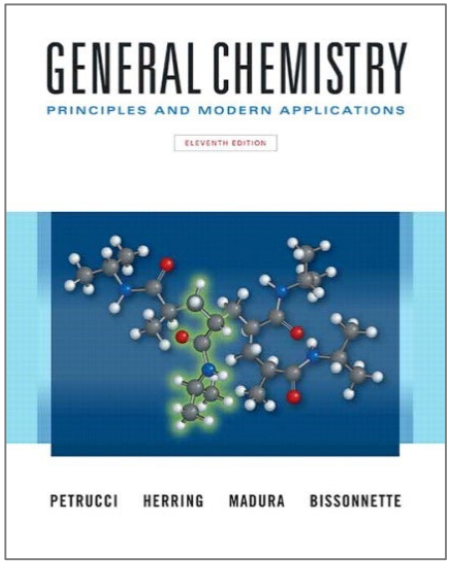


Your reading:

Chapter 14) Solutions and Their Physical Properties

Petrucci; Herring; Madura; Bissonnette:

General Chemistry: Principles and Modern Applications, 11th edition, Pearson



Types of Solutions: Some Terminology



- A **solution** is a ***homogeneous*** (uniform composition and properties) ***mixture*** containing two or more substances in proportions that can be varied.
- The **solvent** is the component present in quantity large enough to determine the state of matter in which the solution exists.
- **Solutes** are other components of a solution. They are said to ***be dissolved*** in the solvent.
- A concentrated solution has a relatively large quantity of dissolved solute(s); a dilute solution has only a small quantity.

- Liquid solutions are most common, but solutions can also exist in gaseous and solid states.

TABLE 14.1 Some Common Solutions

Solution	Components
Gaseous solutions	
Air	N_2, O_2, and several others
Natural gas	CH_4, C_2H_6, and several others
Liquid solutions ← Discussed in this topic	
Seawater	H_2O, NaCl, and many others
Vinegar	H_2O, CH_3COOH (acetic acid)
Soda pop	H_2O, CO_2, $C_{12}H_{22}O_{11}$ (sucrose), and several others
Solid solutions	
Yellow brass	Cu, Zn
Palladium–Hydrogen	Pd, H_2



Solution Concentration

A) Percentage Notation:

- Mass Percent (m/m)
  - 5.00 g of NaCl dissolved in 95.0 g of H_2O in making a 100.0 g solution.
  - Concentration = 5.00**%** NaCl **by *mass***

- Volume Percent (v/v)
  - 25.0 mL of CH_3OH (l) dissolved with water until total solution volume reaches 100.0 mL
  - Concentration = 25.0**%** CH_3OH **by *volume***

solute | solution

- Mass/Volume Percent (m/v)
  - 0.9 g NaCl dissolved in 100.0 mL of water
  - Concentration = 0.9**%** NaCl by ***(mass/volume)***

B) Parts-per Notation:

- ppm: parts per million (μg/g, mg/L)
  - 1 mg of solute dissolved in 1 L (**= 1000 g**) of water
  - Concentration = 1 x 10^{-3} g solute / 10^{3} g solution
    = 1 g solute / 10^{6} g solution = 1 **ppm**

Million = 10^{6}

- ppb: parts per billion (ng/g, μg/L)
  - 1 μg of solute dissolved in 1 L of water
  - Concentration = 1 x 10^{-6} g solute / 10^{3} g solution
    = 1 g solute / 10^{9} g solution = 1 **ppb**

Billion = 10^{9}

- ppt: parts per trillion (pg/g, ng/L)
  - 1 ng of solute dissolved in 1 L of water
  - Concentration = 1 x 10^{-9} g solute / 10^{3} g solution = 1 **ppt**

Trillion = 10^{12}

C) Mole-based Notation:

- Mole Fraction
  - For a solution composed of multiple components (*i, j, k*…), mole fraction of component *i* (designated χ_i) is :

$$\chi_i = \frac{\text{Amount of component } i \text{ (in moles)}}{\text{Total amount of all components (in moles)}}$$

  - where $\chi_i + \chi_j + \chi_k + \ldots = 1$ (the sum of mole fractions of all solution components is 1).

- Mole Percent
  - Mole percent = Mole fraction x 100%

D) Molarity and Molality:

- Molarity
  - No. of mole of solute per unit volume of solution in dm^3 (L):

$$\text{Molarity } (M) = \frac{\text{Amount of solute (in moles)}}{\text{Volume of solution (in liters)}}$$

- Molality
  - When T ↑, volume of solution ↑ (but amount of solute remains constant).
    → Molarity of a solution ***changes*** with temperature.
  - Molality (*m*) is used instead for its temperature independence.

$$\text{Molality } (m) = \frac{\text{Amount of solute (in moles)}}{\text{Mass of solvent (in kilograms)}}$$

  - 1.00 mol of $CO(NH_2)_2$ dissolved in 1.00 kg of water
  - Concentration = 1 ***molal*** → 1.00 ***m*** $CO(NH_2)_2$

Class Practice 1

EXAMPLE 14-1 Expressing a Solution Concentration in Various Units

An ethanol-water solution is prepared by dissolving 10.00 mL of ethanol, CH_3CH_2OH (d = 0.789 g/mL), in a sufficient volume of water to produce 100.0 mL of a solution with a density of 0.982 g/mL. Express the concentration of ethanol in this solution as each of the following.

a) volume percent

b) mass percent

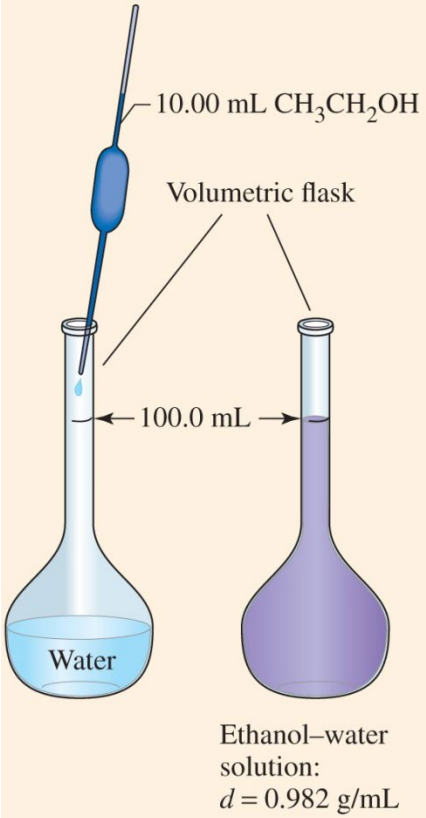
c) mass/volume percent

d) mole fraction

e) mole percent

f) molarity

g) molality



▲ FIGURE 14-1
Preparation of an ethanol–water solution—Example 14-1 illustrated
A 10.00 mL sample of CH_3CH_2OH is added to some water in the volumetric flask. The solution is mixed, and more water is added to bring the total volume to 100.0 mL.



Class Practice 1 (Con't)

Solution:

a) Volume percent $= \dfrac{10.00\ \text{mL ethanol}}{100.0\ \text{mL solution}} \times 100\% = 10.00\%$

b) Mass percent

Mass of ethanol $= 10.00\ \text{mL ethanol} \times \dfrac{0.789\ \text{g ethanol}}{1.00\ \text{mL ethanol}} = 7.89\ \text{g ethanol}$

Mass of solution $= 100.0\ \text{mL soln} \times \dfrac{0.982\ \text{g soln}}{1.0\ \text{mL solution}} = 98.2\ \text{g soln}$

Mass percent $= \dfrac{7.89\ \text{g ethanol}}{98.2\ \text{g solution}} \times 100\% = 8.03\%$

c) Mass/volume percent $= \dfrac{7.89\ \text{g ethanol}}{100.0\ \text{mL solution}} \times 100\% = 7.89\%$

Class Practice 1 (Con't)

d) Mole fraction

No. of mole of ethanol $= 7.89\ \text{g}\ CH_3CH_2OH \times \dfrac{1\ \text{mol}\ CH_3CH_2OH}{46.07\ \text{g}\ CH_3CH_2OH}$

$= 0.171\ \text{mol}\ CH_3CH_2OH$

Mass of water = $98.2\ \text{g soln} - 7.89\ \text{g ethanol} = 90.3\ \text{g water}$

No. of mole of water $= 90.3\ \text{g}\ H_2O \times \dfrac{1\ \text{mol}\ H_2O}{18.02\ \text{g}\ H_2O} = 5.01\ \text{mol}\ H_2O$

Mole fraction of water,

$$x_{CH_3CH_2OH} = \frac{0.171\ \text{mol}\ CH_3CH_2OH}{0.171\ \text{mol}\ CH_3CH_2OH + 5.01\ \text{mol}\ H_2O} = \frac{0.171}{5.18} = 0.0330$$

e) Mole percent $= x_{CH_3CH_2OH} \times 100\% = 0.0330 \times 100\% = 3.30\%$

Class Practice 1 (Con't)

f) Molarity

No. of mole of ethanol = 0.171 mol (from d)

Solution volume = 100.0 mL = 0.1 L

$$\text{Molarity} = \frac{0.171 \text{ mol } CH_3CH_2OH}{0.1000 \text{ L soln}} = 1.71 \text{ M } CH_3CH_2OH$$

g) Molality

Mass of water = 90.3 g (from d) = 0.0903 kg

$$\text{Molality} = \frac{0.171 \text{ mol } CH_3CH_2OH}{0.0903 \text{ kg } H_2O} = 1.89 \, m \, CH_3CH_2OH$$



Class Practice 2

EXAMPLE 14-2 Converting Molarity to Mole Fraction

Laboratory ammonia is 14.8 M NH_3 (aq) with a density of 0.8980 g/mL. What is the mole fraction of NH_3 (χ_{NH_3}) in this solution?

Solution:

Assume: 1L of sample

1. No. of mole of NH_3 $= 1.00\ \text{L} \times \dfrac{14.8\ \text{mol NH}_3}{1\ \text{L}} = 14.8\ \text{mol NH}_3$

2. Mass of solution $= 1000.0\ \text{mL soln} \times \dfrac{0.8980\ \text{g soln}}{1.0\ \text{mL solution}} = 898.0\ \text{g soln}$

3. Mass of NH_3 $= 14.8\ \text{mol NH}_3 \times \dfrac{17.03\ \text{g NH}_3}{1\ \text{mol NH}_3} = 252\ \text{g NH}_3$

4. Mass of H_2O $= 898.0\ \text{g soln} - 252\ \text{g NH}_3 = 646\ \text{g H}_2\text{O}$

5. No. of mole of H_2O $= 646\ \text{g H}_2\text{O} \times \dfrac{1\ \text{mol H}_2\text{O}}{18.02\ \text{g H}_2\text{O}} = 35.8\ \text{mol H}_2\text{O}$

6. Mole fraction of NH_3 $= \dfrac{14.8\ \text{mol NH}_3}{14.8\ \text{mol NH}_3 + 35.8\ \text{mol H}_2\text{O}} = 0.292$

Intermolecular Forces and Solution Process



Why some substances mix to form solutions and others do not?



Oil and water



Water and ethanol

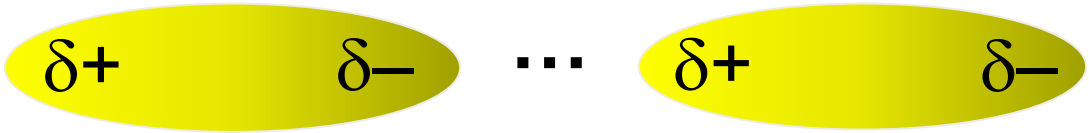
- In the formation of solutions (solution process), heat change is always involved as a result of redistribution of intermolecular forces.

- An enthalpy diagram for solution formation is used to represent the enthalpy change (ΔH_{soln}) in the solution process.

q_p

Quick Review on Intermolecular Forces

- Attractions between the positive end of one molecule and the negative end of another molecule (**dipole interactions**) form the intermolecular forces.



- ***Three*** types of dipoles:
  1) Permanent dipole
  2) Instantaneous dipole
  3) Induced dipole

Van der Waals Forces

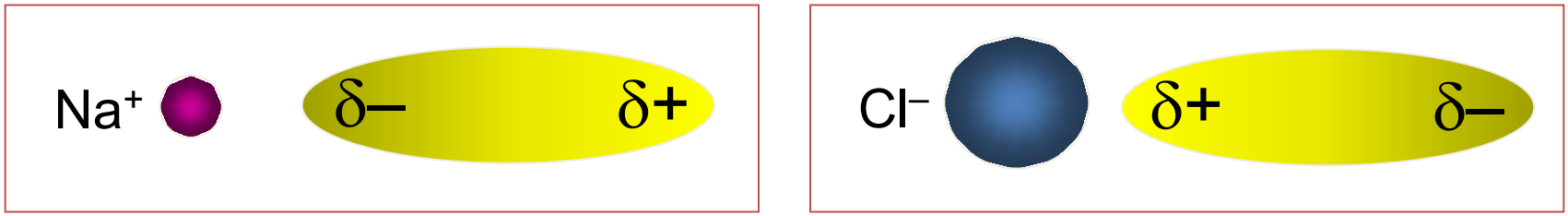
- **Van der Waals forces** refer to a collection of weak attractive forces between groups of atoms or molecules.

- Types of Van der Waals Forces:
  1) Ion – dipole interactions
  2) Dipole – dipole interactions
  3) Ion – induced dipole interactions
  4) Dipole – induced dipole interactions
  5) Instantaneous dipole – induced dipole interactions
     (also known as London forces / Dispersion forces)

Types of Van der Waals Forces

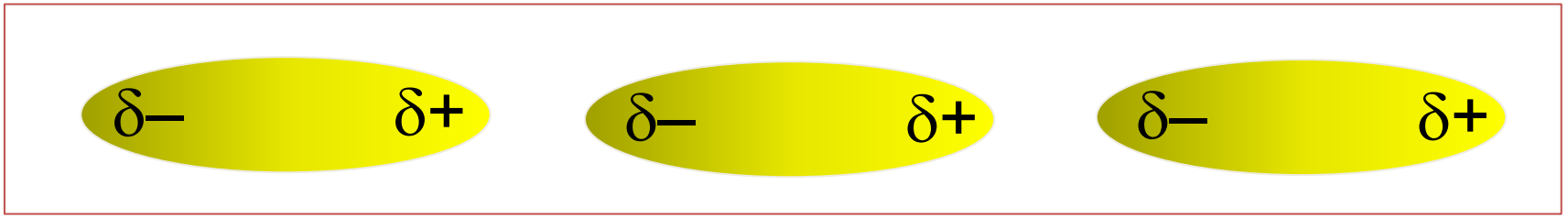
1) Ion – dipole interactions

- Attractions between an ion and a polar molecule (e.g. $Na^+ + H_2O$).



2) Dipole – dipole interactions

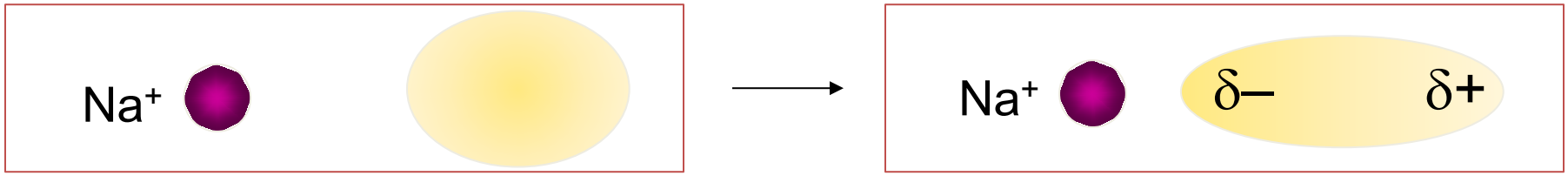
- Attractions between polar molecules (e.g. HCl).



- Molecules tend to line up with the positive end of one dipole directed toward the negative ends of neighboring dipoles.

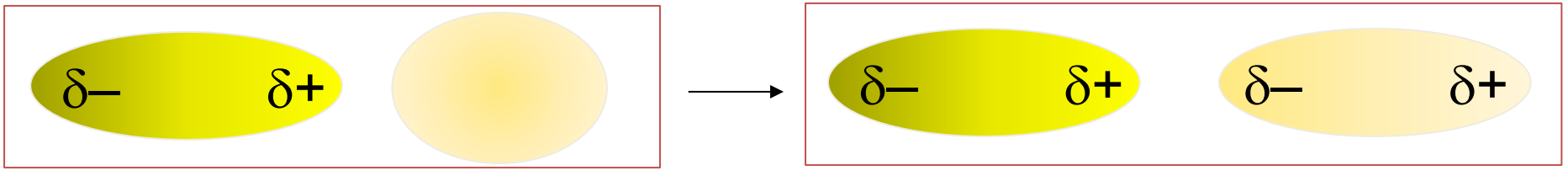
3) Ion – induced dipole interactions

- Attractions between an ion and an atom or a non-polar molecule (e.g. $Na^+ + CCl_4$).
- A dipole is induced in a non-polar molecule.



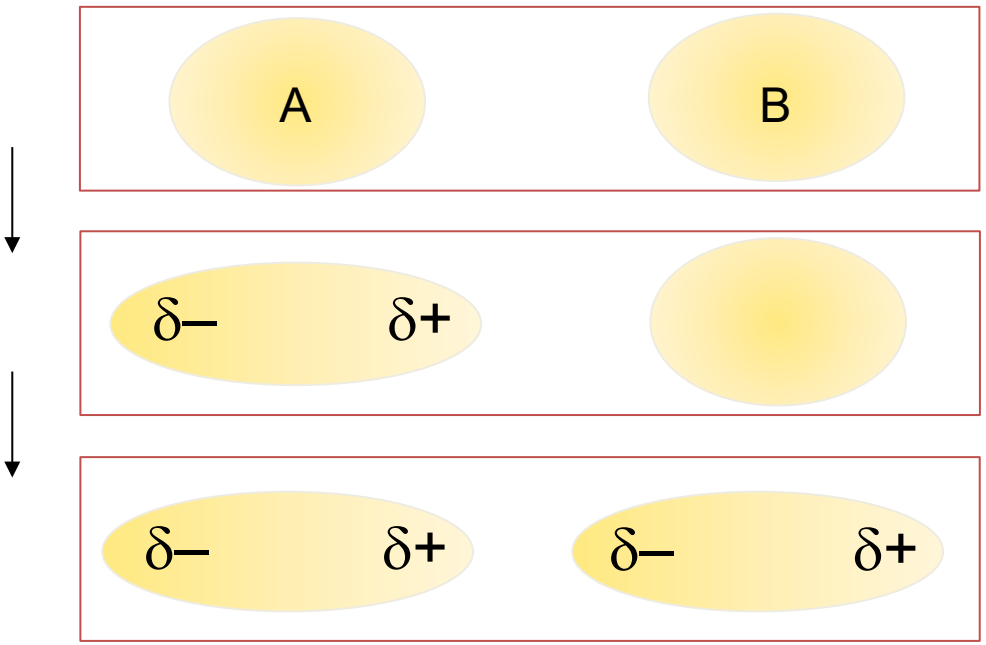
4) Dipole – induced dipole interactions

- Attractions between a polar molecule and an atom or non-polar molecule (e.g. HCl + Ar).



5) Instantaneous dipole – induced dipole interactions

- Attractions between non-polar molecules (e.g. Ar).
- A dipole is induced in a non-polar molecule by the instantaneous dipole of the adjacent molecule.
- These forces exist between **all** molecules.
- Also called London forces / dispersion forces.



- No polarization
- An instantaneous dipole forms in molecule A.
- A dipole is induced in molecule B.

Hydrogen Bonding

- Hydrogen bond is a special type of intermolecular force.

- In a hydrogen bond an H atom is covalently bonded to a highly EN atom, which attracts electron density away from the H nucleus. This allows the H nucleus, a proton, to be exposed and simultaneously attracted to a lone pair on a highly EN atom in a neighboring molecule.



- Hydrogen bonds are possible only with certain **hydrogen-containing compounds**. Only **N**, **O**, and **F** easily meet the requirements for H-bond formation.

Strength of Hydrogen Bond

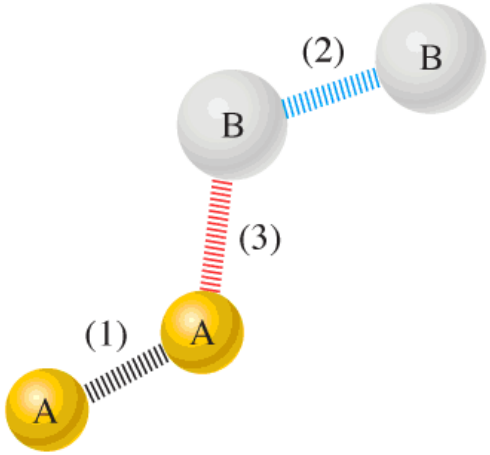
- Compared with other intermolecular forces, hydrogen bonds are relatively strong.

Type of interactions	Energy (kJ /mol)
Hydrogen bond	10 – 40
Dipole – dipole	5 – 25
Dipole – induced dipole	2 – 10
Instantaneous dipole – induced dipole	0.05 – 40

- The strength of hydrogen bond is about **1/10** of that of a typical single covalent bond (150 kJ /mol).

Intermolecular Forces in Mixtures

- ΔH_{soln} depends on the strengths of the ***three*** kinds of intermolecular forces of attraction.



▲ FIGURE 13-3
Intermolecular forces in a solution

The intermolecular forces of attraction, represented here by dashed lines, are between:

(1) **solvent** molecules, A-A;

(2) **solute** molecules, B-B, and

(3) **solvent** and **solute** molecules, A-B.

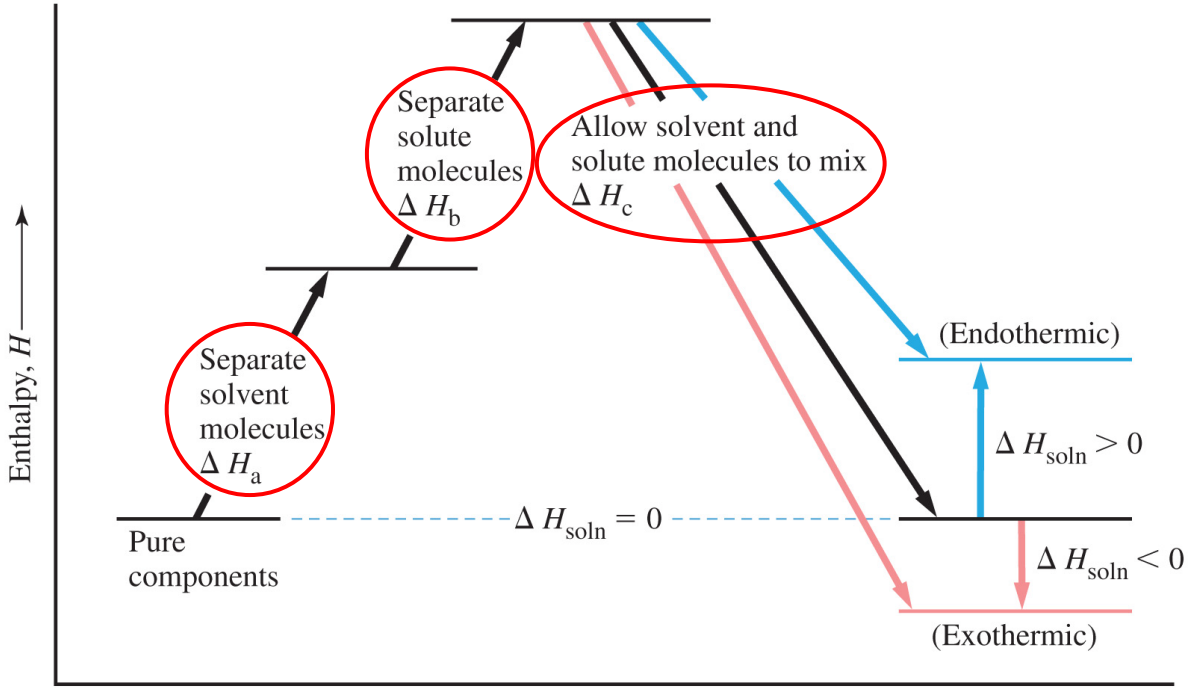
- **Four** scenarios can be resulted according to the relative strengths of these intermolecular forces.

Enthalpy of Solution

- ΔH_{soln} is composed of ***three*** components: ΔH_a, ΔH_b and ΔH_c.

1. ΔH_a: enthalpy change for breaking the **solvent–solvent** attractions; ($\Delta H > 0$).
2. ΔH_b: enthalpy change for breaking the **solute–solute** attractions; ($\Delta H > 0$).
3. ΔH_c: enthalpy change for forming the **solvent–solute** attractions; ($\Delta H < 0$).

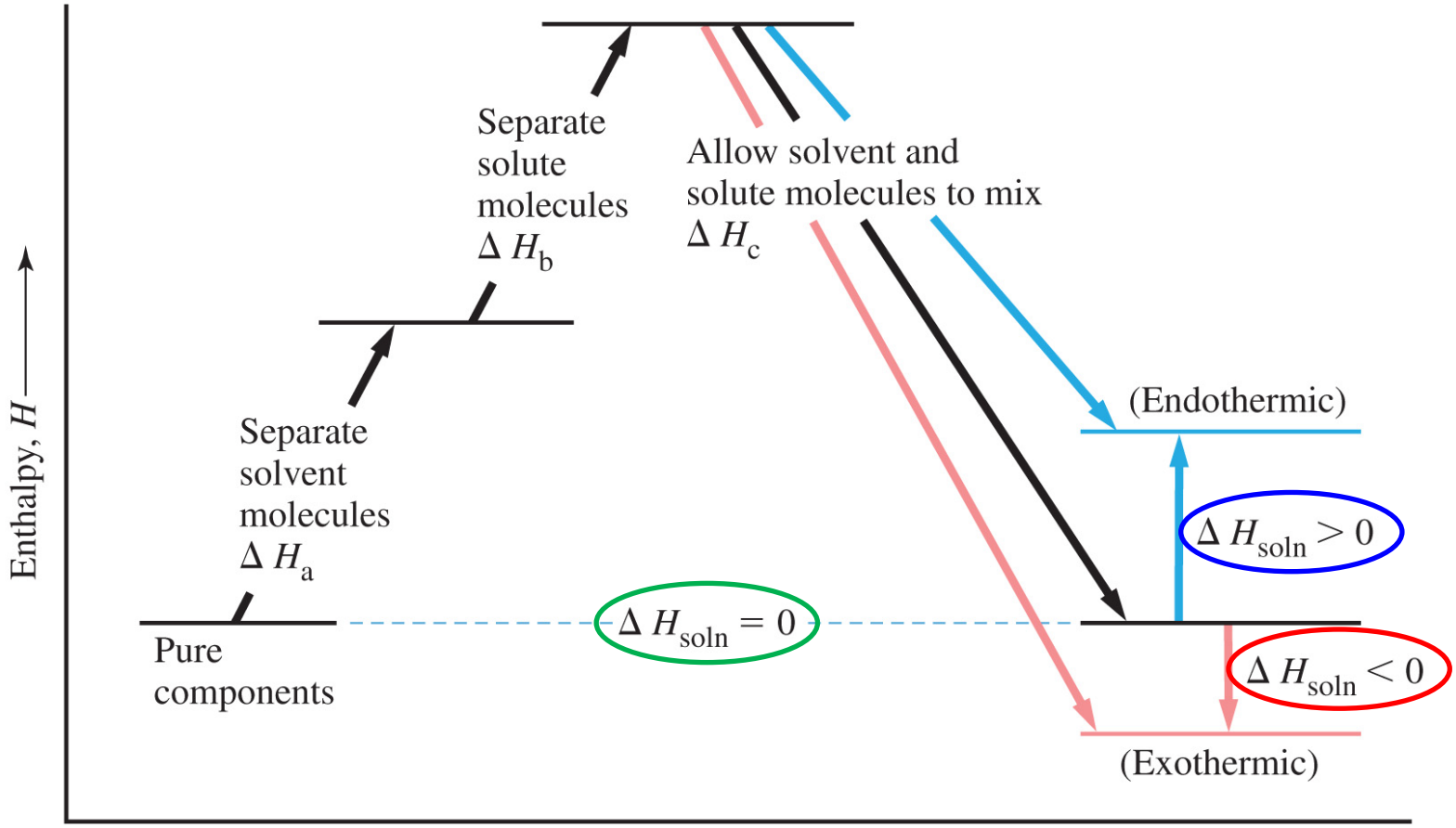
Enthalpy diagram for solution formation



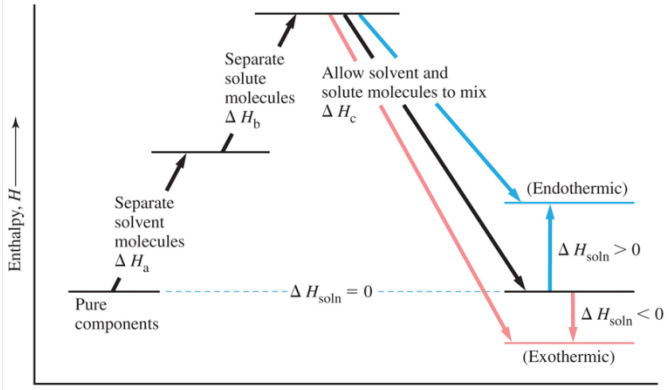
(a)	pure solvent	⟶ separated solvent molecules	$\Delta H_a > 0$
(b)	pure solute	⟶ separated solute molecules	$\Delta H_b > 0$
(c)	separated solvent and solute molecules	⟶ solution	$\Delta H_c < 0$
Overall:	pure solvent + pure solute	⟶ solution	

$$\Delta H_{soln} = \Delta H_a + \Delta H_b + \Delta H_c \qquad (13.1)$$

- ΔH_{soln} can be positive (*endothermic*) or negative (*exothermic*), depending on the values of ΔH_a, ΔH_b and ΔH_c.



Scenario 1: $\Delta H_{soln} = 0$



- Happens when solvent–solute interactions are of **equal strength** with solvent–solvent and solute–solute interactions.

- Heat released from forming the solvent–solute attractions (ΔH_c) equals the heat absorbed in breaking the solvent–solvent and solute–solute attractions ($\Delta H_a + \Delta H_b$).

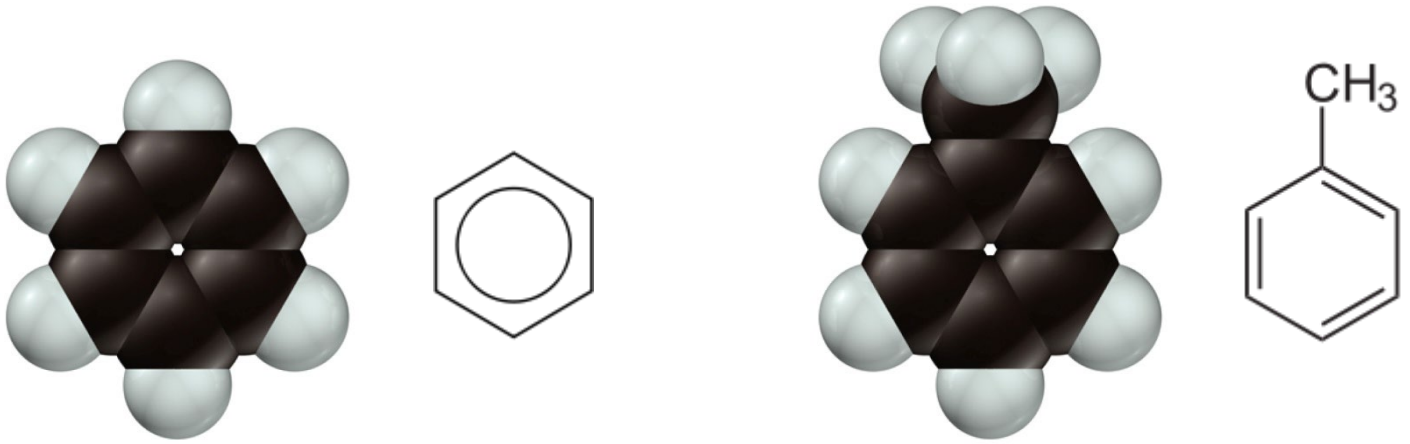
- i.e. $\Delta H_a + \Delta H_b = -\Delta H_c \rightarrow \therefore \Delta H_{soln} = 0$

- A homogeneous mixture or solution results. ∵ properties of this solution can generally be predicted from the properties, such as boiling point and vapor pressure, of its pure components.
  ∴ it is called an ***ideal*** solution.

- Many mixtures of liquid hydrocarbons are of this type or very nearly so (e.g. benzene with toluene). *See next slide.*

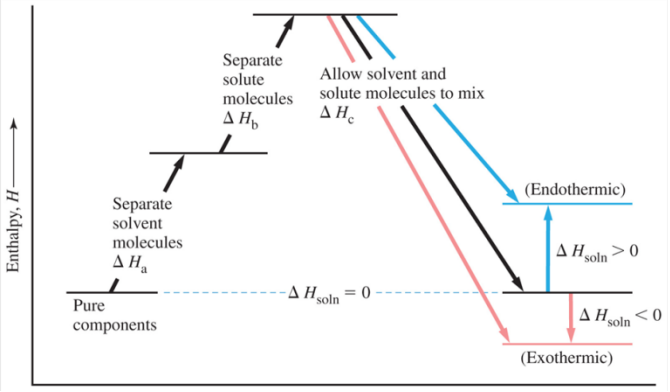
(Con't) Scenario 1: $\Delta H_{soln} = 0$

- Benzene with toluene:



- Both molecules are with **similar** molecular structures.
- Benzene: non-polar (0 *D*)
- Toluene: very slightly polar (0.36 *D*)
- Intermolecular forces for both molecules are of similar strength.

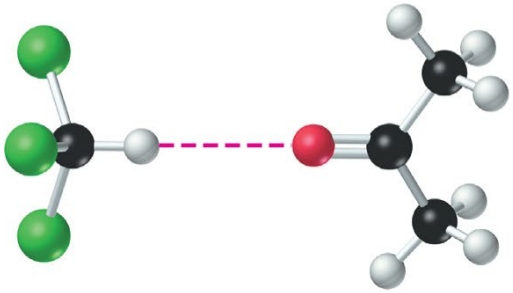
Scenario 2: $\Delta H_{soln} < 0$

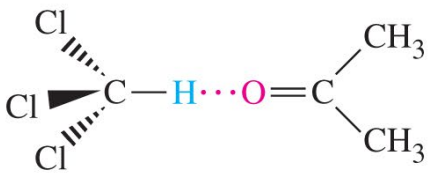


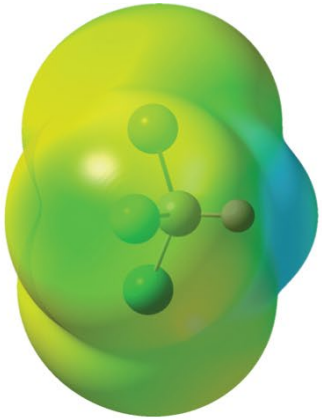
- Happens when solvent–solute interactions are **stronger than** solvent–solvent and solute–solute interactions.
- Heat released from forming the solvent–solute attractions (ΔH_c) is more than the heat absorbed in breaking the solvent–solvent and solute–solute attractions ($\Delta H_a + \Delta H_b$).
- i.e. $\Delta H_a + \Delta H_b < -\Delta H_c \rightarrow \therefore \Delta H_{soln} < 0$
- An non-ideal solution is formed, in which its properties cannot generally be predicted.
- Solution of acetone with chloroform is of this type (An exceptional case to the "like dissolves like" rule).

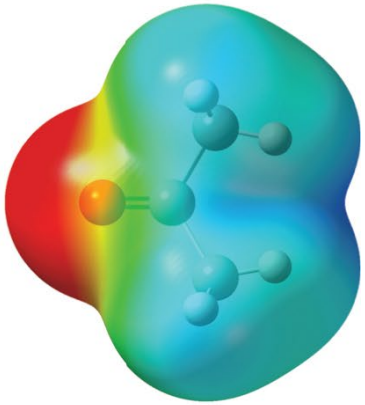
(Con't) Scenario 2: $\Delta H_{soln} < 0$

- Acetone with chloroform:



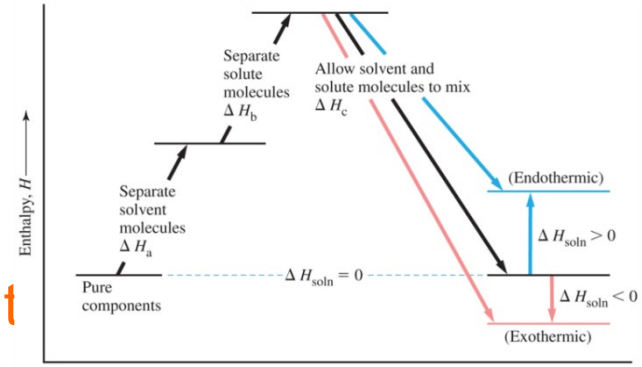






- In $CHCl_3$, three Cl atoms exert strong electron-withdrawing effect on e^- in C–H bond. This makes the H atom very electropositive.
- Hydrogen bond forms between H atom of $CHCl_3$ and O atom of $(CH_3)_2CO$ (but not with Cl atoms in other $CHCl_3$ molecules).
- Acetone–chloroform interactions (H bonds) are stronger than interactions in pure acetone and chloroform (van der Waals force).

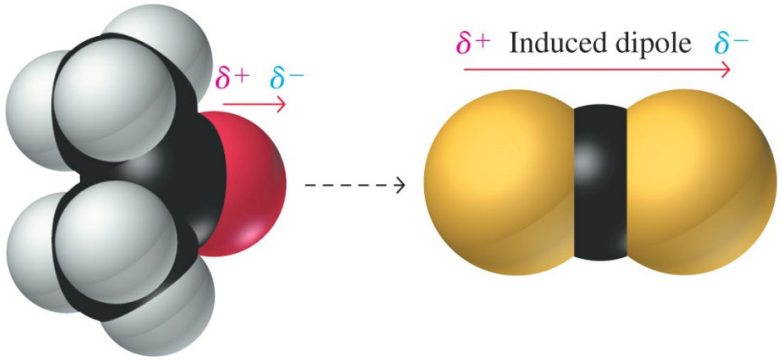
Scenario 3: $\Delta H_{soln} > 0$

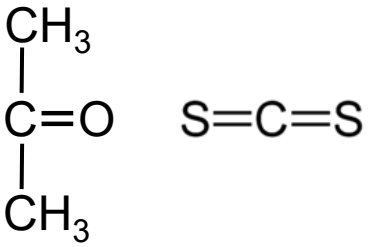


- Happens when solvent–solute interactions are ***somewhat* weaker** than solvent–solvent and solute–solute interactions.
- Heat released from forming the solvent–solute attractions (ΔH_c) is less than the heat absorbed in breaking the solvent–solvent and solute–solute attractions ($\Delta H_a + \Delta H_b$).
- i.e. $\Delta H_a + \Delta H_b > -\Delta H_c \rightarrow \therefore \Delta H_{soln} > 0$
- An non-ideal solution is formed, in which its properties cannot generally be predicted.
- Solution of acetone with carbon disulfide (CS_2) is of this type.

(Con't) Scenario 3: $\Delta H_{soln} > 0$

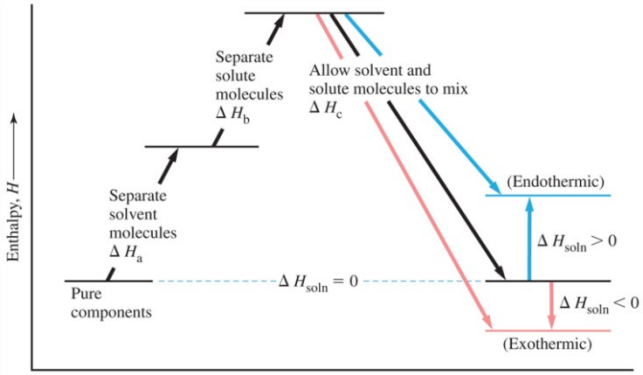
- Acetone with carbon disulfide:





- Acetone: polar (2.88 *D*)
- Carbon disulfide: non-polar (0 *D*)
- Acetone–carbon disulfide interactions (dipole–induced dipole forces) are ***weaker*** than the interactions in pure acetone (dipole–dipole forces).

Scenario 4: $\Delta H_{soln} >> 0$



- Happens when solvent–solute interactions are ***much* weaker** than solvent–solvent and solute–solute interactions.

- Heat released from forming the solvent–solute attractions (ΔH_c) is much less than the heat absorbed in breaking the solvent–solvent and solute–solute attractions ($\Delta H_a + \Delta H_b$).

- i.e. $\Delta H_a + \Delta H_b >> -\Delta H_c \rightarrow \therefore \Delta H_{soln} >> 0$

- Components remain segregated in a heterogeneous mixture. Dissolution does **not** occur to any significant extent.

- "Mixtures" of water (with strong hydrogen bonds) and octane (with weak van der Waals forces) are of this type.



Class Practice 3

EXAMPLE 14-3 Using Intermolecular Forces to Predict Solution Formation

Predict whether or not a solution will form in each of the following mixtures and whether the solution is likely to be ideal:

(a) CH_3CH_2OH and H_2O

(b) $CH_3(CH_2)_4CH_3$ and $CH_3(CH_2)_6CH_3$

(c) $CH_3(CH_2)_6CH_2OH$ and H_2O

(a) A non-ideal solution is formed.

(Both compounds have hydrogen bonding, but of different strength because of different molecular structure.)

(b) A nearly ideal solution is formed.

(Both compounds are non-polar. They differ only in carbon chain length by two units, intermolecular attractive forces are similar.)

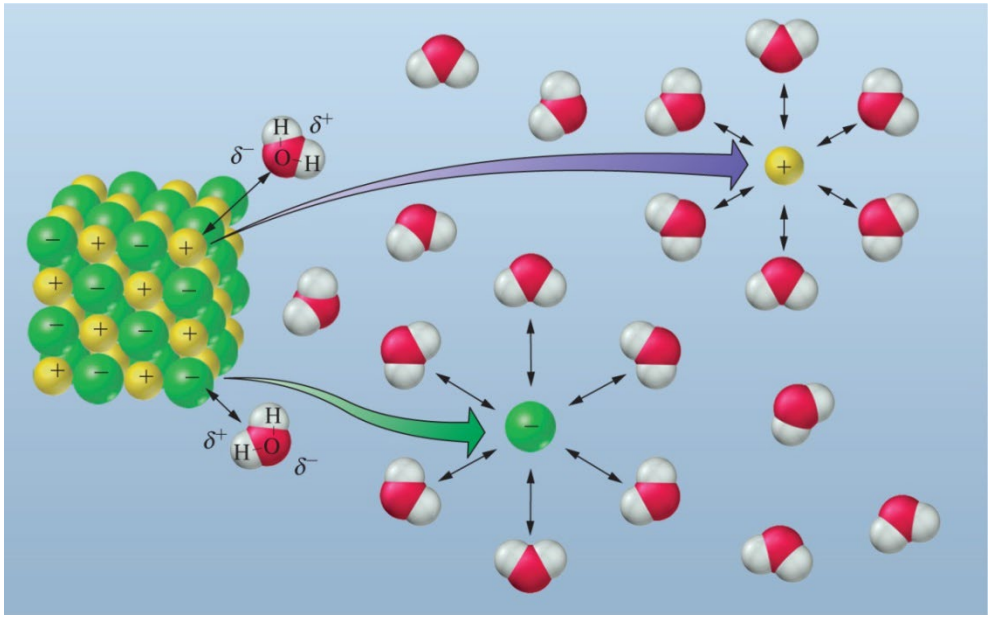
(c) No solution formation.

Dissimilar compounds. The long carbon chain in octanol limits its solubility (due to –OH group) in water.

Hydrophobicity ↑as the length of carbon chain ↑

Formation of Ionic Solutions

- The formation of ***aqueous*** solutions of ionic compounds involves interactions between ions and water dipoles:



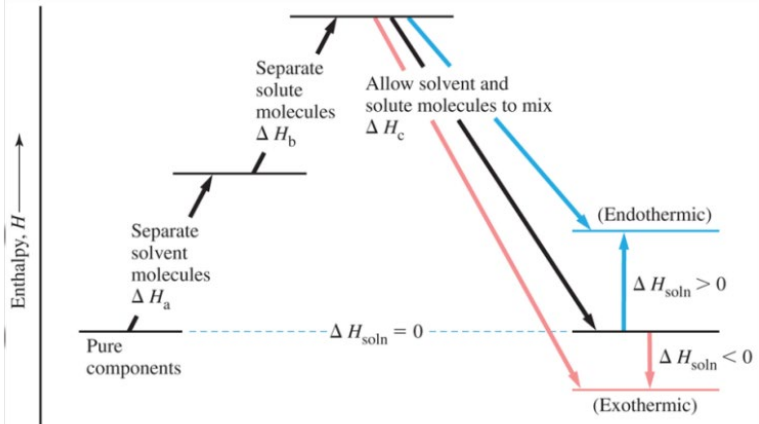
- Negative ends of water dipoles point toward the positive ions.
- Positive ends of water dipoles point toward the negative ions.

- These interactions are known as the **ion–dipole forces**. Ions surrounded by clusters of water molecules are said to be hydrated and energy is released ($\Delta H_{hydration}$).

($\uparrow \Delta H_{hydration} \Rightarrow \uparrow$ chance of the ionic solid to dissolve in water)

- ΔH_{soln} for ionic solutions has three components: ΔH_1, ΔH_2 and ΔH_3.
  - ΔH_1: Enthalpy change for dissociation of an ionic solid into separated gaseous ions, i.e. $-\Delta H_{lattice}$ ($\Delta H_1 > 0$).
  - ΔH_2: Enthalpy change for hydration of gaseous cations ($\Delta H_2 < 0$).
  - ΔH_3: Enthalpy change for hydration of gaseous anions ($\Delta H_3 < 0$).

- Example: Enthalpy of solution for NaCl



$$NaCl(s) \longrightarrow Na^+(g) + Cl^-(g) \qquad \Delta H_1 =$$

$$Na^+(g) \xrightarrow{H_2O} Na^+(aq) \qquad \Delta H_2 =$$

$$Cl^-(g) \xrightarrow{H_2O} Cl^-(aq) \qquad \Delta H_3 = \text{(hydration energy of } Cl^-\text{)} < 0$$

$$NaCl(s) \xrightarrow{H_2O} Na^+(aq) + Cl^-(aq) \qquad \Delta H_{soln} = \Delta H_1 + \Delta H_2 + \Delta H_3 \approx +5 \text{ kJ/mol}$$

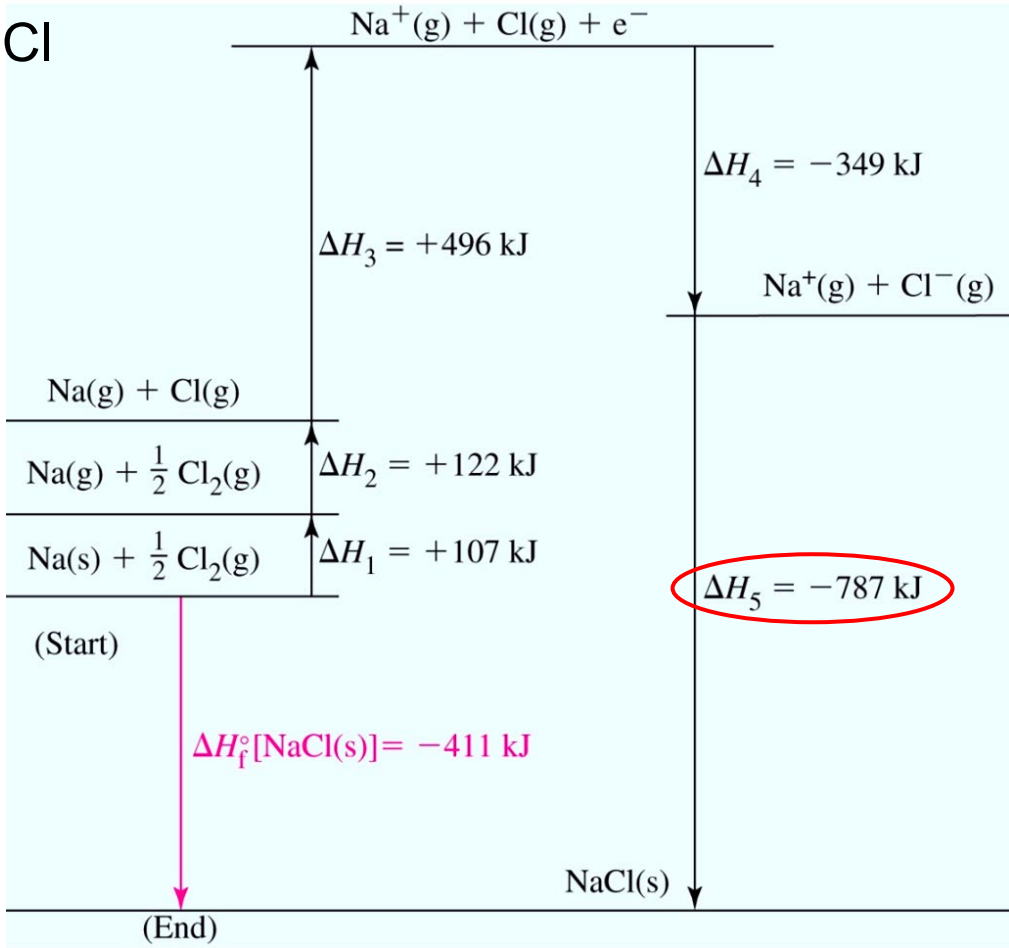
- ***Lattice energy*** is the energy released when separated gaseous cations and anions come together to form one mole of a solid ionic compound.

  - e.g. $Na^+(g) + Cl^-(g) \rightarrow NaCl\ (s)$ $\Delta H = \Delta H_{lattice}$ [NaCl(s)]
  - An exothermic process for ionic bonds are formed between the separated gaseous ions.
  - Lattice enthalpy is a measure of ionic bond strength. *(Greater the charges of ions involved, the more negative is the lattice enthalpy.)*
  - Determined indirectly through the Born-Haber cycle *(see next slide for an example)*.

Determination of lattice energy of NaCl:

Born-Haber cycle

- ΔH_f = enthalpy of formation of NaCl
  = –411 kJ/mol
- $\Delta H_1 = \Delta H_{Sublimation}$ of Na
  = +107 kJ/mol
- ΔH_2 = ½ Cl–Cl bond energy
  = +122 kJ/mol
- ΔH_3 = 1st I. E. of Na
  = +496 kJ/mol
- ΔH_4 = E. A. of Cl
  = –349 kJ/mol
- ΔH_5 = lattice energy of NaCl



$$\Delta H_{overall} = -411\,kJ = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4 + \Delta H_5$$

$$-411\,kJ = 107\,kJ + 122\,kJ + 496\,kJ - 349\,kJ + \Delta H_5$$

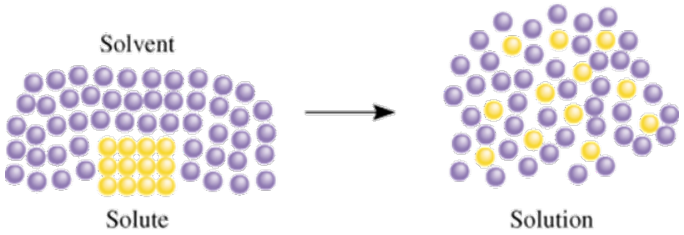
$$\Delta H_5 = \text{lattice energy} = (-411 - 107 - 122 - 496 + 349)\,kJ = -787\,kJ/mol$$

- In the case of dissolution of NaCl (s) in water in forming NaCl(aq),
  $\Delta H_{soln} \approx +5$ kJ/mol $\rightarrow \Delta H > 0$



Why is the dissolution of NaCl (s) spontaneous then?

- The driving force for the dissolution process is **entropy**.



- Gibbs Free Energy (ΔG):

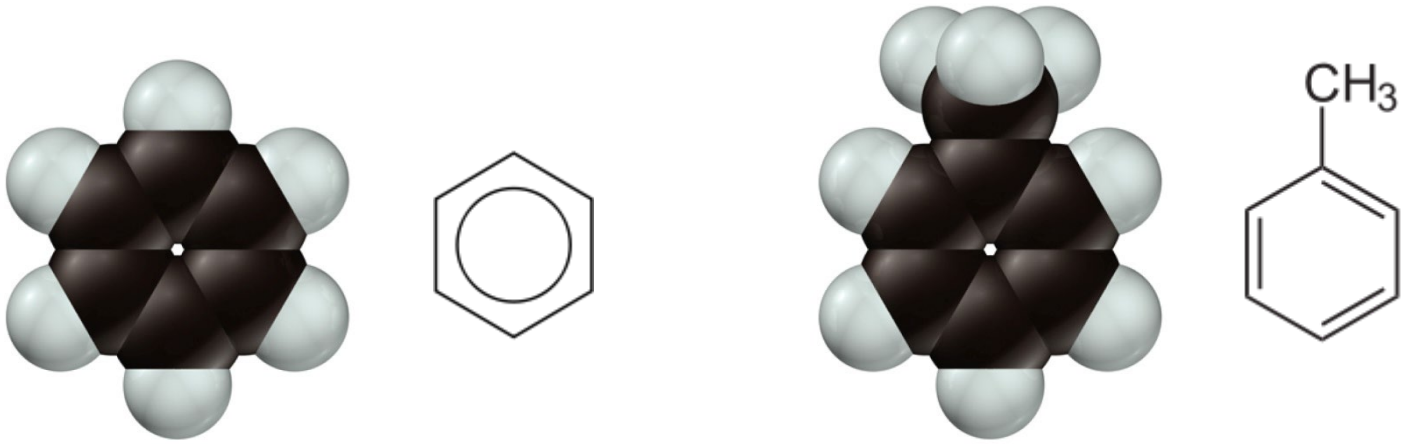
$\Delta G = \Delta H - T\Delta S \rightarrow$ $\Delta G < 0$ spontaneous;
$\Delta G > 0$ not spontaneous;
$\Delta G = 0$ at equilibrium

$\Delta G = \Delta H - T\Delta S \rightarrow$ $\Delta G < 0$ spontaneous;
$\Delta G > 0$ not spontaneous;
$\Delta G = 0$ at equilibrium

- For $\Delta H > 0$, as long as the term $T\Delta S$ is large enough, $\Delta G < 0$ and the solution process is spontaneous.
- 1) Relatively small value for ΔH and 2) large increase in entropy in the mixing of solvent and solute molecules make the solution process for ionic compounds possible.

(Con't) Scenario 1: $\Delta H_{soln} = 0$

- Benzene with toluene:



- Both molecules are with **similar** molecular structures.
- Benzene: non-polar (0 *D*)
- Toluene: very slightly polar (0.36 *D*)
- Intermolecular forces for both molecules are of similar strength.

Common Recrystallization Solvents

Dielectric constant is a measure of a substance's ability to insulate charges from each other.

One measure of polarity. ↑ ε ⇒ ↑ capability of dissolving polar/ionic cpds

Solvent	Boiling pt (°C)	Miscibility in water	Solvent Polarity	Dielectric constant (ε)
Diethyl ether	34.6	-	intermediate	4.3
Acetone	56	+	intermediate	20.7
Petroleum ether	60-80	-	non-polar	~ 2
Chloroform	61	-	intermediate	4.8
Methanol	65	+	polar	32.6
Hexane	69	-	non-polar	1.9
Ethyl acetate	77	-	intermediate	6.0
Ethanol	78.5	+	polar	24.3
Water	100		very polar	80
Toluene	110.6	-	non-polar	2.4
Acetic acid	118	+	intermediate	6.15

ε: 2-3 (non-polar), >10 (polar)

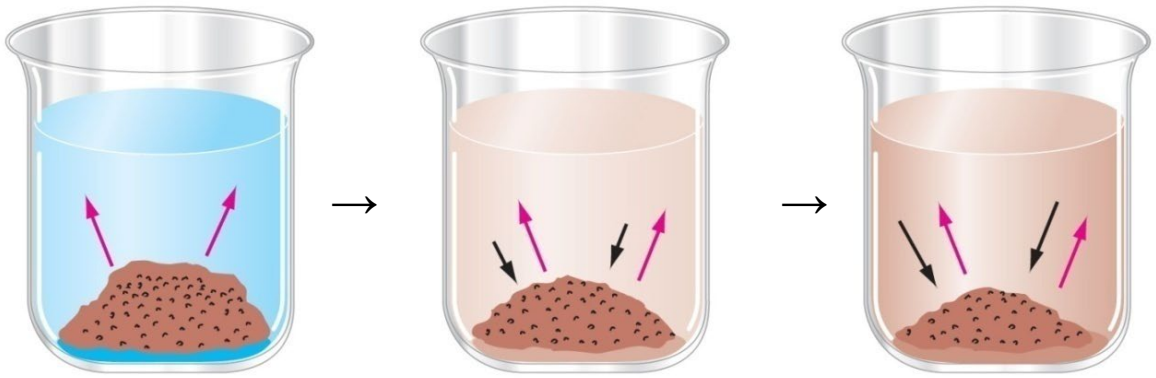
Solution Formation and Equilibrium

- When a solid solute and liquid solvent are mixed:

  Stage 1: Only dissolution occurs, no crystallization takes place

  Stage 2: Dissolution rate > Crystallization rate

  Stage 3: Dissolution rate = Crystallization rate → **dynamic equilibrium**



Stage 1 Stage 2 Stage 3

↑ = rate of dissolution ↓ = rate of crystallization

Saturated **solution:** Solution having solute concentration *equals* its ***solubility***.

- The solution at stage 3 is said to be *saturated*.

- The concentration of the *saturated* solution is called the ***solubility*** of the solute in the given solvent. Solubility varies with temperature.

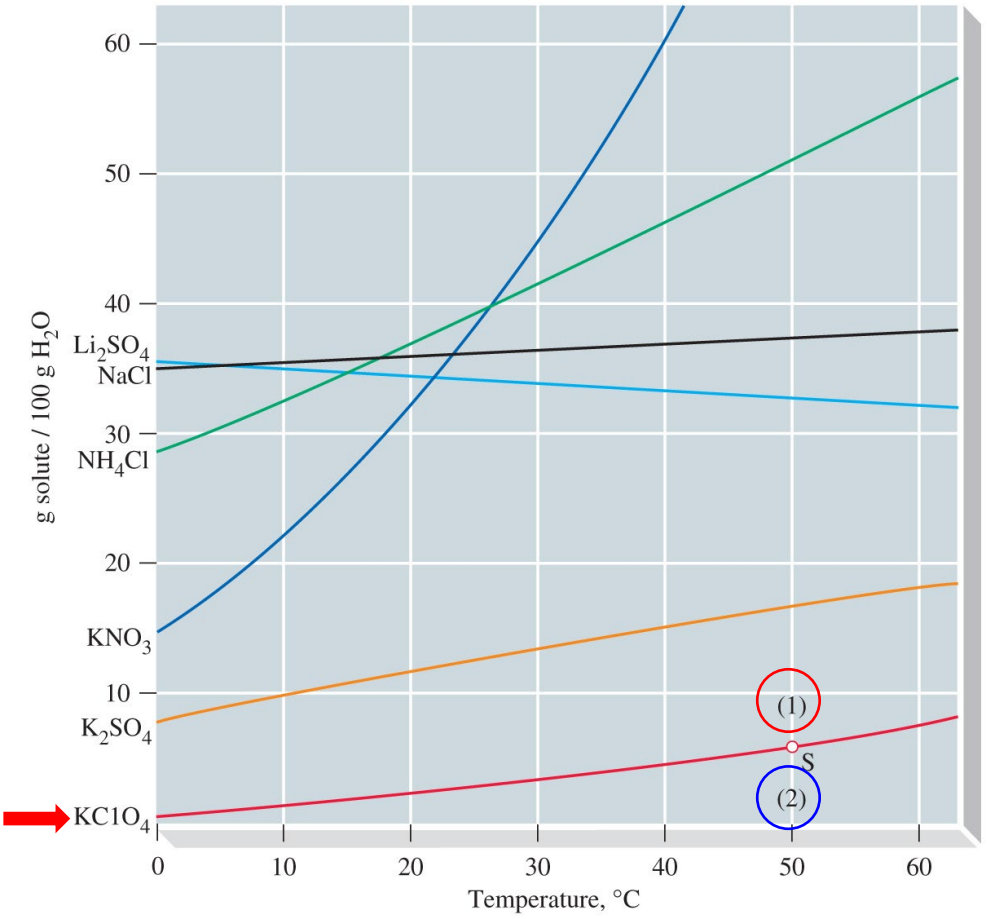
- By this definition:
  - *Saturated* solution:
    Solution having solute concentration equals its solubility.
  - *Unsaturated* solution:
    Solution having solute concentration less than in the saturated solution.
  - *Supersaturated* solution:
    Solution containing quantity of solute greater than in a saturated solution.



Why is it possible to form a supersaturated solution?

Solubility as a Function of Temperature

- In most cases (~95% of ionic substances), solubility increases with increasing temperature (positive slope of the solubility curves).



Aqueous solubility of several salts as a function of temperature.

For $KClO_4$ curve:

- Each point on the curve represents solubility at a given temperature.
- Regions above the curve (1) correspond to supersaturated solutions.
- Regions below the curve (2) correspond to unsaturated solutions.

- Trend of solubility curve can be explained by the Le Châtelier's principle.

- For solute with $\Delta H_{soln} > 0$,
  - Raising temperature stimulates dissolution. This *increases* the solubility of the solute.
  - e.g. $KClO_4$, KNO_3 (those with +ve slope of solubility curve)

- For solute with $\Delta H_{soln} < 0$,
  - Raising temperature stimulates crystallization. This *decreases* the solubility of the solute.
  - e.g. Li_2SO_4, Na_2SO_4 (those with -ve slope of solubility curve)

Recrystallization

Fractional crystallization

- The increased solubility of a solute in a solvent with increased temperature enables the use of recrystallization for purification.

- Recrystallization is a common method for separating a relatively pure, homogeneous solid from impurities.

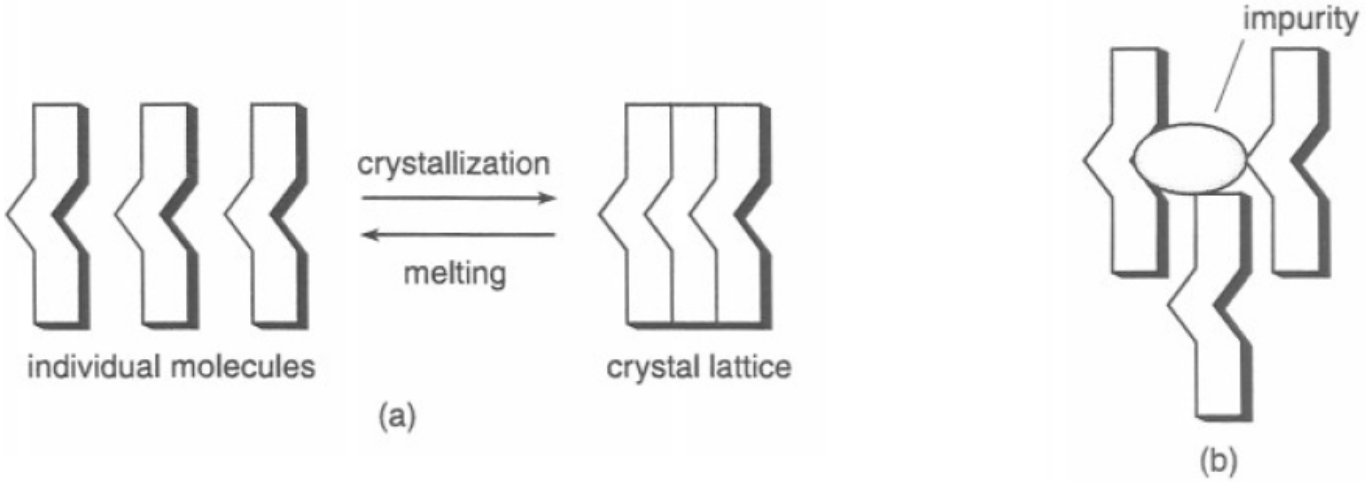


Recrystallization of KNO_3

Colourless crystals of KNO_3 separated from an aqueous solution of KNO_3 and $CuSO_4$ (an impurity in this case).

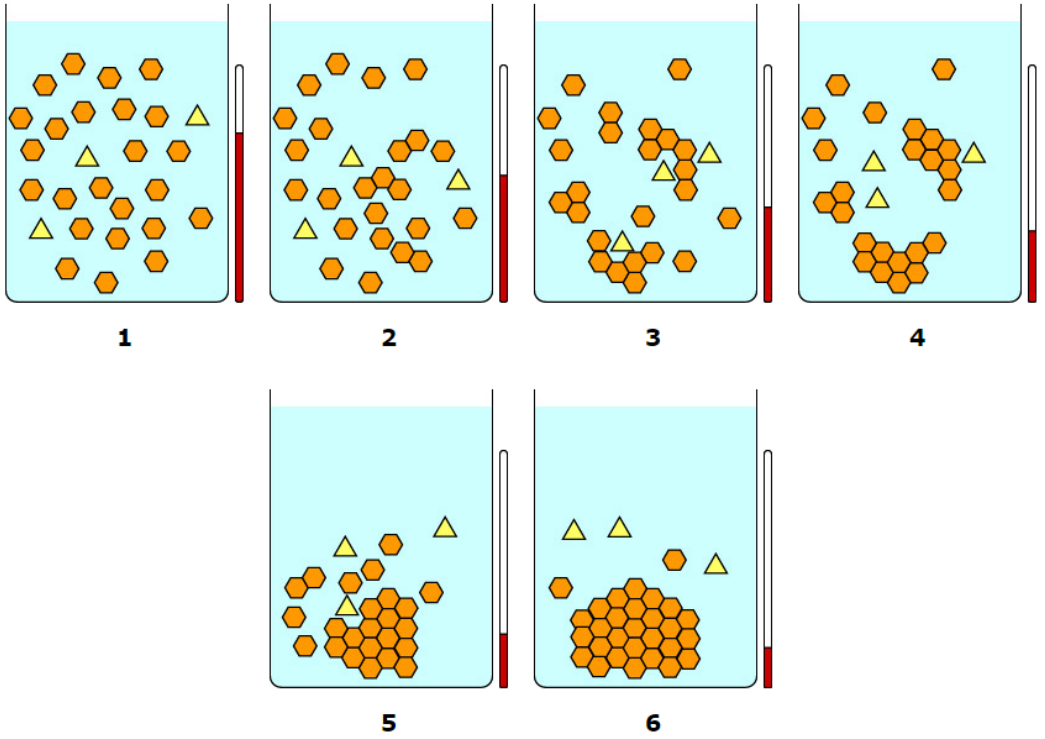
- Crystallization is a selective process. When a solid crystallizes at the right speed under the right conditions (conc & solvent), an almost perfectly pure crystalline material is formed.
  (∵ only molecules of the right shape fit into the crystal lattice)

Figure 1
(a) Identical molecules attached to one another, forming a crystal lattice.
(b) Impurities have different shapes or sizes and do not layer.



Crystals start growing by a process called "nucleation".

- *Nucleation* is the step where the solute molecules dispersed in the solvent start to gather into clusters. If the clusters are stable (i.e. reach a critical size), they constitute the nuclei for subsequent crystal growth. However, when the clusters are not stable, they dissolve.



Recrystallization

Fractional crystallization

- For the solution of an impure solid:

  [desired solute] >> [impurities]

- This means, as temperature decreases, the solution can always become saturated for desired solute (but not for impurities).

- When the concentrated solution cools, the impurities remain in solution while the desired solute crystallizes.



Recrystallization of KNO_3

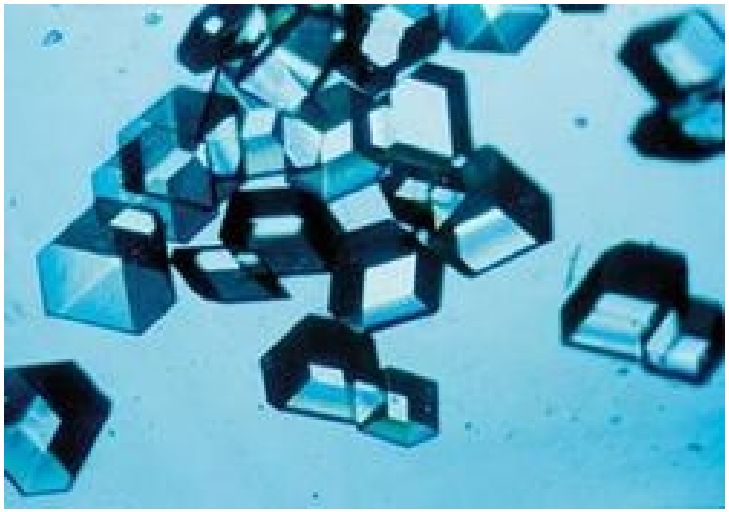
Colourless crystals of KNO_3 separated from an aqueous solution of KNO_3 and $CuSO_4$ (an impurity in this case).

Principle of Recrystallization

Impure crystal lattice → Dissolution in a Hot solvent → Slow cooling → **Pure crystal lattice is selectively produced**

Impurities either

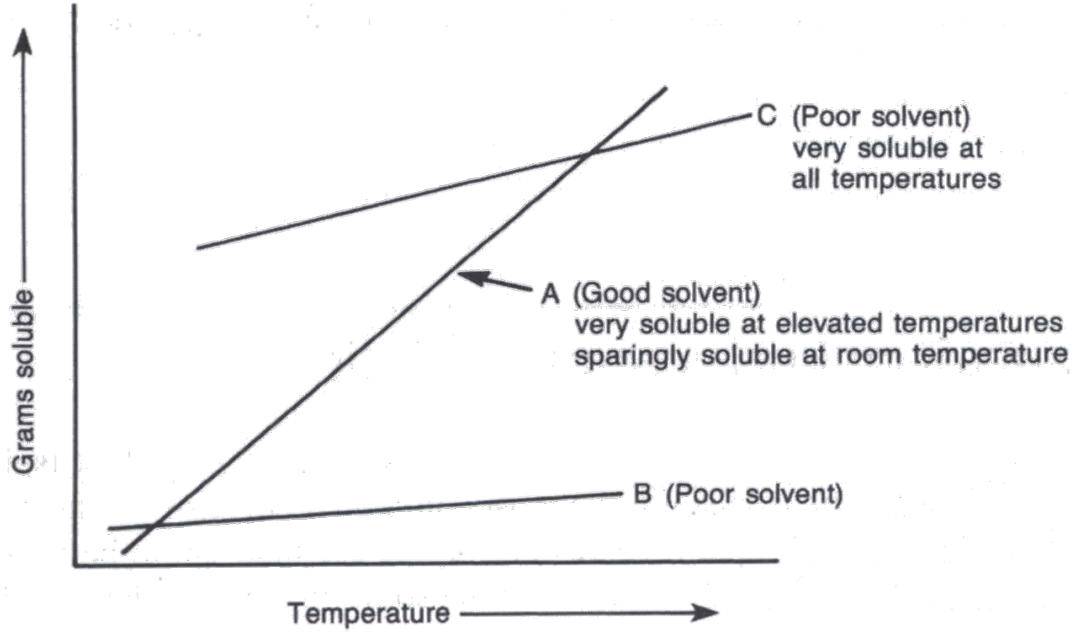
- being insoluble even in hot solution, or
- remain soluble in the cold solution



Recrystallization Technique

1. Selection of an appropriate solvent

- **Most crucial part**
- The solute should have a max. solubility in the hot solvent (~ 1g/20mL), and a min. solubility in the cold solvent (permits recovery of the purified product in high yield).



- The choice of solvent – a trial and error process ('like dissolves like').
- The impurities should either be insoluble in the solvent at all T or retain soluble in the cold solvent.
- The solvent should be of low boiling point (lower than melting point of the solid being purified).

Common Recrystallization Solvents

Dielectric constant is a measure of a substance's ability to insulate charges from each other.

One measure of polarity. ↑ ε ⇒ ↑ capability of dissolving polar/ionic cpds

Solvent	Boiling pt (°C)	Miscibility in water	Solvent Polarity	Dielectric constant (ε)
Diethyl ether	34.6	-	intermediate	4.3
Acetone	56	+	intermediate	20.7
Petroleum ether	60-80	-	non-polar	~ 2
Chloroform	61	-	intermediate	4.8
Methanol	65	+	polar	32.6
Hexane	69	-	non-polar	1.9
Ethyl acetate	77	-	intermediate	6.0
Ethanol	78.5	+	polar	24.3
Water	100		very polar	80
Toluene	110.6	-	non-polar	2.4
Acetic acid	118	+	intermediate	6.15

ε: 2-3 (non-polar), >10 (polar)



- First take a small sample (20-30 mg) of the compound to be recrystallized, place it in a test tube, and add 5-10 drops of a trial solvent.

- Shake the tube for mixing.
  - If the compound dissolves immediately → no good (too soluble).
  - If no solubility is observed, heat the solvent to its b.p., if solubility is then observed, cool the solution to observe crystallization.

- The formation of crystals in 10-20 min suggests that the solvent is a good one.

For example,

Which solvent(s) of the following should be chosen for purifying naphthalene?

The solubilities of naphthalene in each of the solvents are shown below. (+ means soluble; – means insoluble)

Solvent	Cold	Hot
Water	–	–
Ethanol	–	+
Acetone	+	+
Toluene	+	+

No good.
Undissolved
in both conditions

☑

No good.
Solubility is too high;
difficult to recover
the crystal

2. Dissolving the Solid

- Carefully pour the solid to be recrystallized into an Erlenmeyer flask.

anti-bumping granules

- Add 1-2 boiling chips or a boiling stick and heat an appropriate volume of the solvent in another Erlenmeyer flask.

- Then add small portion of hot solvent (just below boiling) to the solid (to avoid using too much solvent).

- Begin heating the solid-solvent mixture, allowing it to boil briefly between additions, until the solid dissolves; then add a little excess solvent.

Mixed Solvent Pairs

e.g.) ethanol & pet. ether;
ethyl acetate & hexane

- A pair of **miscible** solvents can be used when no single solvent seems to work.
- If the solute is very soluble in one solvent but marginally soluble in another solvent,

1. First add the solute into a flask and add a warm solvent, which dissolves it better than the other solvent. The solute should not be fully dissolved at this step.

2. Warm the solution nearly to the b.p. If the solute is not fully dissolved, add more solvent dropwise until it is fully dissolved.

3. Add the other solvent dropwise, until a slight cloudiness appears (saturated).

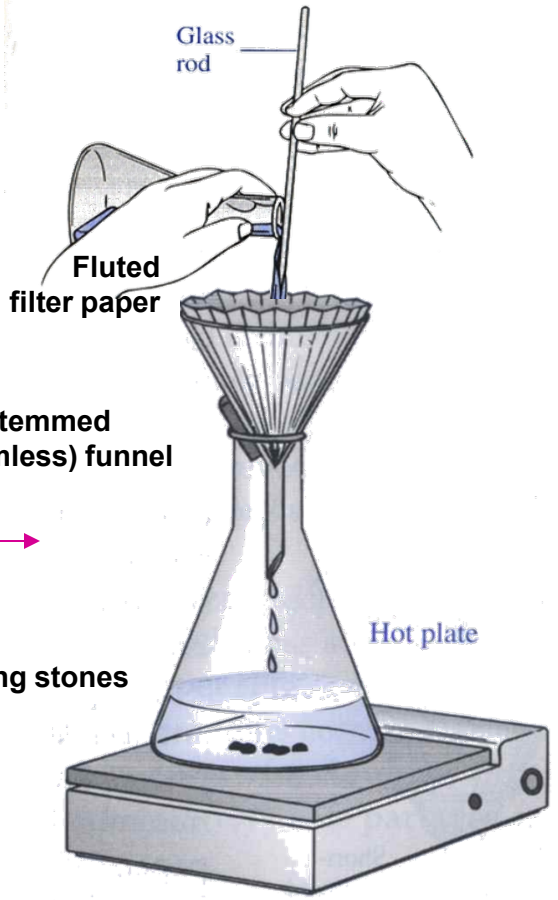
3. Add some of the first solvent again until the cloudiness just disappears (then add a few more drops).

4. Let the solution cool slowly to initiate the crystallization.

3. Filtering Insoluble Impurities (if any)

- **Gravity filtration** is necessary if there are **insoluble** material.
- Fluted filter paper and a short-stemmed funnel are used (suction filtration is not suitable in this application).

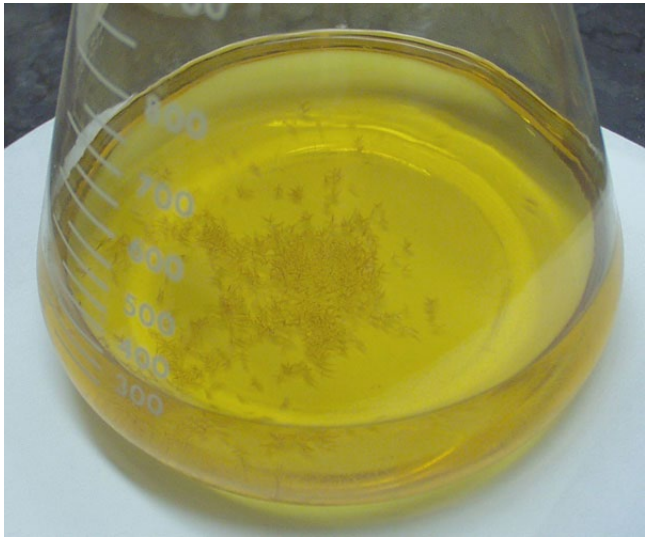
The hot solution should be added in small quantities → prevent cooling

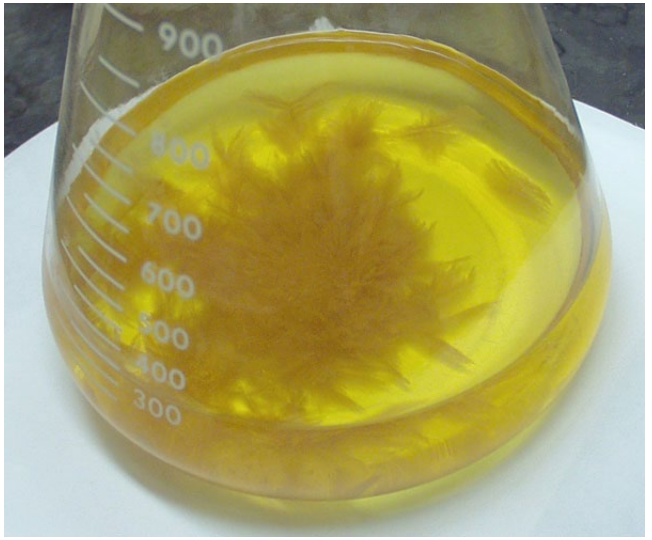


Keeping warm → Prevent premature crystallization

4. Cooling the Solution

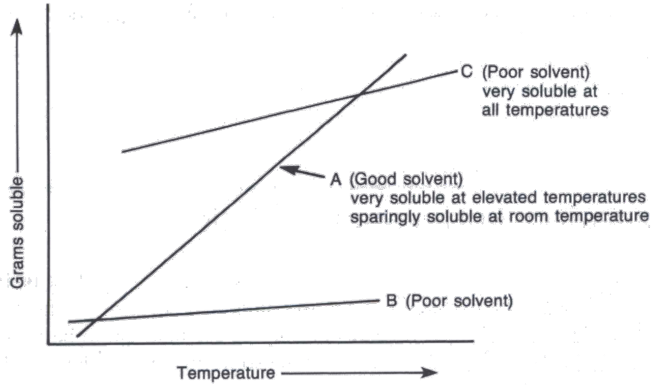
- Allow the solution to cool **SLOWLY**. The hot solution is allowed to stand on the bench top.
- **The size and purity of the crystals obtained depends on the rate of cooling.**
- Upon cooling, crystal formation will begin and the flask reaches room temp.
- Finally cooling in an ice-water bath for 15 mins.





Failure of crystallization

→ Too much solvent has been used





How to induce crystallization if the cooled solution does not crystallize?

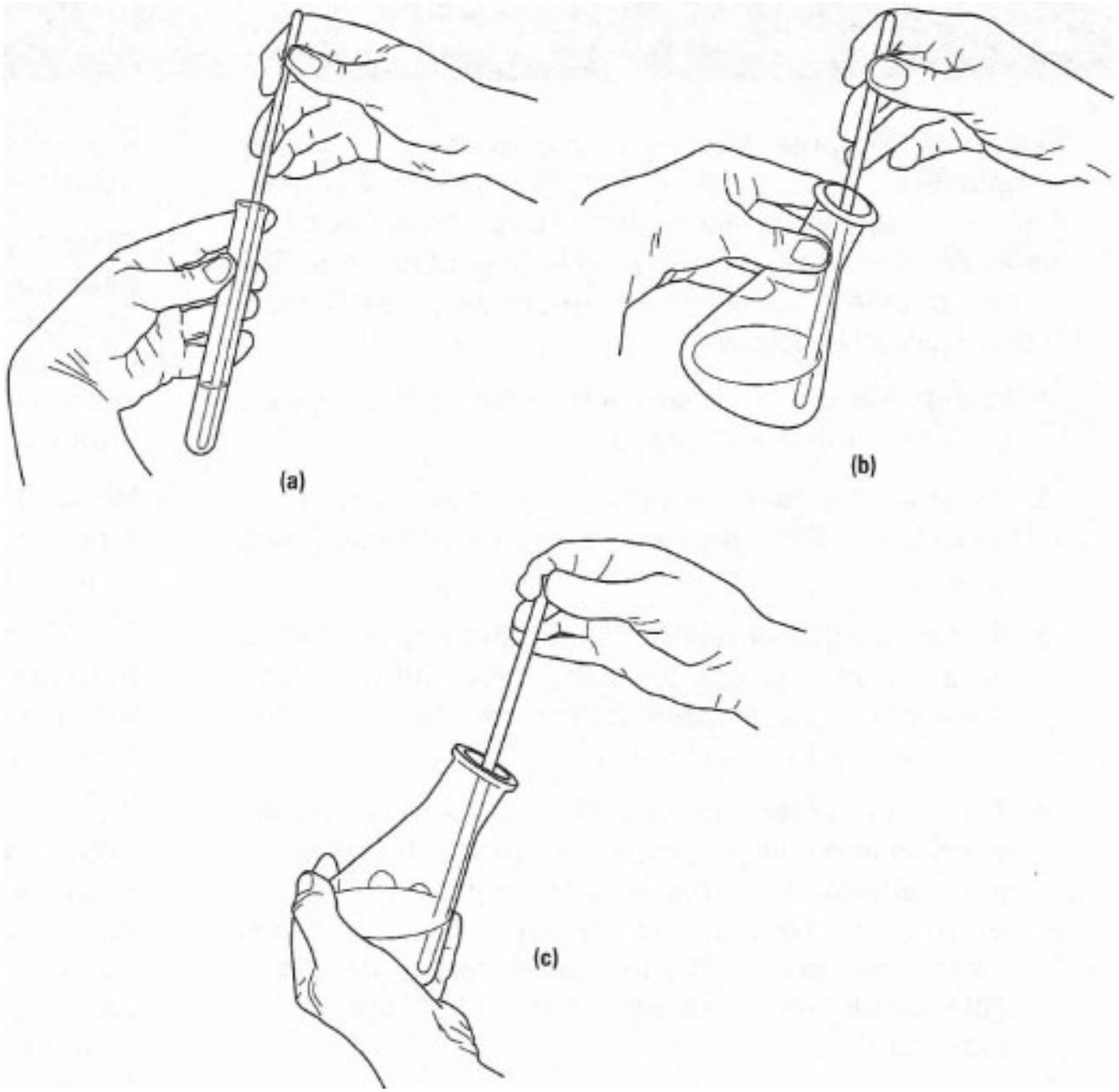
✓ Use a glass rod to scratch the inside surface of the flask vigorously *(be cautious)*

Vertical motion,
in and out of the solution

Or

✓ 'Seeding' (add a seed crystal which has been retained at the beginning)

Seed crystal – crystal of the pure compounds


(a)
(b)
(c)

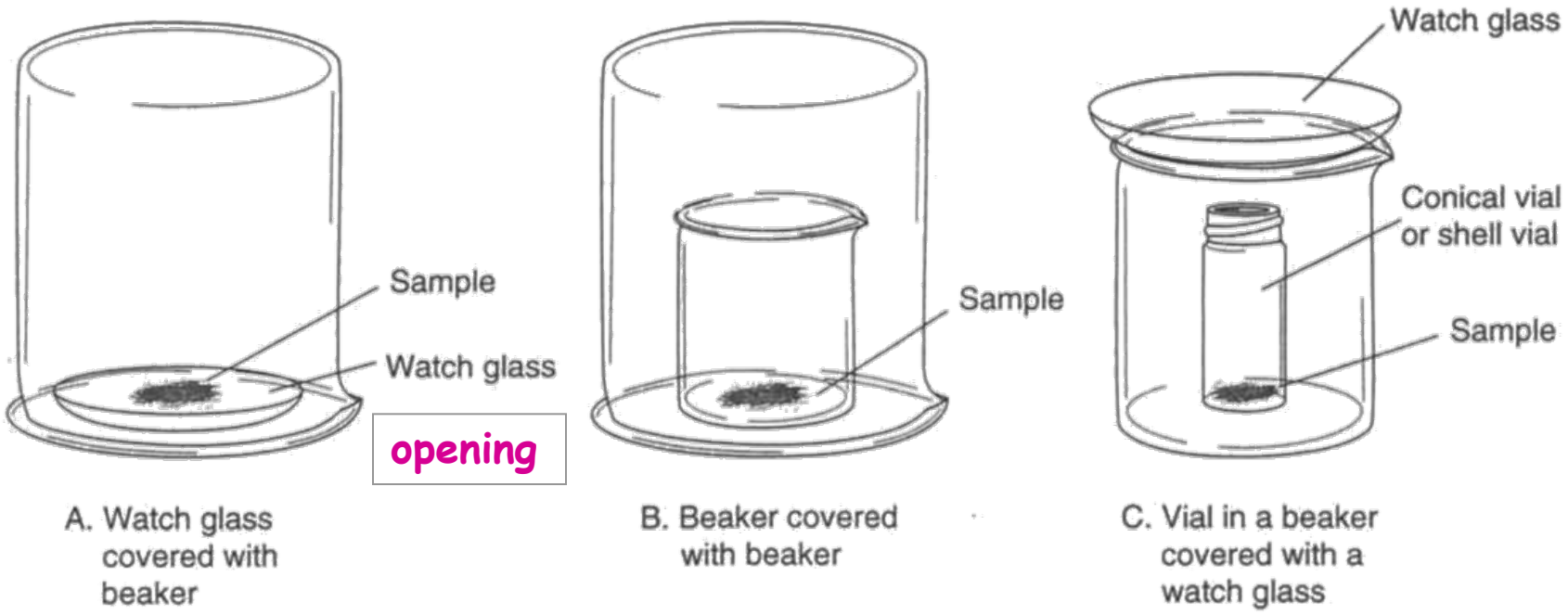
5. Collecting the Crystals



- By vacuum filtration – using a Buchner funnel (for macrcoscale), filter flask, and trap bottle or flask.
- After wetting the filter paper with the solvent with vacuum on, pour the slurry of crystals and solvent into the funnel.
- Wash the crystals with a small amount of **cold** solvent (1-5 mL, depends) with vacuum off.
- Then reconnect the vacuum, draw the liquid off.
- Initiate the crystal drying process by pulling air through them for a few minutes. Then disconnect the vacuum, leave the crystals open to the air for drying.

6. Drying the Crystals

- Most common method: To allow the crystals to dry in air.
  [advantage: no heat is required → prevent decomposition;
  disadvantage: may cause hydration (hygroscopic substance)]



A. Watch glass covered with beaker

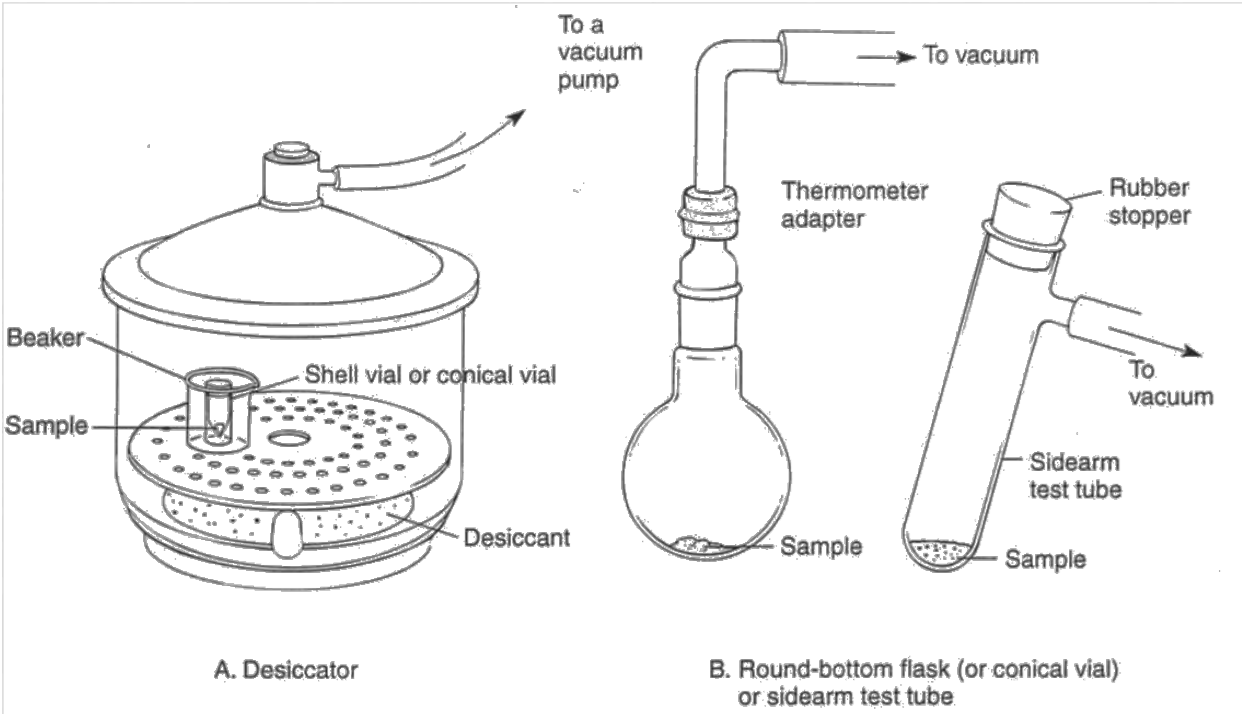
B. Beaker covered with beaker

C. Vial in a beaker covered with a watch glass

- Another method: To place the crystals in an oven (temp of the oven must be 20-30 °C below the m.p.). Crystals that sublime or decompose should NOT be dried using this method.

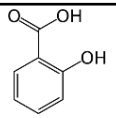
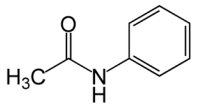
- Third method – **drying in *vacuo***

Beware of sublimation



Class Practice 4

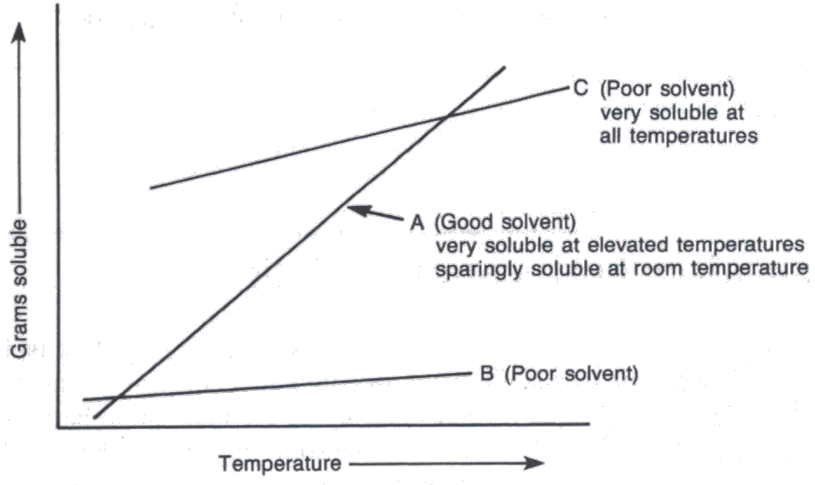
Consider using recrystallization to purify a 5.00 g sample of salicylic acid, which has been contaminated with 0.25 g of acetanilide (Total 5.25 g).

	Salicylic acid 	Acetanilide 
Solubility in water at 100 °C	7.5 g / 100 mL	
Solubility in water at 20 °C	0.20 g / 100 mL	0.5 g / 100 mL

(a) What is the minimum volume of hot water (at 100 °C) required to purify the sample?

(b) How many grams of salicylic acid can be recovered from this recrystallization?

(c) What is the percent recovery?





Class Practice 4 (Con't)

	Salicylic acid	Acetanilide
Solubility in water at 100 °C	7.5 g / 100 mL	
Solubility in water at 20 °C	0.20 g / 100 mL	0.5 g / 100 mL

Solution:

(a) Minimum volume of hot water (at 100 °C) required to purify the sample

= (100 mL/7.5 g) × 5.00 g = **67 mL**

(b) At 20 °C, the amount of salicylic acid remains in the solvent (water)

= (0.20 g/100 mL) × 67 mL

= 0.13 g

∴ 5.00 g – 0.13 g = **4.87 g** of salicylic acid is recovered.

(c) % recovery = (4.87 g/5.00 g) × 100 % = **97.4 %**

Class Practice 5

Fractional crystallization = recrystallization

EXAMPLE 14-4 Applying Solubility Data in Fractional Crystallization

A solution is prepared by dissolving 95 g NH_4Cl in 200.0 g H_2O at 60 °C.

(a) What mass of NH_4Cl will recrystallize when the solution is cooled to 20°C?

(b) How might we improve the yield of NH_4Cl?

Given the solubilities of NH_4Cl at the two corresponding temperatures as follows:

20°C	37 g NH_4Cl / 100 g H_2O
60°C	55 g NH_4Cl / 100 g H_2O

Solution:

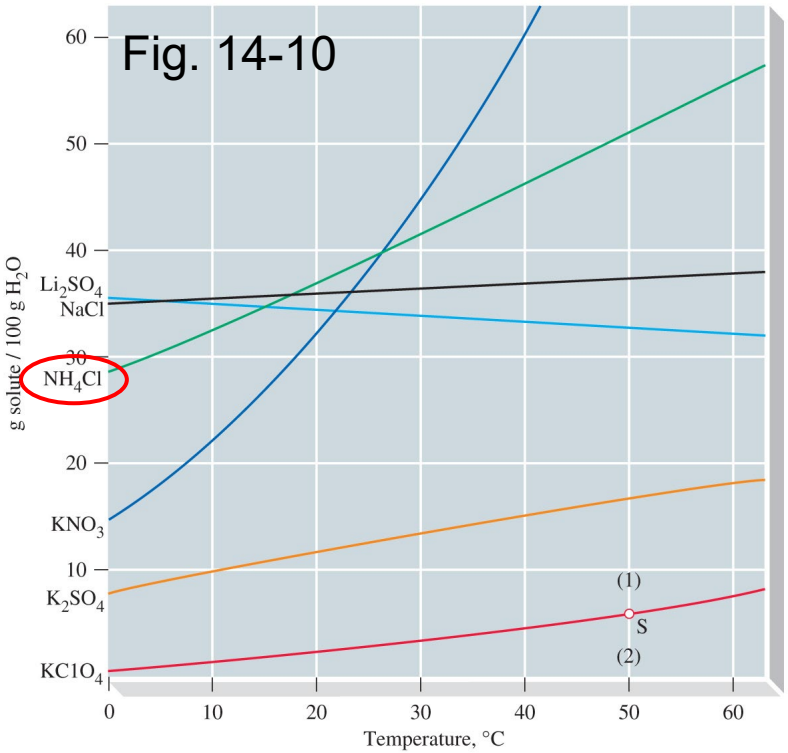
(a) Quantity of NH_4Cl in the saturated solution at 20 °C:

$$200.0\ \text{g}\ H_2O \times \frac{37\ \text{g}\ NH_4Cl}{100\ \text{g}\ H_2O} = 74\ \text{g}\ NH_4Cl$$

∴ Mass of NH_4Cl recrystallized = 95 g – 74 g = 21 g

Class Practice 5 (Con't)

(b) The yield of NH_4Cl in (a) is rather poor—21 g out of 95 g, or 22%. We can do better: (1) The solution at 60 °C, although concentrated, is not saturated. Using Figure 14-10, we estimate that a saturated solution at 60 °C has 55 g NH_4Cl/100 g H_2O. Thus, the 95 g NH_4Cl requires less than 200.0 g H_2O to make a saturated solution. At 20 °C, a smaller quantity of saturated solution would contain less NH_4Cl than in (a), and the yield of recrystallized NH_4Cl would be greater. (2) Instead of cooling the solution to 20 °C, we might cool it to 0 °C. Here the solubility of NH_4Cl is less than at 20 °C, and more solid would recrystallize. (3) Still another possibility is to start with a solution at a temperature higher than 60 °C, say closer to 100 °C. The mass of water needed for the saturated solution would be less than at 60 °C. Note that options (1) and (3) both require changing the conditions by using a different amount of water from that originally specified.



Solubilities of Gases







Why does soda go flat after a time?

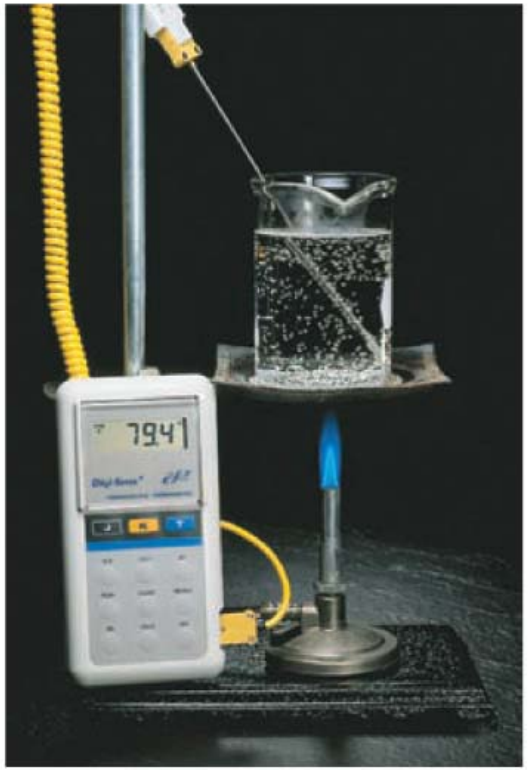
Effect of Temperature

- The effect of temperature on the solubility of gases is generally different from that on solid solutes. The pressure of a gas strongly affects its solubility.

- In water, the solubilities of most gases [e.g. $N_2(g)$ and $O_2(g)$] ***decrease*** with an increase in temperature. (the solution process is exothermic)



- However, in organic solvents, gases may become more soluble at higher temperatures. (endothermic)



Effect of temperature on the solubilities of gases
Dissolved air is released as water is heated, even at temperatures well below the boiling point.

Effect of Pressure of a Gas

- **Pressure has a stronger effect** on the solubility of a gas than temperature does.
- William Henry found that the solubility of a gas ***increases with increasing pressure.***

Henry's law: $C = k \times P_{gas}$

where C = solubility of a gas in a particular solvent at a fixed T,
P_{gas} = partial pressure of the gas above the solution, and
k = proportionality constant.

Henry's law: $C = k \times P_{gas}$

where C = Concentration of a gas in a particular solvent at a fixed T,
P_{gas} = partial pressure of the gas above the solution, and
k = proportionality constant.

E.g.) The aqueous solubility of $N_2(g)$ at 0 °C and 1.00 atm is 23.54 mL N_2 per litre, then k would be:

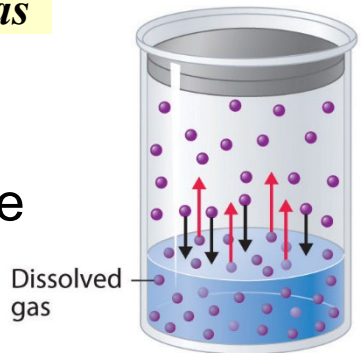
$$k = \frac{C}{P_{gas}} = \frac{23.54 \text{ mL } N_2/\text{L}}{1.00 \text{ atm}}$$

Suppose we want to increase the solubility of $N_2(g)$ to 100.00 mL N_2 per litre. we must then increase the pressure of $N_2(g)$ above the solution to:

$$P_{N_2} = \frac{C}{k} = \frac{100.0 \text{ mL } N_2/\text{L}}{(23.54 \text{ mL } N_2/\text{L})/1.00 \text{ atm}} = 4.25 \text{ atm}$$

Rationalisation of Henry's law

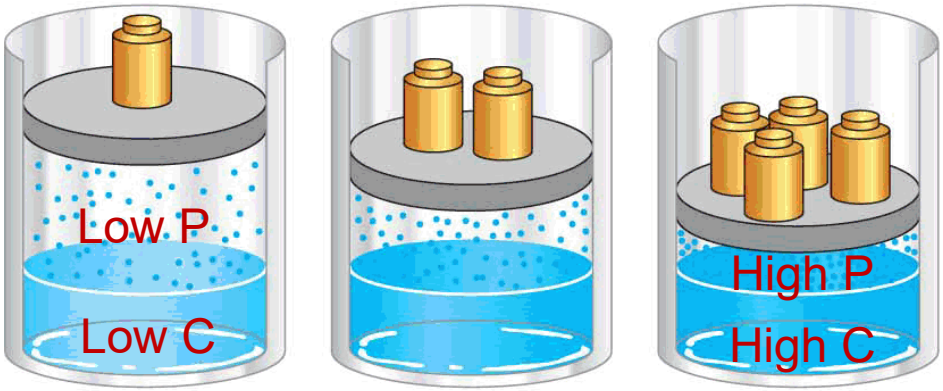
$$C = k \times P_{gas}$$

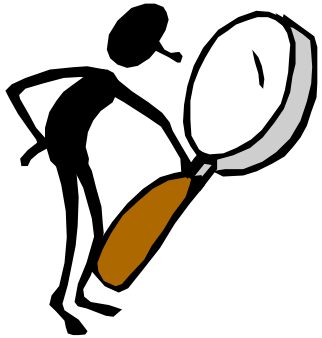


- In a saturated solution, rate of evaporation of gas molecules from solution = rate of condensation of gas molecules into the solution.
- Both rates depend on the no. of molecules per unit volume.
- With increasing pressure on the system,
  1. No. of molecules per unit volume in the gaseous state increases (∵ increase in gas pressure).
  2. No. of molecules per unit volume in the solution must also increase (∵ increase in concentration).

Effect of pressure on the solubility of a gas

The concentration of dissolved gas (suggested by the depth of color) is proportional to the pressure of the gas above the solution (suggested by the density of the dots).







$$C = k \times P_{gas}$$

Think about what would happen when a bottle of soda water is opened.

According to Henry's law, ↑ P_{gas} ⇒ ↑ C.

……...



Deep-sea diving is another example of Henry's law.

The compressed air, that divers breathe, is high-pressure air and much more soluble in the blood and other body fluids than is air at normal pressures. When a diver returns to the surface, excess dissolved $N_2(g)$ is released as tiny bubbles from body fluids. When the ascent to the surface is made too quickly, N_2 diffuses out of the blood too quickly, causing severe pain in the limbs and joints. This dangerous condition, known as the bends, can be avoided if the diver ascends very slowly or spends time in a decompression chamber.

More about Henry's law

- Note that Henry's law fails for gases at high pressures; it also fails if the gas ionizes in water or reacts with water.

  E.g.) at 20 °C and with P_{HCl} = 1 atm, a saturated solution of HCl (aq) is about 20 M. But to prepare 10 M HCl, we do not need to maintain P_{HCl} = 0.5 atm above the solution, nor is P_{HCl} = 0.05 atm above 1 M HCl. We cannot even detect HCl(g) above 1 M HCl by its odor. The reason we cannot is that HCl ionizes in aqueous solutions, and in dilute solutions there are almost no molecules of HCl.

$$HCl\,(g) \xrightarrow{H_2O} H^+(aq) + Cl^-(aq)$$

- Henry's law applies only to the equilibrium between molecules of a gas and the **same** molecules in solution.



Class Practice 6

EXAMPLE 14-5 Using Henry's Law

At 0 °C and an O_2 pressure of 1.00 atm, the aqueous solubility of $O_2(g)$ is 48.9 mL O_2 per litre. What is the molarity of O_2 in a saturated water solution when the O_2 is under its normal partial pressure in air, 0.2095 atm?

Note that when working problems involving gaseous solutes in a solution in which the solute is at very low concentration, use Henry's law.

Solubility

Volume of O_2 per litre of water ⟺ Molarity (i.e. no. of mol of O_2 per litre of water)



Class Practice 6 (Con't)

Analyze

Think of this as a two-part problem. (1) Determine the molarity of the saturated O_2 solution at 0 °C and 1 atm. (2) Use Henry's law in the manner just outlined.

Solve

Determine the molarity of O_2 at 0 °C when $P_{O_2} = 1$ atm. We are given the information that, at an O_2 pressure of 1.00 atm, a saturated solution of O_2 in water contains 48.9 mL (0.0489 L) of O_2. We also know that, at 0 °C and 1.00 atm, 1 mol O_2 occupies a volume of 22.4 L. Therefore,

$$\text{molarity} = \frac{0.0489\ \text{L} \times \dfrac{1\ \text{mol}}{22.4\ \text{L}}}{1\ \text{L soln}} = 2.18 \times 10^{-3}\ \frac{\text{mol}}{\text{L soln}} = 2.18 \times 10^{-3}\ \text{M}$$

Evaluate the Henry's law constant.

$$k = \frac{C}{P_{\text{gas}}} = \frac{2.18 \times 10^{-3}\ \text{M}}{1.00\ \text{atm}}$$

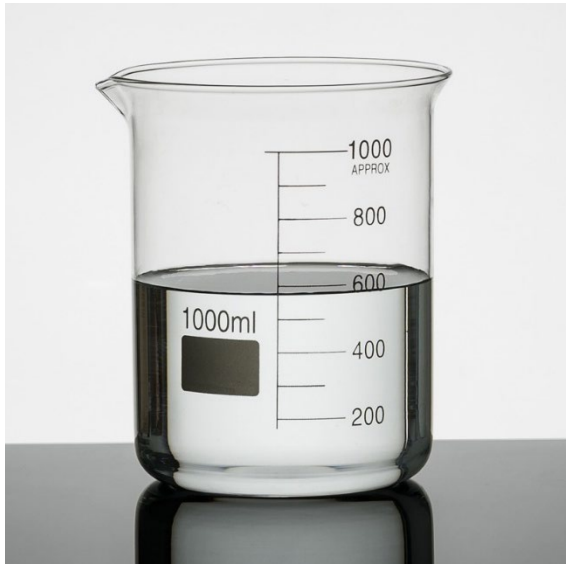
Apply Henry's law.

$$C = k \times P_{\text{gas}} = \frac{2.18 \times 10^{-3}\ \text{M}}{1.00\ \text{atm}} \times 0.2095\ \text{atm} = 4.57 \times 10^{-4}\ \text{M}$$

Assess

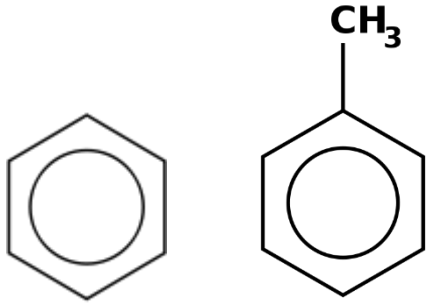
When working problems involving gaseous solutes in a solution in which the solute is at very low concentration, use Henry's law.

Vapour Pressures of Solutions

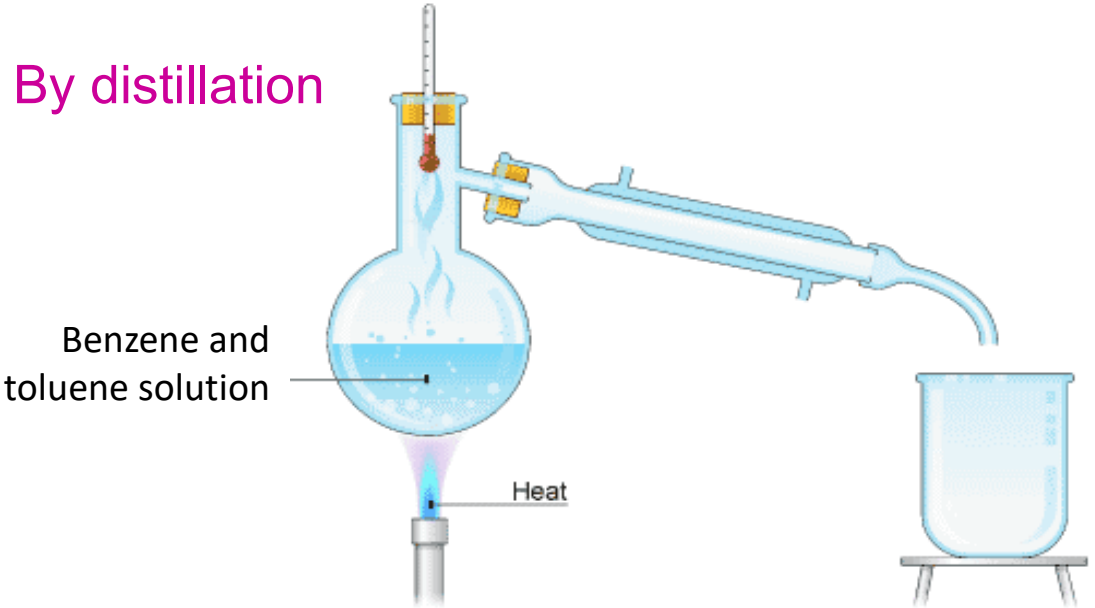


Consider a mixture of benzene and toluene, how could you separate the two components?





By distillation

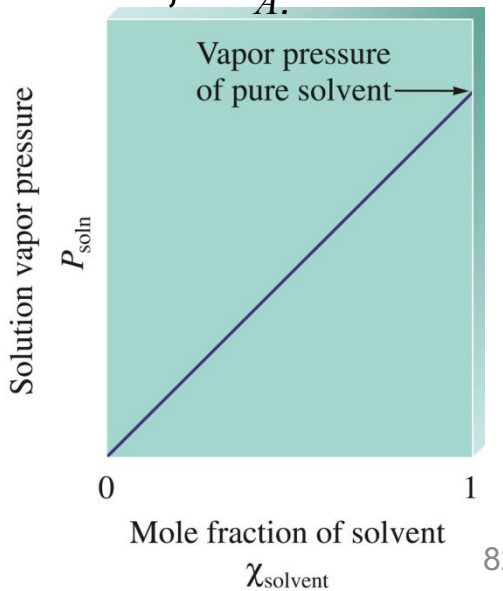


Vapour Pressures of Solutions

- Consider only solutions with two components, **solvent A and solute B**, and both components are ***volatile***. In the 1880s, F. M. Raoult found that a dissolved solute ***lowers*** the vapour pressure of the solvent.

- **Raoult's law** states that the partial pressure, P_A, exerted by solvent vapour above an **ideal solution** is the product of the mole fraction of solvent in the solutions, x_A, and the vapour pressure of the pure solvent at the given temperature, P^o_A.

Raoult's law: $$P_A = x_A P_A^\circ$$



Raoult's law: $P_A = x_A P_A^\circ$

- Consider a solution with two components, solvent A and solute B, the dissolved solute lowers the vapour pressure of the solvent because if $x_A + x_B = 1.00$, x_A must be less than 1.00, and P_A must be smaller than $P^o{}_A$.
- Raoult's law applies only to ideal solutions and to all volatile components of the solutions. For non-ideal solutions, the law often works reasonably well for the solvents in dilute solutions, e.g. solutions of $x_{solv} > 0.98$.

Class Practice 7

EXAMPLE 14-6 Predicting Vapor Pressures of Ideal Solutions

The vapor pressures of pure benzene and pure toluene at 25 °C are 95.1 and 28.4 mmHg, respectively. A solution is prepared in which the mole fractions of benzene and toluene are both 0.500. What are the partial pressures of the benzene and toluene above this solution? What is the total vapor pressure?

Note that a benzene-toluene solution should be ideal and so Raoult's law can be applied here.

Solution:

$$P_{benz} = x_{benz}P^*_{benz} = 0.500 \times 95.1\ mmHg = 47.6\ mmHg$$
$$P_{tol} = x_{tol}P^*_{tol} = 0.500 \times 28.4\ mmHg = 14.2\ mmHg$$
$$P_{total} = P_{benz} + P_{tol} = 47.6\ mmHg + 14.2\ mmHg = 61.8\ mmHg$$

Assess

In this example we assumed these to be ideal solutions, which allowed us to use Raoult's law. We observe that the vapor pressure of each component is lowered because of the presence of the other component.

Class Practice 8

EXAMPLE 14-7 Calculating the Composition of Vapor in Equilibrium with a Liquid Solution

What is the composition of the vapor in equilibrium with the benzene–toluene solution of Example 14-6?

Class Practice 7

Analyze

We are being asked to find the mole fraction of benzene and of toluene in the vapor. From Example 14-6 we know the vapor pressure of pure benzene and pure toluene. We have already calculated the partial vapor pressures; now we need to apply the definition of mole fraction.

Solution:

$$\frac{n_A}{n_{tot}} = \frac{P_A}{P_{tot}} = \frac{V_A}{V_{tot}} = x_A$$

The ratio of each partial pressure to the total pressure is the mole fraction of that component in the vapor. (This is another application of equation 6.17.) The mole-fraction composition of the vapor is

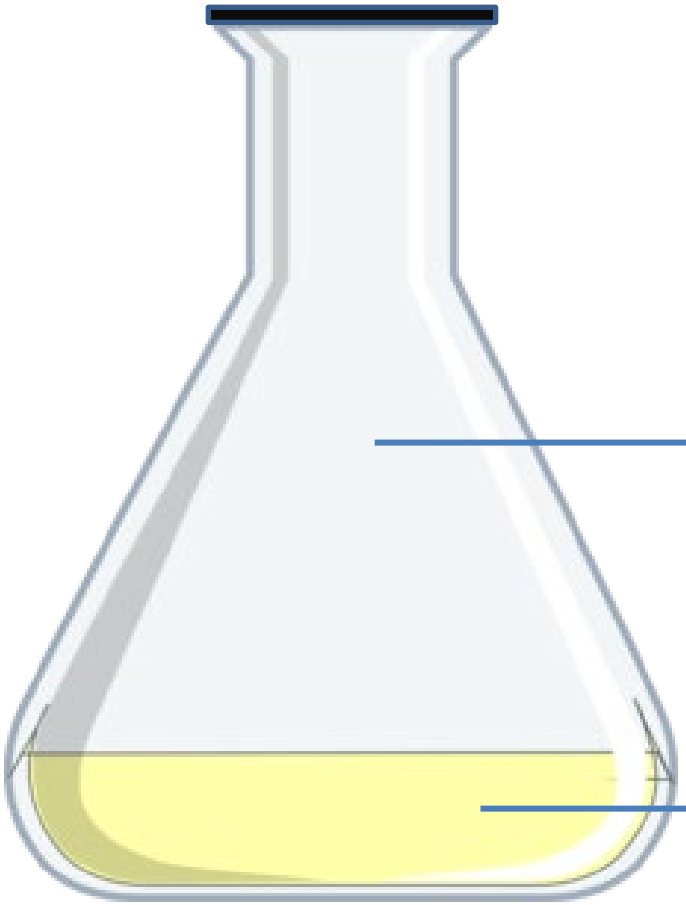
$$x_{benz} = \frac{P_{benz}}{P_{total}} = \frac{47.6\ \text{mmHg}}{61.8\ \text{mmHg}} = 0.770$$

$$x_{tol} = \frac{P_{tol}}{P_{total}} = \frac{14.2\ \text{mmHg}}{61.8\ \text{mmHg}} = 0.230$$

Assess

The mole fraction of benzene in the vapor is 0.770, whereas in the liquid the mole fraction of benzene is 0.5. For toluene the mole fraction in the vapor is 0.230, whereas in the liquid the mole fraction of toluene is 0.5. This difference in mole-fraction vapor composition caused by the difference in vapor pressures of the two components is the central concept of fractional distillation, which is discussed next.

A benzene-toluene liquid in Class Practices 7 & 8



Vapor composition

$x_{benzene} = 0.770$;

$x_{toluene} = 0.230$

Liquid composition

$x_{benzene} = 0.500$;

$x_{toluene} = 0.500$

Class Practice 7

EXAMPLE 14-6 Predicting Vapor Pressures of Ideal Solutions

The vapor pressures of pure benzene and pure toluene at 25 °C are 95.1 and 28.4 mmHg, respectively. A solution is prepared in which the mole fractions of benzene and toluene are both 0.500. What are the partial pressures of the benzene and toluene above this solution? What is the total vapor pressure?

Note that a benzene-toluene solution should be ideal and so Raoult's law can be applied here.

Solution:

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$$P_{tol} = x_{tol}P^*_{tol} = 0.500 \times 28.4\ mmHg = 14.2\ mmHg$$
$$P_{total} = P_{benz} + P_{tol} = 47.6\ mmHg + 14.2\ mmHg = 61.8\ mmHg$$

Assess

In this example we assumed these to be ideal solutions, which allowed us to use Raoult's law. We observe that the vapor pressure of each component is lowered because of the presence of the other component.

Class Practice 8

EXAMPLE 14-7 Calculating the Composition of Vapor in Equilibrium with a Liquid Solution

What is the composition of the vapor in equilibrium with the benzene–toluene solution of Example 14-6?

Class Practice 7

Analyze

We are being asked to find the mole fraction of benzene and of toluene in the vapor. From Example 14-6 we know the vapor pressure of pure benzene and pure toluene. We have already calculated the partial vapor pressures; now we need to apply the definition of mole fraction.

Solution:

$$\frac{n_A}{n_{tot}} = \frac{P_A}{P_{tot}} = \frac{V_A}{V_{tot}} = x_A$$

The ratio of each partial pressure to the total pressure is the mole fraction of that component in the vapor. (This is another application of equation 6.17.) The mole-fraction composition of the vapor is

$$x_{benz} = \frac{P_{benz}}{P_{total}} = \frac{47.6\ \text{mmHg}}{61.8\ \text{mmHg}} = 0.770$$

$$x_{tol} = \frac{P_{tol}}{P_{total}} = \frac{14.2\ \text{mmHg}}{61.8\ \text{mmHg}} = 0.230$$

Assess

The mole fraction of benzene in the vapor is 0.770, whereas in the liquid the mole fraction of benzene is 0.5. For toluene the mole fraction in the vapor is 0.230, whereas in the liquid the mole fraction of toluene is 0.5. This difference in mole-fraction vapor composition caused by the difference in vapor pressures of the two components is the central concept of fractional distillation, which is discussed next.

Liquid-Vapour Equilibrium: Ideal Solutions

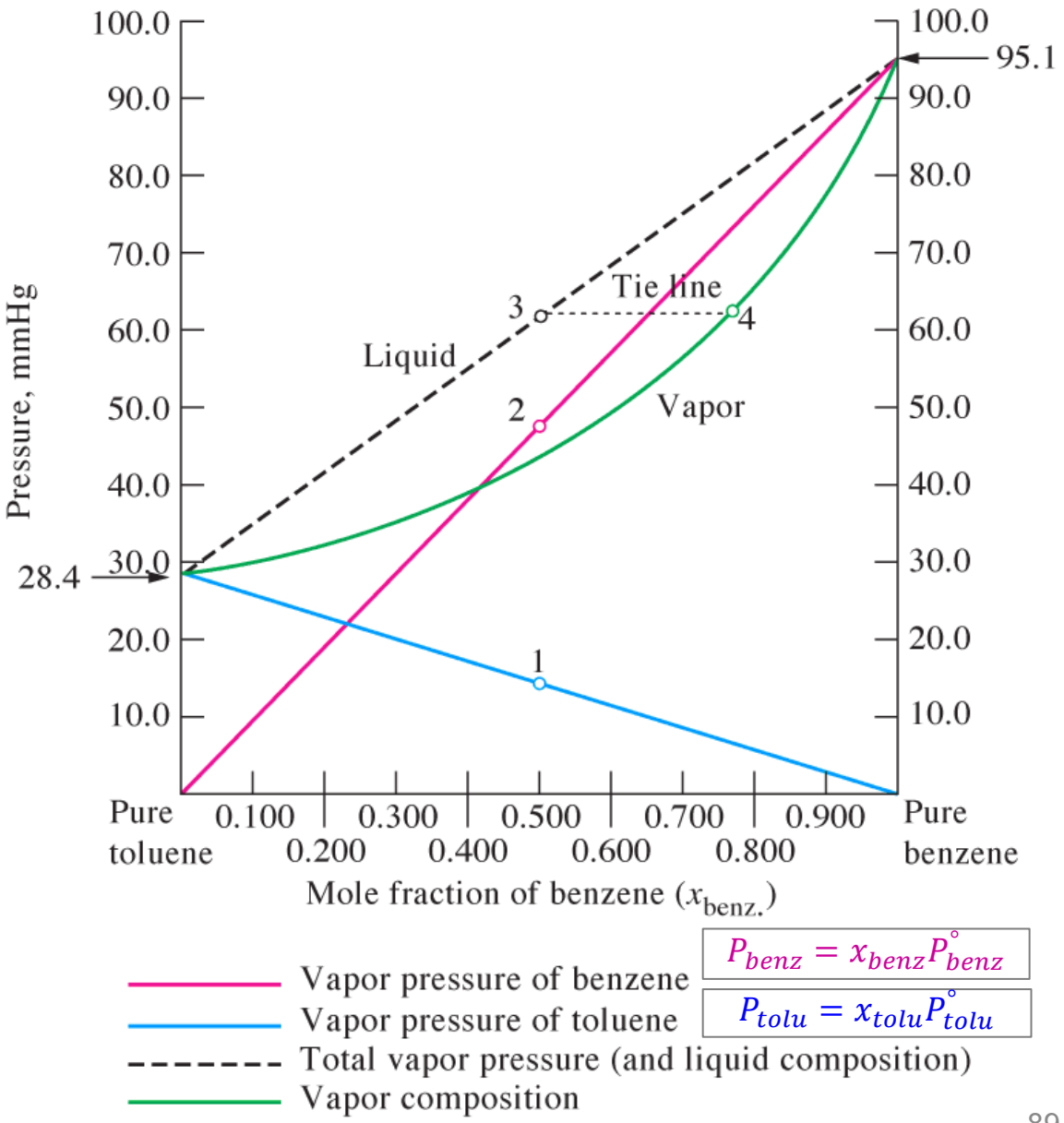
Fig. 14-16) Liquid vapour equilibrium for benzene-toluene mixtures at 25 °C

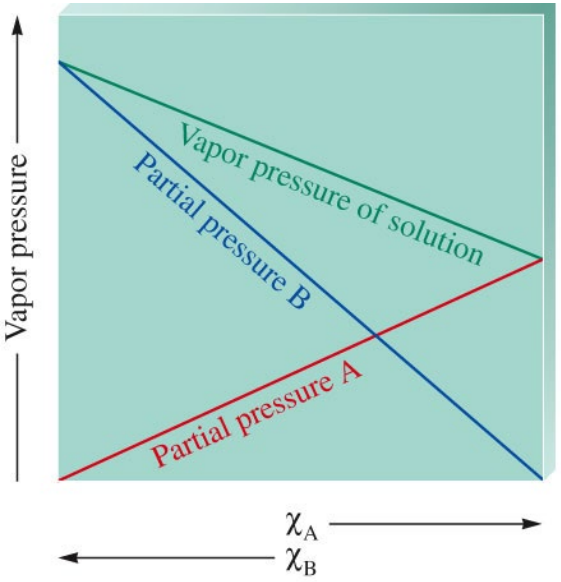
In this diagram, partial pressures and the total pressure of the vapour are plotted as a function of the solution and vapour compositions.

Note:

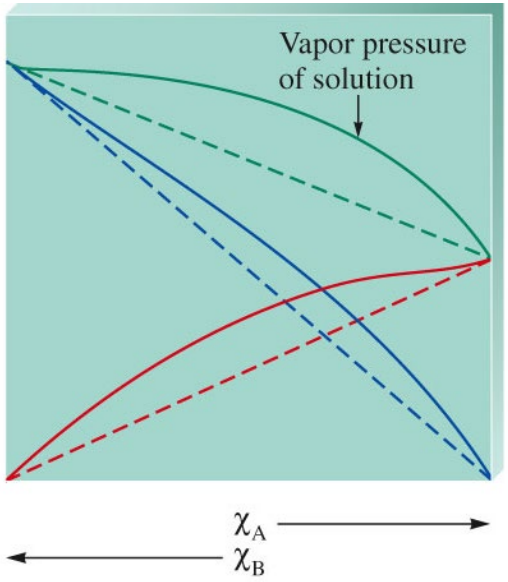
(1) Each pressure on the black dashed line is the sum of the pressures on the two straight lines that lie below it [i.e. point 3 = point 1 + point 2 (see CP7)]. 61.8 = 14.2 + 47.6

(2) The vapour phase is richer in the more volatile component than is the liquid phase (as calculated in CP8).

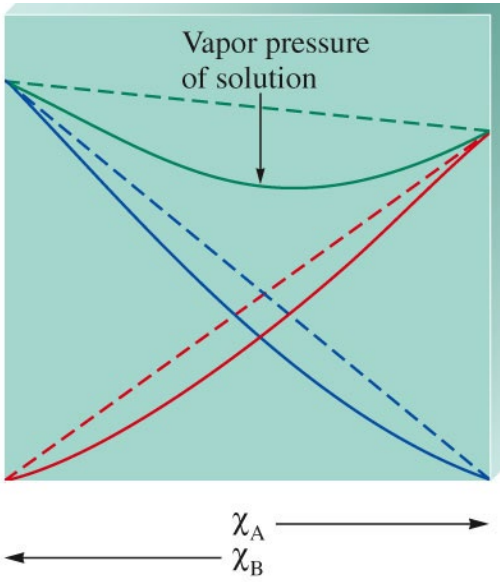




(a) Ideal solution



(b) Weak solute–solvent interactions



(c) Strong solute–solvent interactions

Interactive Forces Between Solute (A) and Solvent (B) Particles	ΔH_{soln}	ΔT for Solution Formation	Deviation from Raoult's Law	Example
(a) A ↔ A, B ↔ B ≡ A ↔ B	Zero	Zero	None (ideal solution)	Benzene-toluene
(b) A ↔ A, B ↔ B > A ↔ B	Positive (endothermic)	Negative	Positive (non-ideal solution)	Ethanol-hexane
(c) A ↔ A, B ↔ B < A ↔ B	Negative (exothermic)	Positive	Negative (non-ideal solution)	Acetone-chloroform

Fractional Distillation

- Instead of plotting vapour pressures as a function of the solution and vapour composition (as in Fig. 14-16), let us look at the normal boiling temperatures (at which the total vapour pressure of the solution is 1 atm) of the benzene-toluene mixtures (see Fig. 14-17 next slide). This graph is useful in explaining fractional distillation, a procedure for separating volatile liquids from one another.

Fig. 14-16) Liquid vapour equilibrium for benzene-toluene mixtures at 25 °C

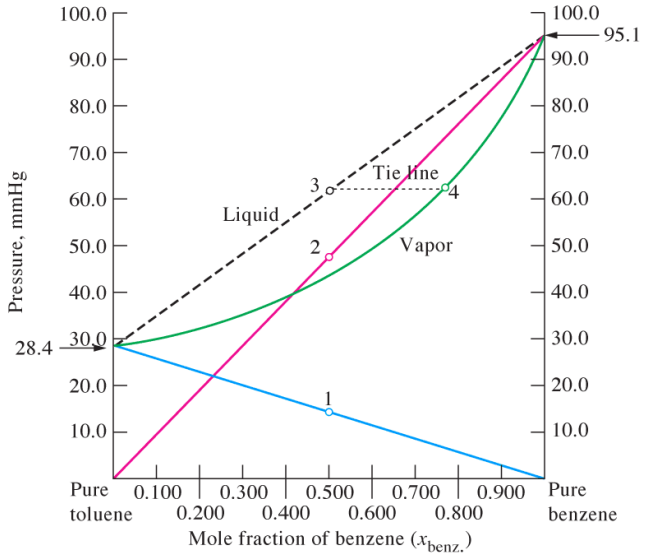


Fig. 14-17) Liquid vapour equilibrium for benzene-toluene mixtures at 1 atm

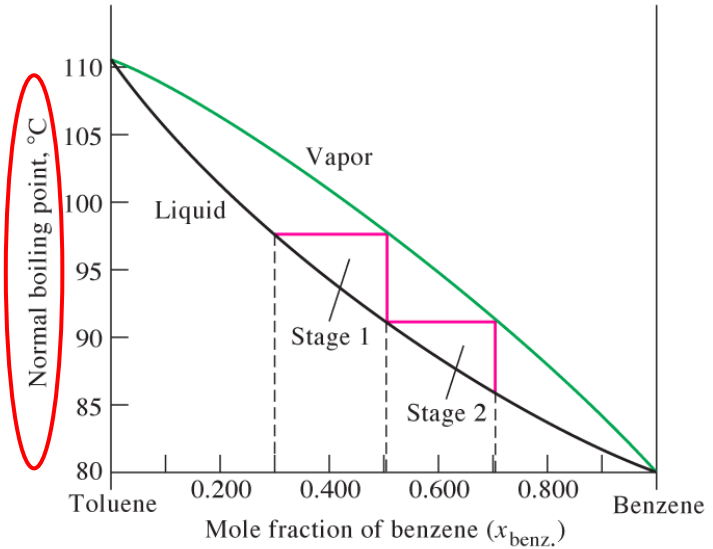
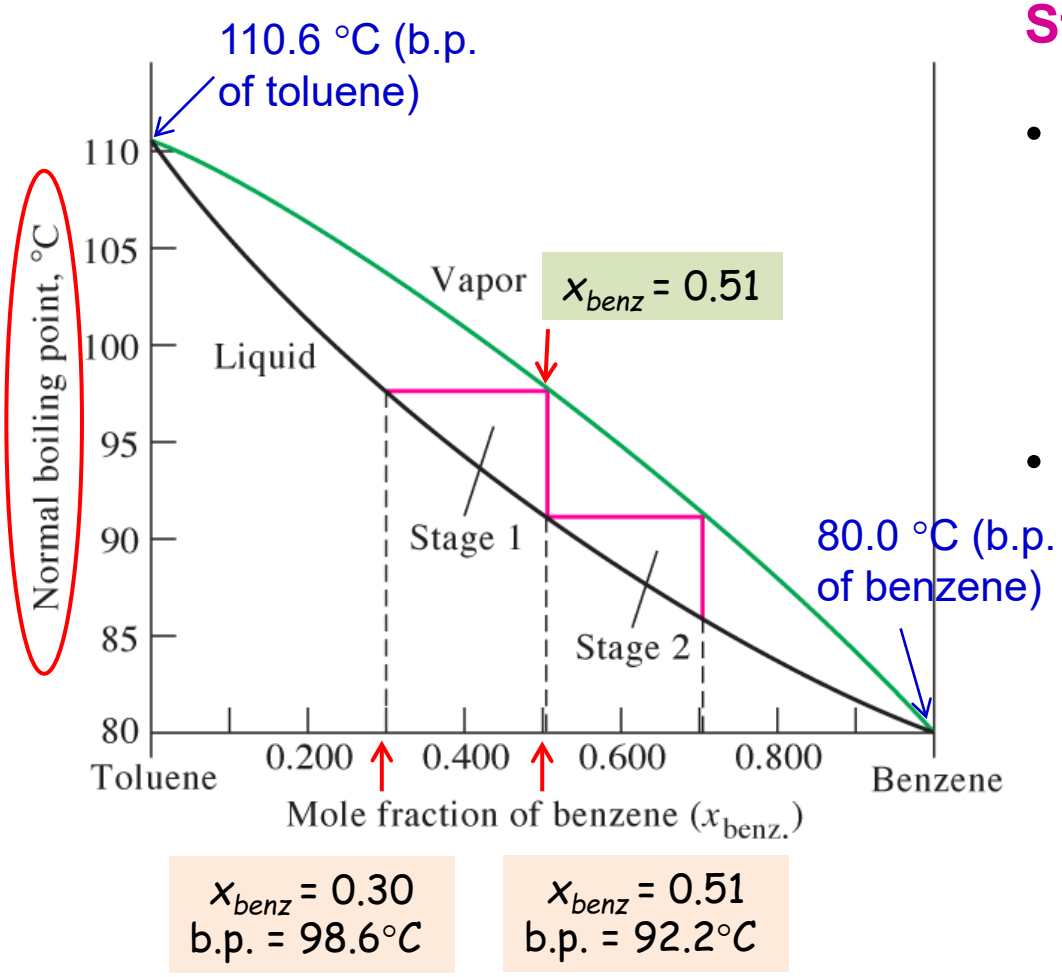


Fig. 14-17) Liquid vapour equilibrium for benzene-toluene mixtures at 1 atm

In this diagram, the normal boiling points of solutions are plotted as a function of solution and vapour compositions.

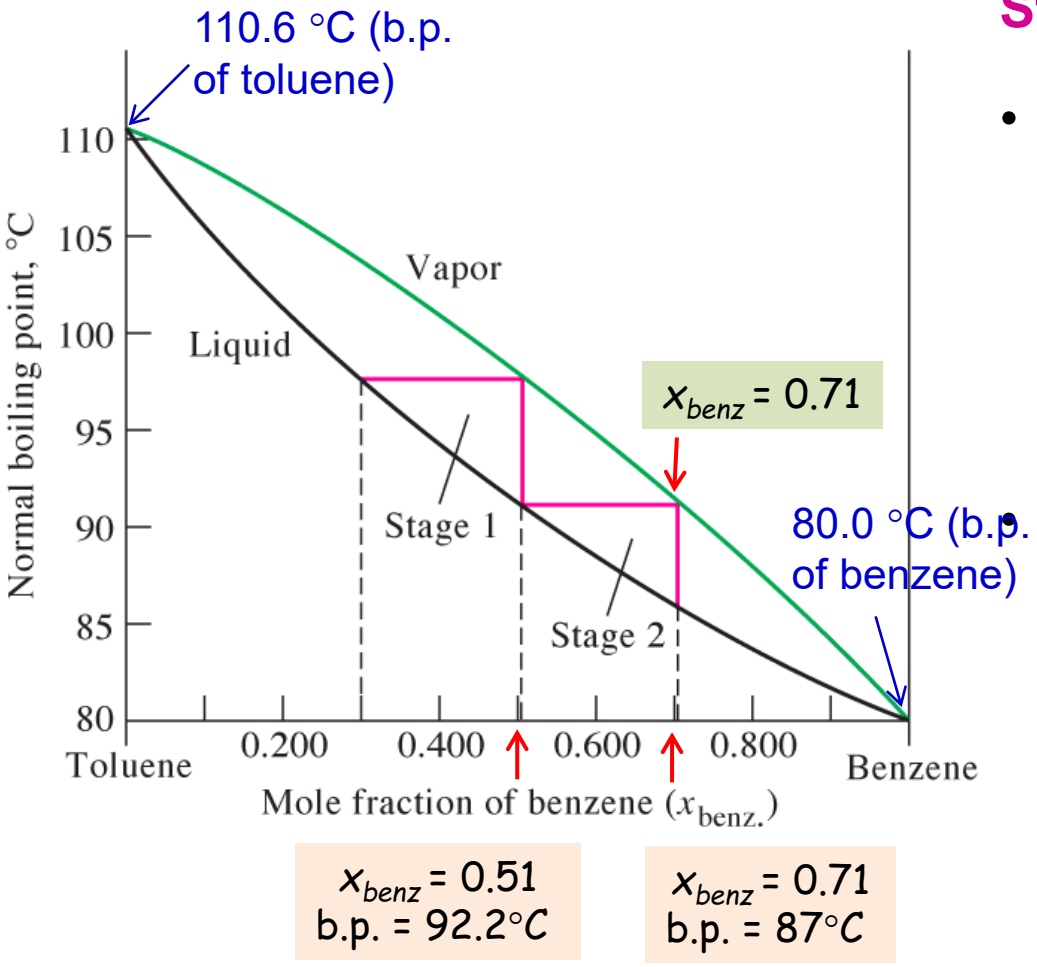


Stage 1:

- A benzene-toluene solution with x_{benz} = 0.30 boils at 98.6 °C and is in equilibrium with a vapour in which x_{benz} = 0.51.
- Imagine extracting some of this vapour and cooling it until it condenses to liquid. The new liquid will have x_{benz}= 0.51 which is the conclusion of stage 1.

Fig. 14-17) Liquid vapour equilibrium for benzene-toluene mixtures at 1 atm

In this diagram, the normal boiling points of solutions are plotted as a function of solution and vapour compositions.

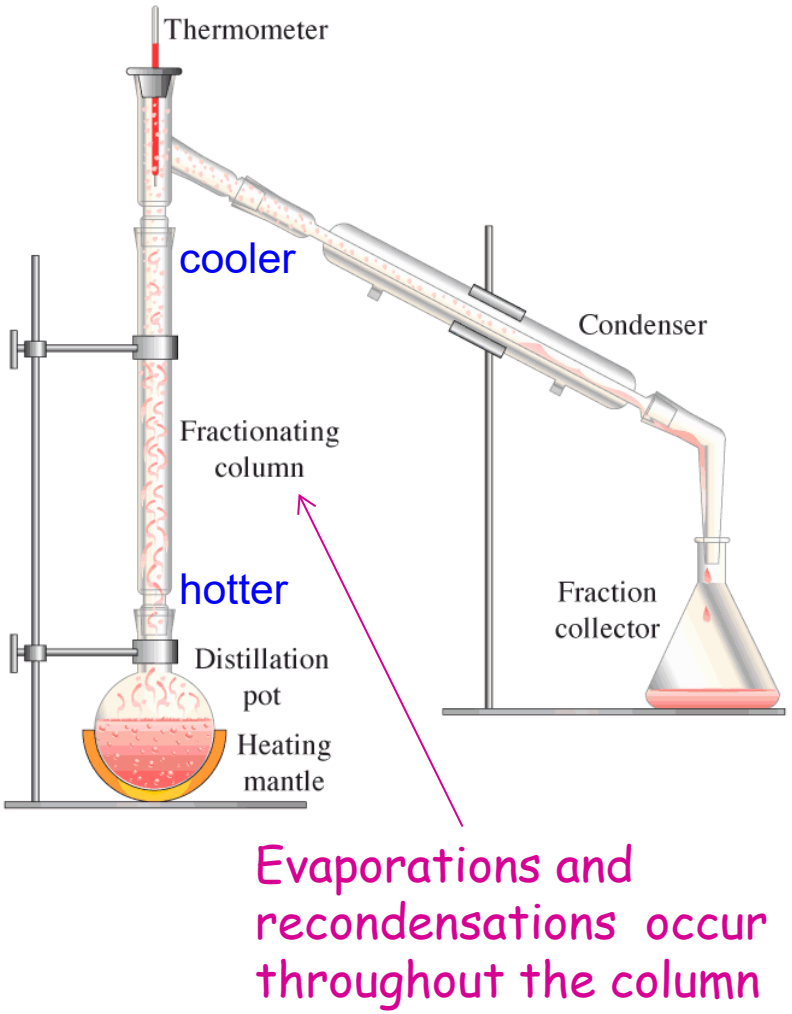


Stage 2:

- Imagine repeating the process, i.e. vaporising a solution with x_{benz} = 0.51 and condensing the vapour. The new liquid at the end of stage 2 has x_{benz} = 0.71.
- By repeating the cycle ##, the vapour becomes progressively richer in benzene.

a fractionating column can be used (see next slide).

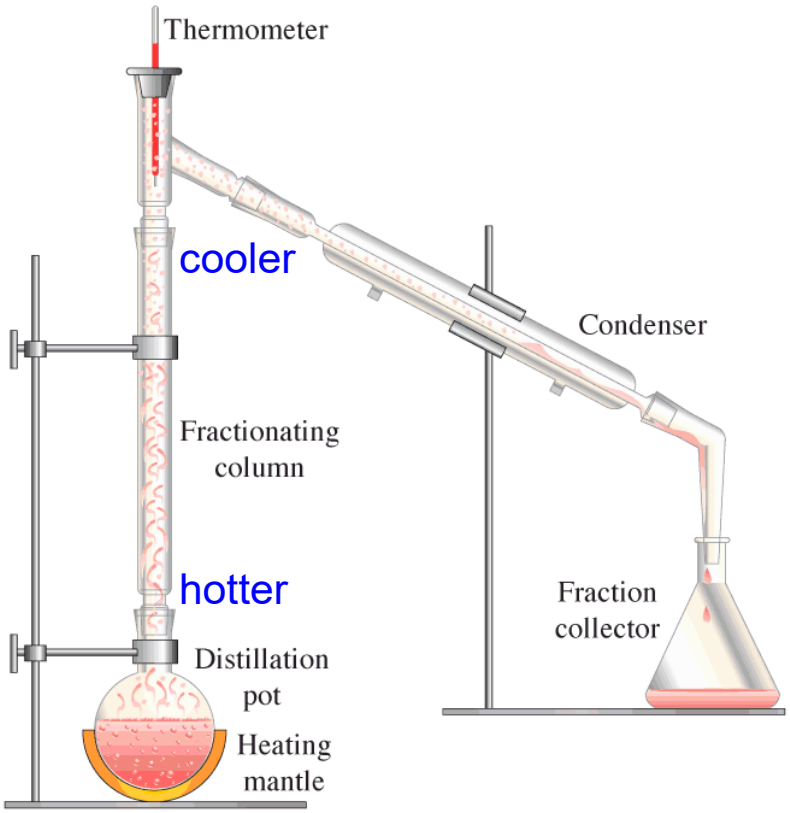
A fractionating column



The fractionating column is packed with **glass beads or stainless steel turnings**.

- Initially, as vapor rises from the pot and encounters these cooler objects, it condenses to a liquid. As the beads or turnings heat up, the liquid-vapor equilibrium front moves progressively up the column.
- Soon, liquid-vapor equilibrium occurs throughout the column, but with the equilibrium temperature changing continuously from the hottest regions at the bottom of the column to the coolest at the top. The vapor emerging from the top of the column is condensed to a liquid in the water-cooled condenser.

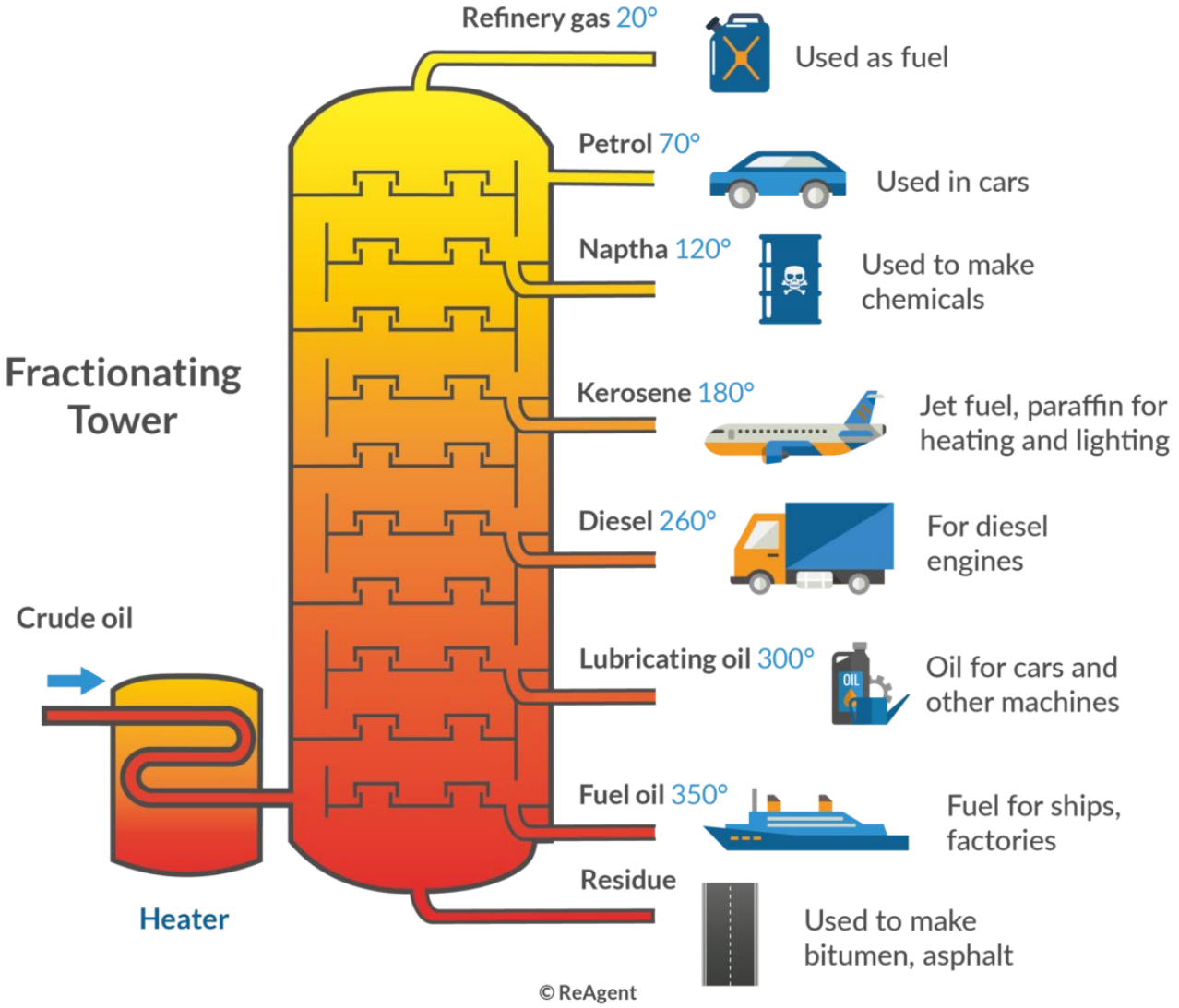
A fractionating column (Con't)



- The first fraction collected contains the **most volatile** component (lowest boiling point). Later fractions are less volatile liquids. The least volatile (highest boiling point) components remain as a residue in the distillation pot.

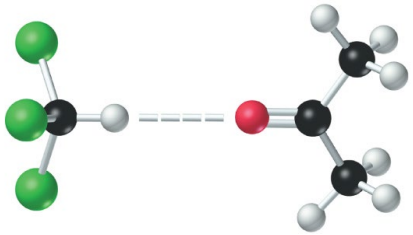
Fractional distillation is used in many industrial processes (e.g. the production of fuel and petroleum products from crude oil).

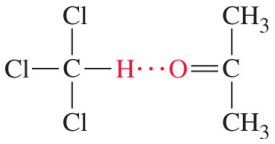
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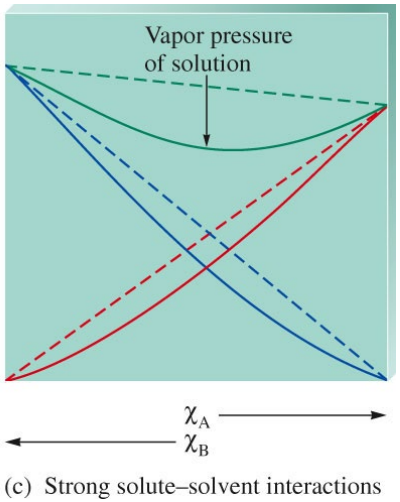


Liquid-Vapour Equilibrium: Non-ideal Solutions

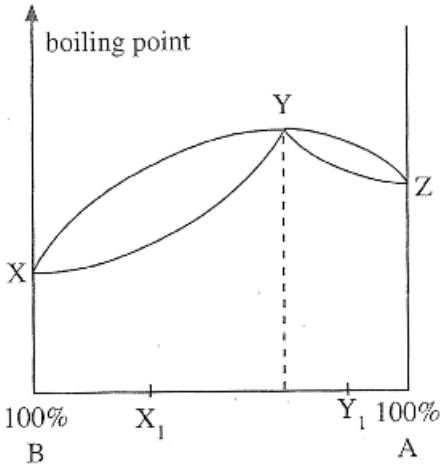
- The liquid-vapour equilibrium diagram for non-ideal solutions cannot be easily constructed. Consider acetone-chloroform solutions for example, the vapour pressures are ***lower*** than predicted and boiling temperatures are correspondingly ***higher***.







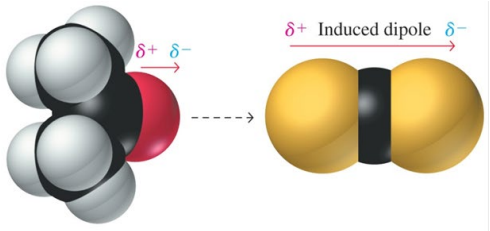
(c) Strong solute–solvent interactions

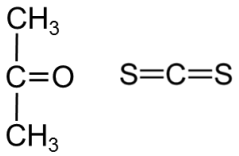


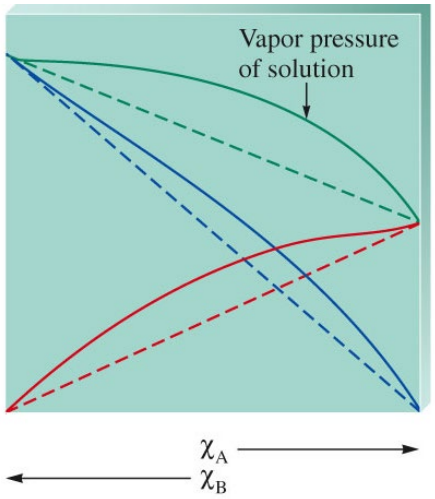
Acetone–chloroform interactions are ***stronger*** than interactions in pure acetone and chloroform.

⇒ A **reduced** tendency for the components in such solutions to vaporise (**lower**-than-predicted vapour pressures).

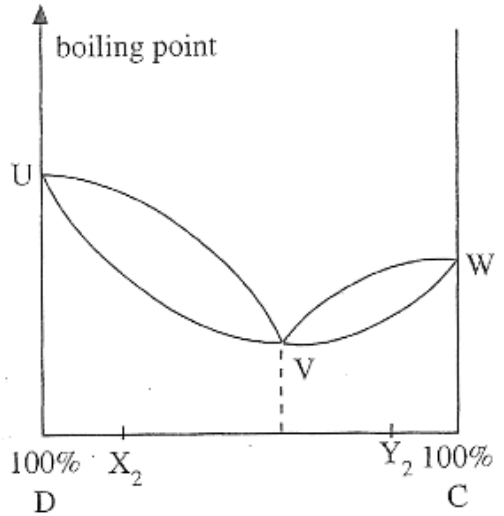
- Conversely in acetone-carbon disulphide solutions, vapour pressures are ***higher*** than predicted and boiling temperatures are correspondingly ***lower***.







(b) Weak solute–solvent interactions



Acetone–carbon disulfide interactions are ***weaker*** than the interactions in pure acetone and carbon disulfide.

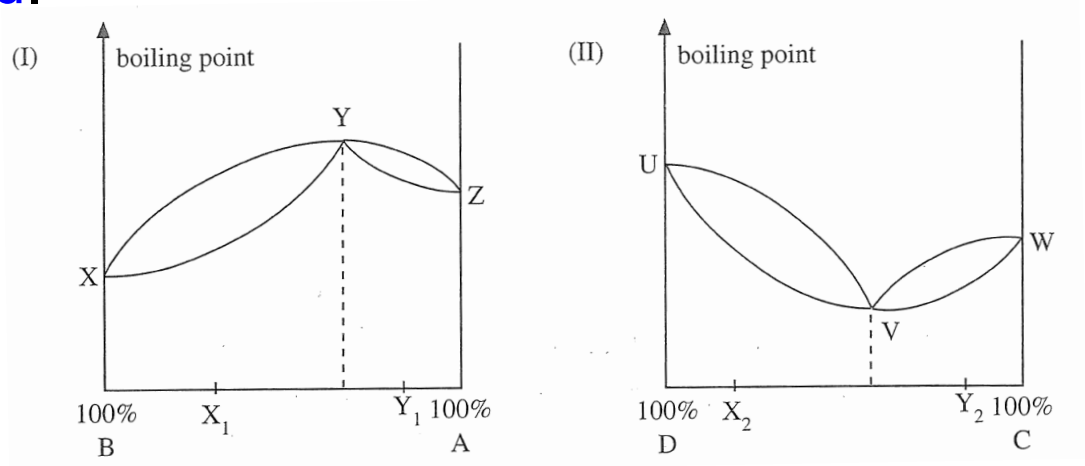
⇒ A **greater** tendency for vaporisation (**higher**-than-predicted vapour pressures).

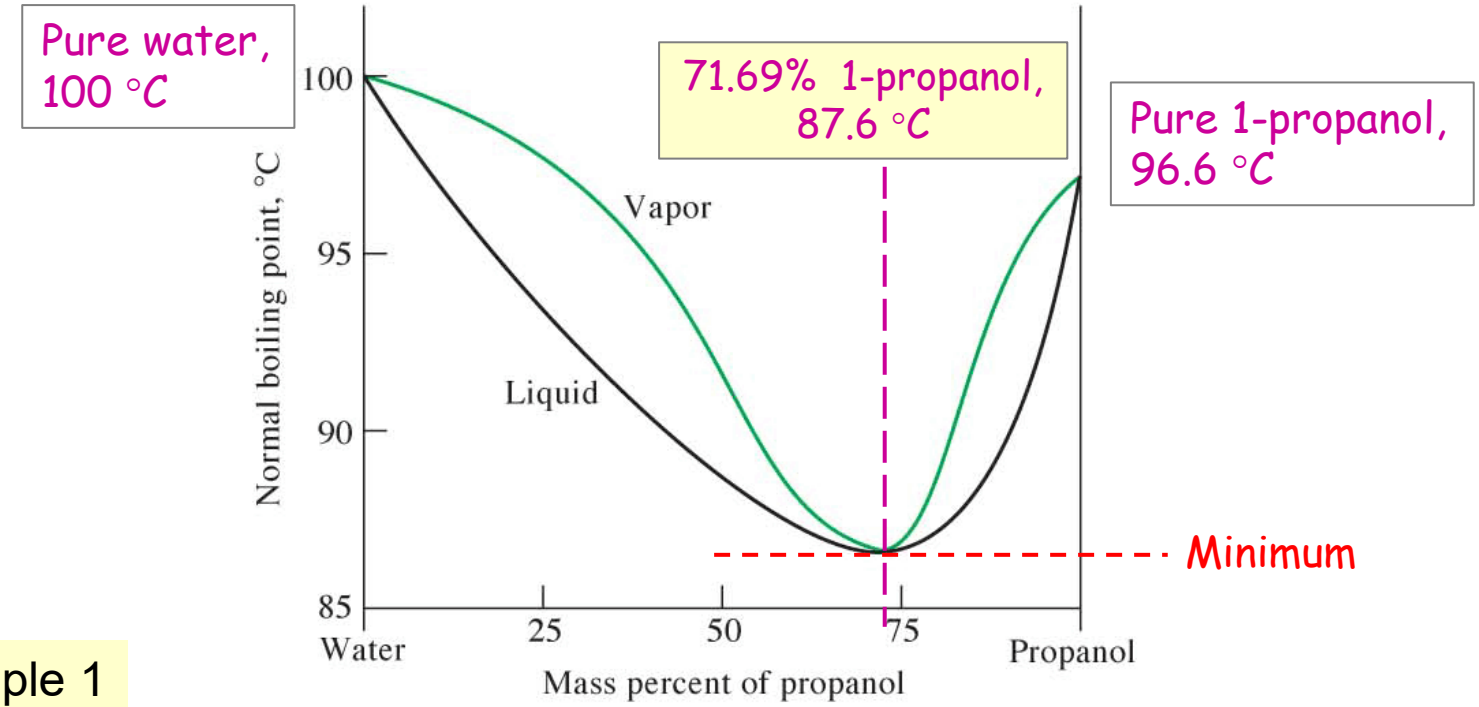
- Solutions having **great** departures from ideal solution behaviour may have vapour pressures pass through either a maximum or a minimum in vapour pressure-composition graphs. Correspondingly, their boiling points pass through either a minimum or maximum in boiling-point-composition graphs.

constant-boiling mixtures

- These solutions are called **azeotropes**, They boil at a constant temperature and produce a vapour having the same composition as the liquid.

Azeotropes do not exist for all non-ideal solvent solutions. All solvent solutions are more or less non-ideal, but not all solvent combinations form azeotropes. E.g. ethanol and isopropanol form azeotropes with water, but methanol does not.





Example 1

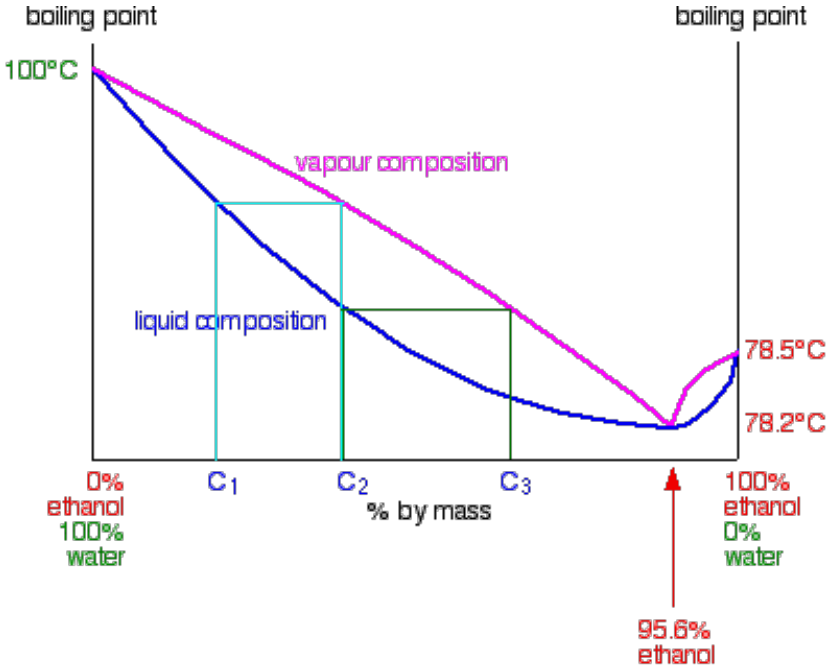
A minimum boiling-point azeotrope

A solution of **propanol in water** having 71.69% $CH_3CH_2CH_2OH$ by mass – an azeotrope – has a lower boiling point than any other solution of these two components. In fractional distillation, solutions having less than 71.69% of the alcohol yield the azeotrope and water as ultimate products. Solutions with more than 71.69% of the alcohol yield the azeotrope and propanol. In each case, the azeotrope is drawn off through the condenser, and the other component remains in the pot.

More example

Example 2

In the case of mixtures of ethanol and water, this minimum occurs with 95.6% by mass of ethanol in the mixture. The boiling point of this mixture is 78.2°C, compared with the boiling point of pure ethanol at 78.5°C, and water at 100°C.



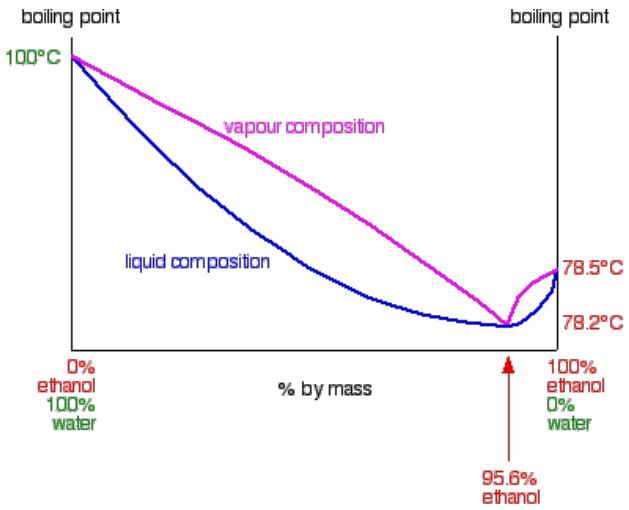
Suppose you are going to distill a mixture of ethanol and water with composition C_1 as shown on the next diagram. It will boil at a temperature given by the liquid curve and produce a vapour with composition C_2. When that vapour condenses it will, of course, still have the composition C_2. If you reboil that, it will produce a new vapour with composition C_3. You can see that if you carried on with this boiling-condensing-reboiling sequence, you would eventually end up with a vapour with a composition of 95.6% ethanol. If you condense that you obviously get a liquid with 95.6% ethanol.

Class Practice 9

What would you collect in the collecting flask and the boiling flask respectively (given that a suitable fractionating column is used and you are careful with the temperature control) in each of the following cases:

(i) If you distill a mixture of ethanol and water containing less than 95.6% of ethanol by mass.

(ii) If you distill a mixture containing more than 95.6% ethanol by mass.

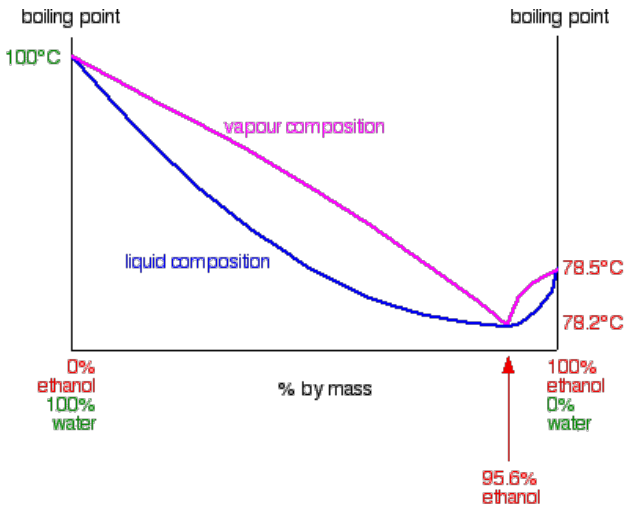


Class Practice 9

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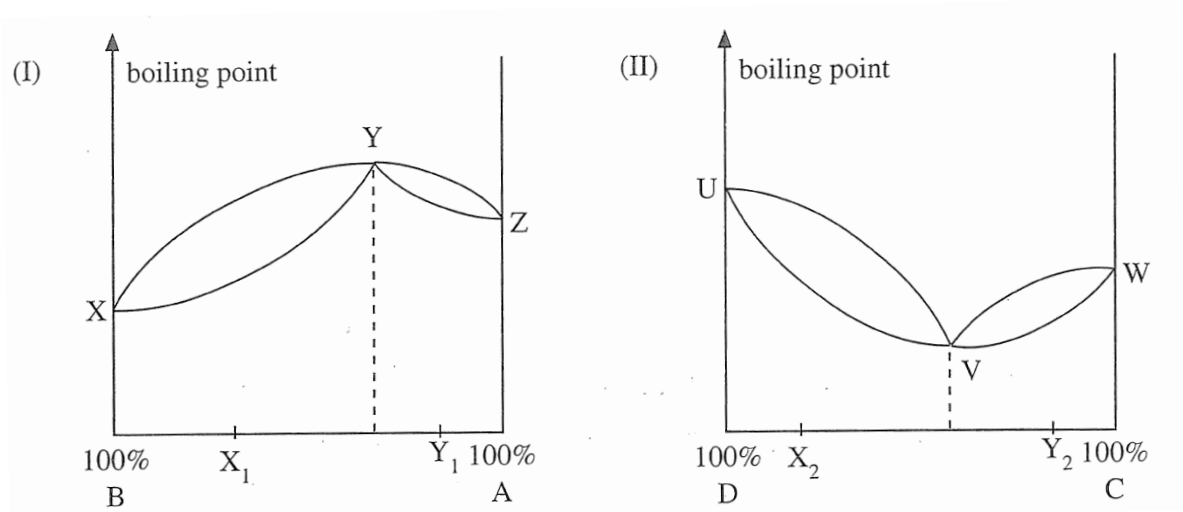
Solution:

(i) A distillate containing 95.6% of ethanol in the collecting flask and pure water in the boiling flask.

(ii) A distillate containing 95.6% of ethanol in the collecting flask and pure ethanol in the boiling flask.

Class Practice 10

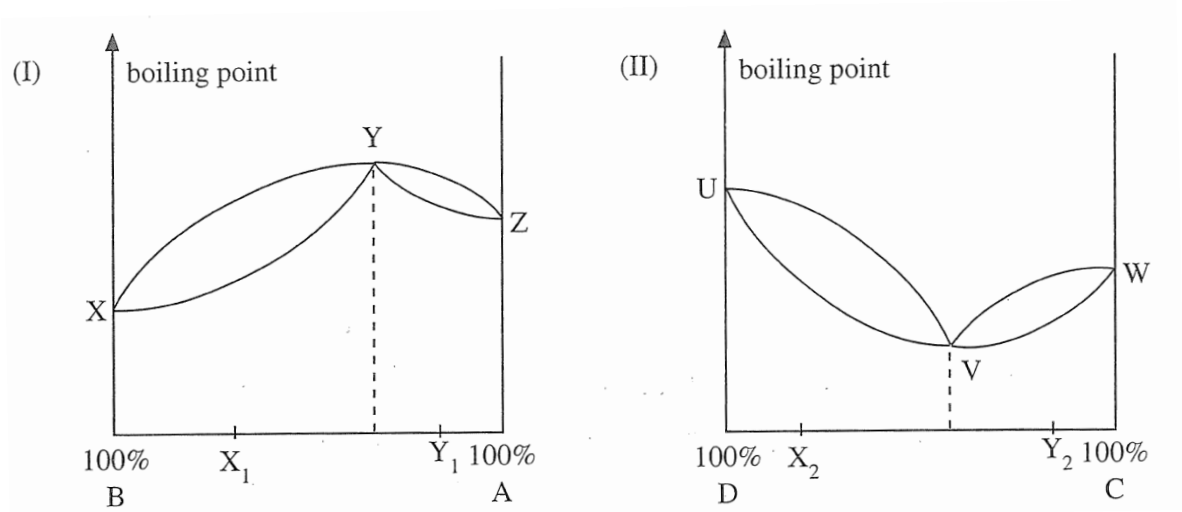
Referring to figures (I) and (II),



(a) identify the type (maximum or minimum boiling-point) of azeotropes and also the type of deviations from Raoult's law in each case.

Class Practice 10

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(a) identify the type (maximum or minimum boiling-point) of azeotropes and also the type of deviations from Raoult's law in each case.

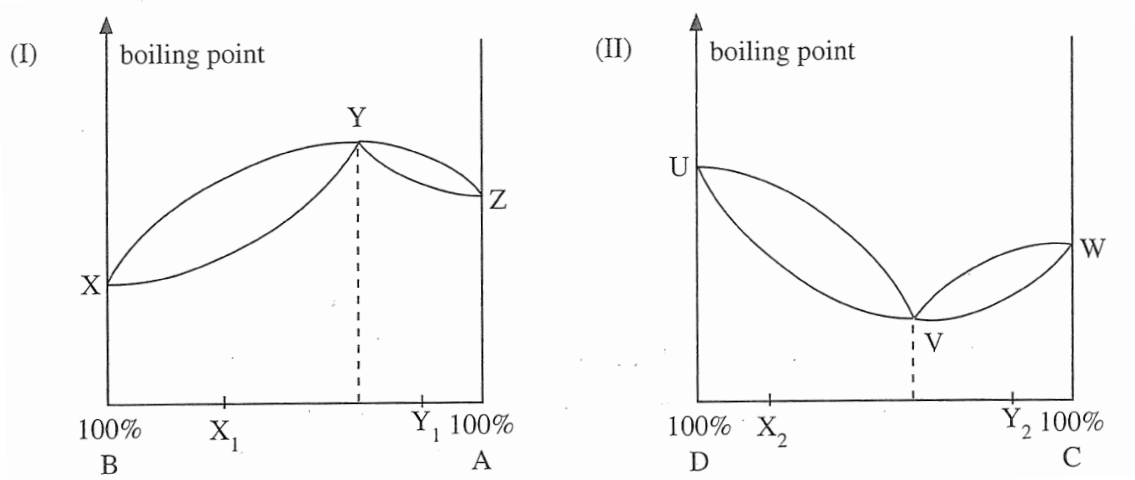
Figure I: Maximum boiling-point azeotrope
(negative deviations from Raoult's law)

Figure II: Minimum boiling-point azeotrope
(positive deviations from Raoult's law)

Class Practice 10 (Con't)

(b) Give the result of distillation (i.e. identify the distillate and residue) for

(i) case (I) started at composition X_1	(iii) case (II) started at composition X_2
(ii) case (I) started at composition Y_1	(iv) case (II) started at composition Y_2

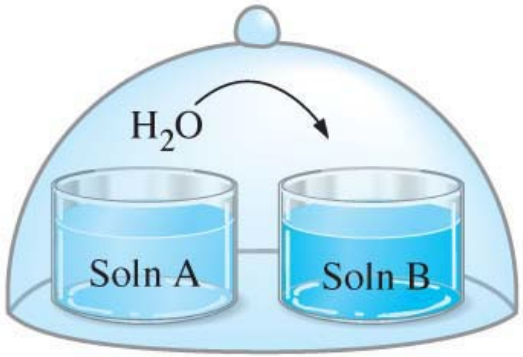


(i) Distillate: Pure B Residue: Azeotrope Y	(iii) Distillate: Azeotrope V Residue: Pure D
(ii) Distillate: Pure A Residue: Azeotrope Y	(iv) Distillate: Azeotrope V Residue: Pure C

Osmotic Pressure

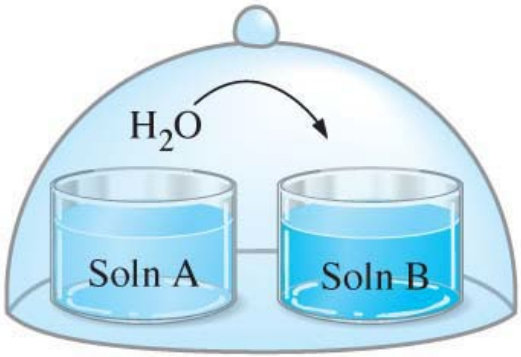
Colligative properties

- Another type of solutions is one with **a volatile solvent** (e.g. water), but with **one or more non-volatile solutes** (e.g. glucose).
- Raoult's law still applies to these solutions, i.e. the vapour pressure of the solvent is lowered.

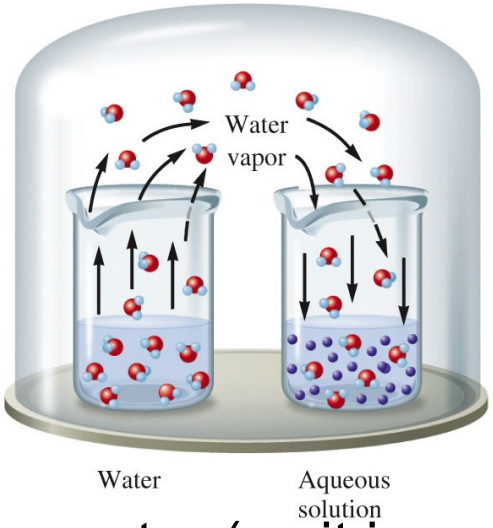


- Referring to the picture, you see two aqueous solutions (A & B) of a non-volatile solute within the same enclosure, water vaporises from A and condenses into B. *What is the driving force for this?*

A colligative property is a property of a solution that is dependent on the ratio between the total number of solute particles (in the solution) to the total number of solvent particles. Colligative properties are not dependent on the chemical nature of the solution's components.



Water vaporises from A and condenses into B. *What is the driving force for this?*



A is **more dilute** than B, so P_{H2O} above A is **greater**.

It is because …

- the vapour pressure of H_2O above A is greater (as it is more dilute), and so it has a larger mole fraction of H_2O.
- During the transfer of water from A to B, the mole fraction of H_2O ↓ in A and the mole of fraction of H_2O ↑ in B.
- The net transfer of water from A to B will stop when the mole fraction of H_2O is the same in both solutions.

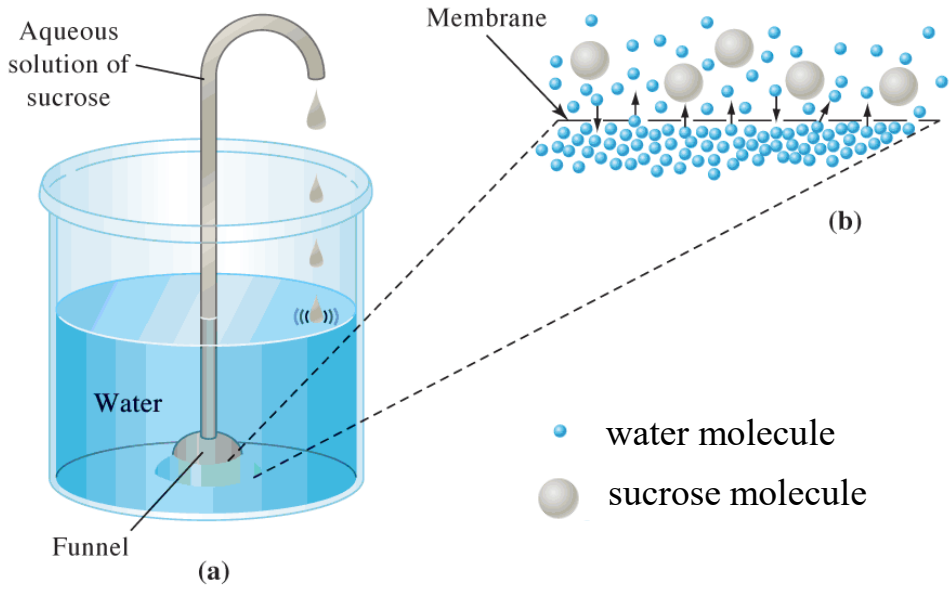




A related phenomenon occurs

- when $CaCl_2 \cdot 6H_2O(s)$ is exposed to air, water vapour in air condenses on the solid and the solid begins to dissolve. This phenomenon is known as ***deliquescence (becoming liquid)***.
- For deliquescence to occur, the partial pressure of water vapour in air must be greater than the vapour pressure of water above a saturated aqueous solution of the solid.

- An aqueous sucrose solution in a long glass tube is separated from pure water by a semi-permeable membrane (permeable to water only). Water molecules can pass through the membrane in either direction.



- ∵ the concentration of water molecules in the pure water > that in the solution, ∴ there is a net flow from the pure water into the solution. This net flow is called ***osmosis***.

- The solution will rise in the tube. The more concentrated the sucrose solution, the higher the solution level rises.

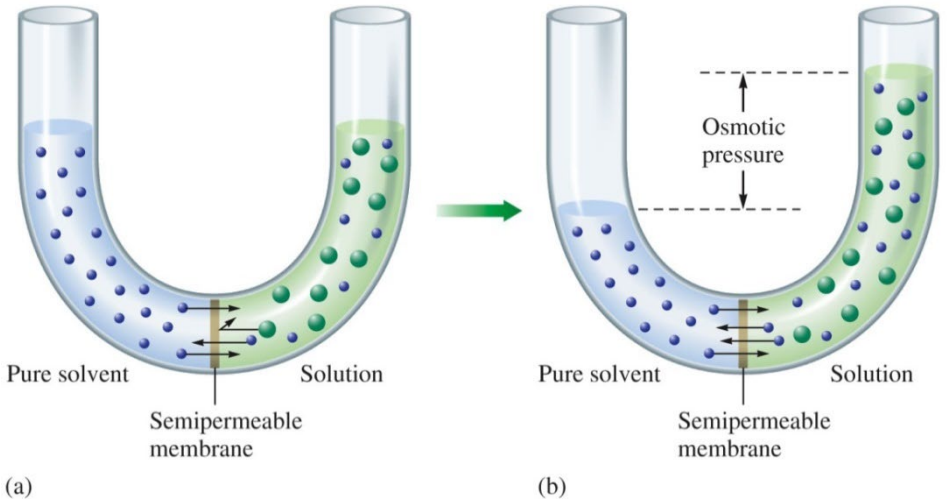
- Applying pressure to the sucrose solution slows down the net flow of water across the membrane into the solution.

- Under a high pressure, the net influx of water can be stopped altogether. The necessary pressure to stop osmotic flow is called the **osmotic pressure** of the solution. E.g.) For a 20% sucrose solution, the osmotic pressure is about 15 atm.

Colligative properties

- The magnitude of osmotic pressure depends only on the ***number*** (but **not** the *identity*) of solute particles per unit volume of solution.

A colligative property is a property of a solution that is dependent on the ratio between the total number of solute particles (in the solution) to the total number of solvent particles. Colligative properties are not dependent on the chemical nature of the solution's components.



- For ***dilute*** solutions of non-electrolytes, the osmotic pressure (π) can be calculated by the following equation:

$$\pi V = nRT$$

$$\pi = \frac{n}{V}RT = M \times RT$$

M is molarity of solution

where π = osmotic pressure of *dilute* solution,
R = gas constant = 0.08206 $L \cdot atm \cdot mol^{-1} \cdot K^{-1}$,
T = temperature in Kelvin,
n = no. of mole of solute, and
V = volume in litres of solution.



Class Practice 11

EXAMPLE 14-8 Calculating Osmotic Pressure

What is the osmotic pressure at 25 °C of an aqueous solution that is 0.0010 M $C_{12}H_{22}O_{11}$ (sucrose)?

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

Class Practice 11

EXAMPLE 14-8 Calculating Osmotic Pressure

What is the osmotic pressure at 25 °C of an aqueous solution that is 0.0010 M $C_{12}H_{22}O_{11}$ (sucrose)?

Solution:

$$\pi = \frac{n\mathrm{RT}}{V}$$

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

$$\pi = \frac{0.0010\ \text{mol} \times 0.08206\ \text{atm L mol}^{-1}\ \text{K}^{-1} \times 298\ \text{K}}{1\ \text{L}}$$

$$\pi = 0.024\ \text{atm (18 mmHg)}$$

Assess

At a very low concentration, there is an appreciable amount of osmotic pressure. This fact is used when measuring the molar mass of polymers and biopolymers.



Class Practice 12

EXAMPLE 14-9 Establishing a Molar Mass from a Measurement of Osmotic Pressure

A 50.00 mL sample of an aqueous solution contains 1.08 g of human serum albumin, a blood-plasma protein. The solution has an osmotic pressure of 5.85 mmHg at 298 K. What is the molar mass of the albumin?

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
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Class Practice 12

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Solution:

Express osmotic pressure in atmospheres.

$$\pi = 5.85 \text{ mmHg} \times \frac{1 \text{ atm}}{760 \text{ mmHg}} = 7.70 \times 10^{-3} \text{ atm}$$

Modify the basic equation for osmotic pressure, by showing moles of solute (n) as the mass of solute (m) divided by its molar mass (M), and solve for M.

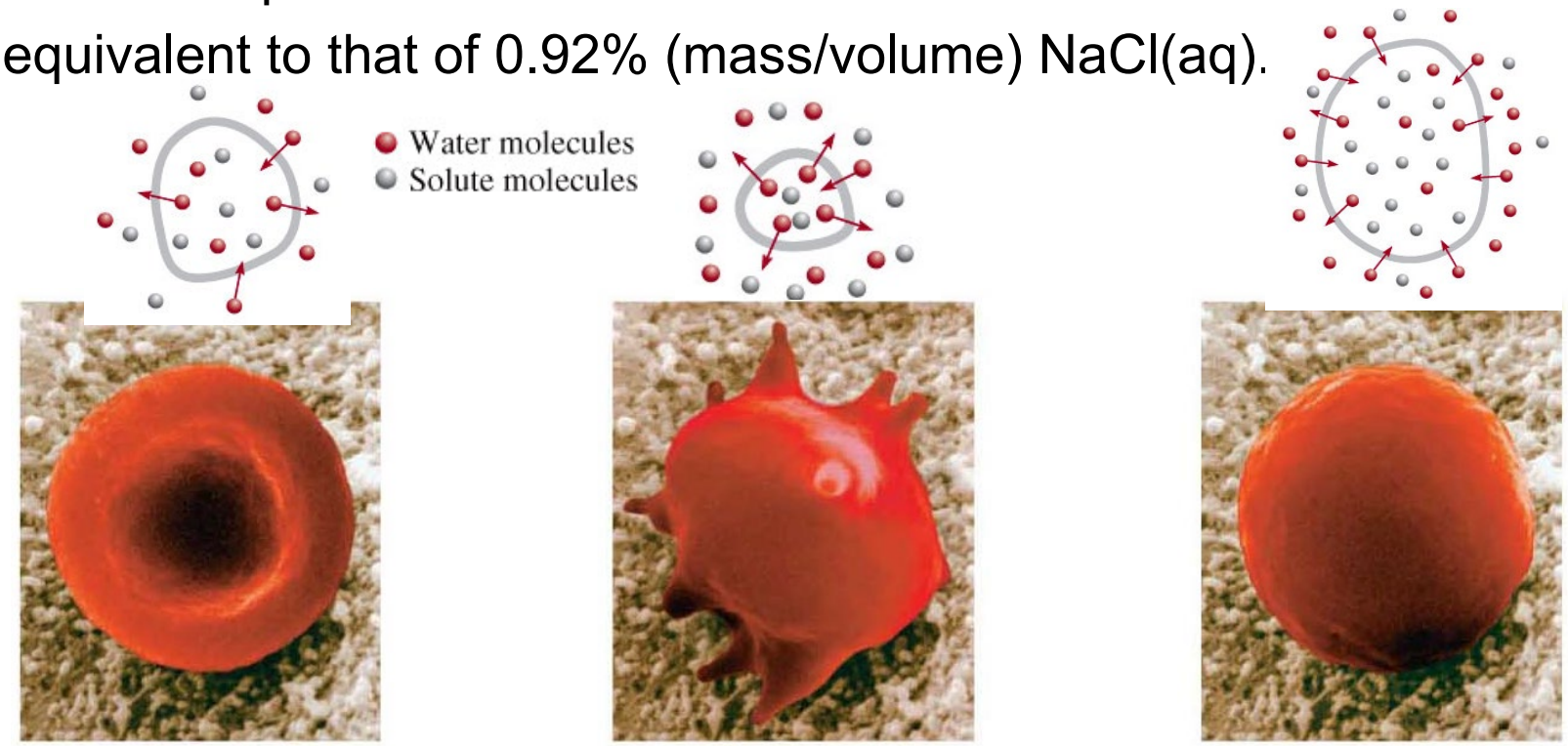
$$\pi V = nRT \qquad \pi V = \frac{m}{M}RT \qquad M = \frac{mRT}{\pi V}$$

Obtain the value of M by substituting the given data into the last equation above, ensuring that units cancel to yield g/mol as the unit for M.

$$M = \frac{1.08 \text{ g} \times 0.08206 \text{ atm L mol}^{-1} \text{ K}^{-1} \times 298 \text{ K}}{7.70 \times 10^{-3} \text{ atm} \times 0.0500 \text{ L}} = 6.86 \times 10^{4} \text{ g/mol}$$

Practical applications of osmosis

- The osmotic pressure associated with the fluid inside the red blood cell is equivalent to that of 0.92% (mass/volume) NaCl(aq).



(a) **Isotonic** solution
A red blood cell in a 0.92% (m/v) NaCl(aq), there is no net flow of water and the cell remains stable.

(b) **Hypertonic** solution
If conc. **>** 0.92% NaCl, water flows out of the cell and the cell shrinks.

(c) **Hypotonic** solution
If conc. **<** 0.92% NaCl, water flows into the cell and the cell swells.

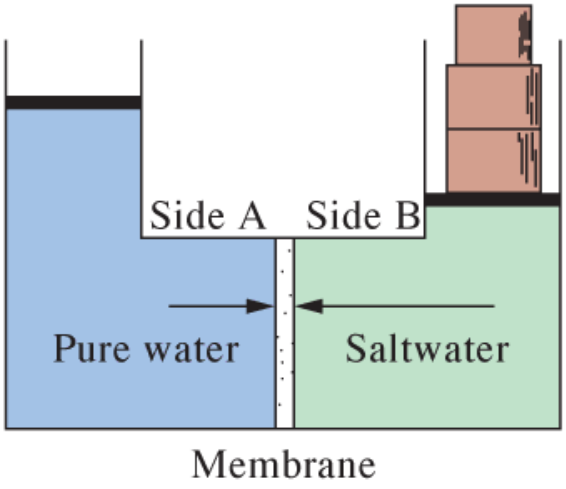
∴ Fluids that are intravenously injected into patients to combat dehydration or to supply nutrients must be adjusted so that they are isotonic with blood.

Practical applications of osmosis

Reverse osmosis

With a larger pressure in B

- Pressure applied to side B < the osmotic pressure of saltwater, ordinary osmosis (water flows from A to B) occurs.
- Pressure applied to side B > the osmotic pressure of saltwater, **reverse osmosis** occurs. A *net* flow of water occurs in the reverse direction from the saltwater into the pure water.



Reverse osmosis: flow of water is B → A.

The membrane is permeable to water only.

- ***Applications***:

(1) Desalination of seawater to supply drinking water as a source of municipal water.

(2) Removal of dissolved materials from industrial or municipal wastewater.

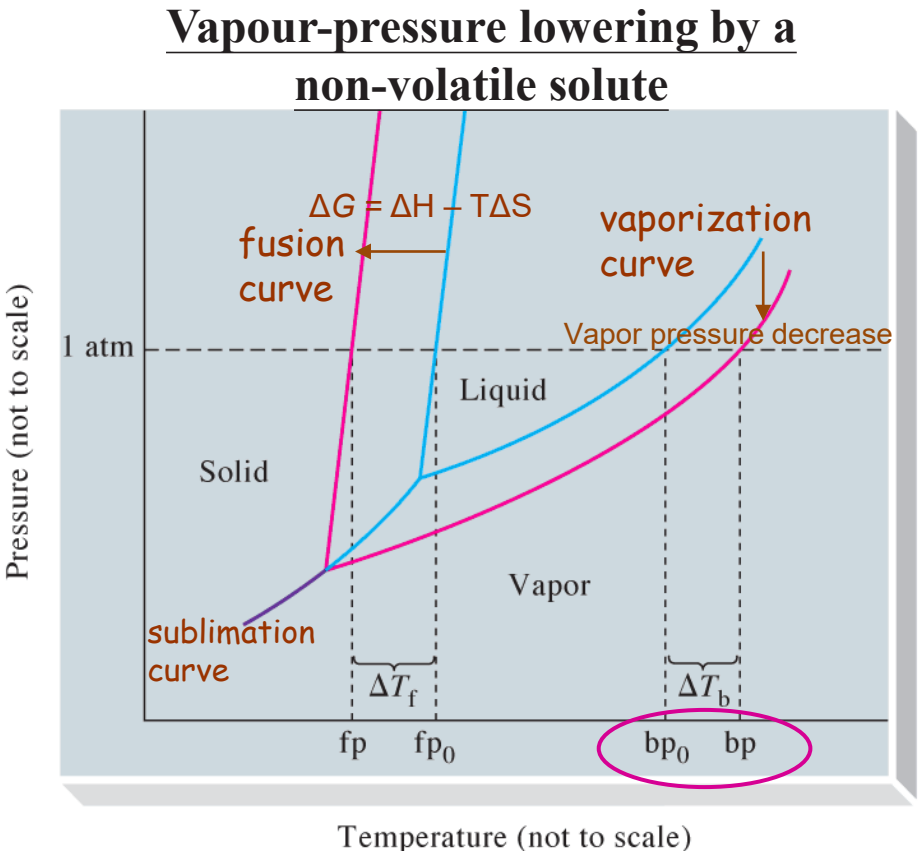
Freezing-point Depression and Boiling-point Elevation of Nonelectrolyte Solutions

Colligative properties

- We already know the vapour pressure of a solvent is ***lowered*** by a solute.

What would happen to the boiling temperature of the solution then?

- In the figure, the **blue** curves represent the vapour pressure, fusion, and sublimation curves in the phase diagram for a pure solvent. The **red** curves represent the vapour pressure and fusion curves of the solvent in a solution. The sublimation curve for the solid solvent that freezes from the solution is shown in purple.



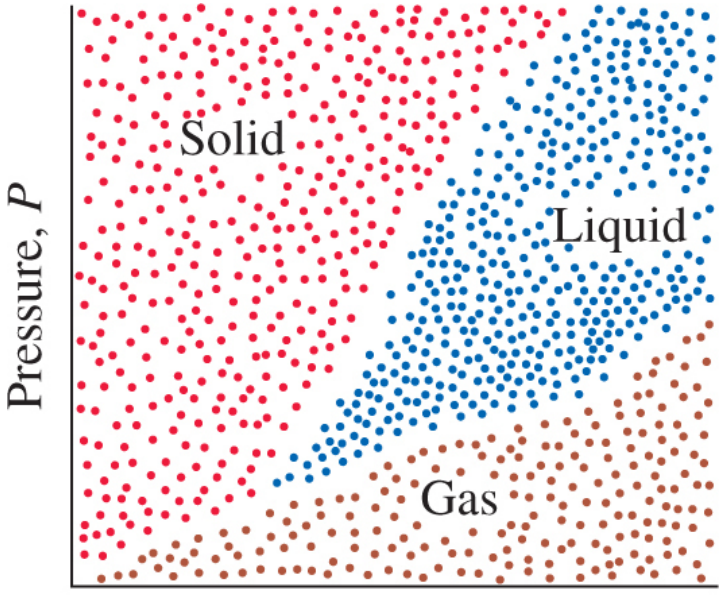
- Two assumptions:

(i) the solute is non-volatile.

(ii) the solid that freezes from a solution is pure solvent.

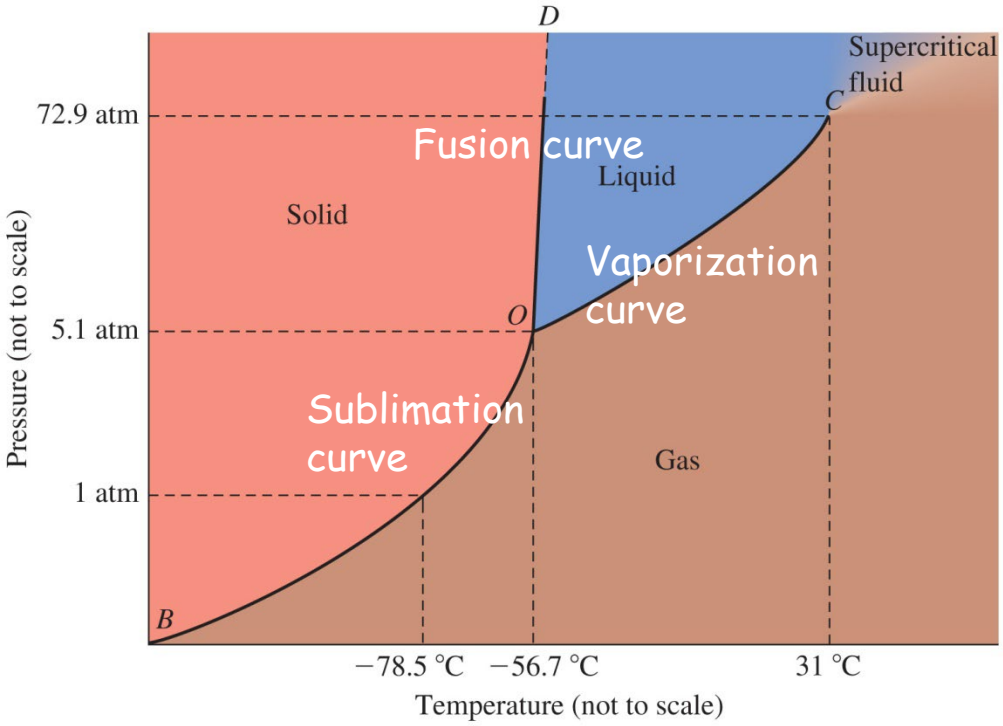
Supplementary Information on Phase Diagrams

- A ***phase diagram*** is a graphical representation of the conditions of T and P at which solids, liquids, and gases (vapors) exist, either as single phases (states) of matter or as two or more phases in equilibrium with one another. Straight/curved lines where single-phase regions adjoin represent two phases in equilibrium.



Temperature, pressure, and states of matter

Phase Diagram for Carbon Dioxide

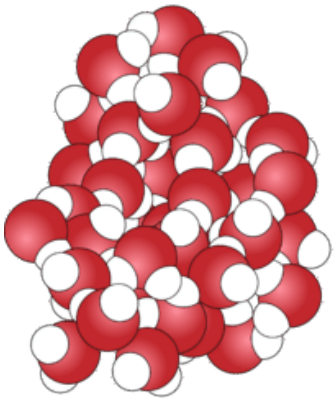


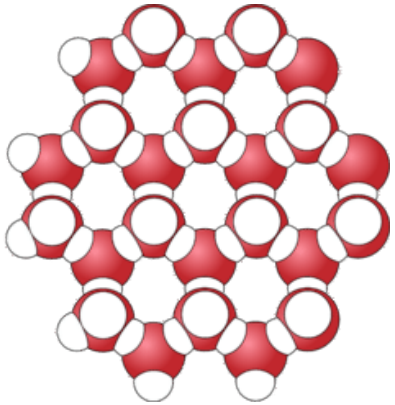
- The triple point of CO_2 is above 1 atm. A line at P = 1 atm intersects the sublimation curve (but not the vapor pressure curve). CO_2 (s) does not melt at atmospheric pressure but sublimes away.
- Liquid CO_2 an be obtained at P > 5.1 atm.

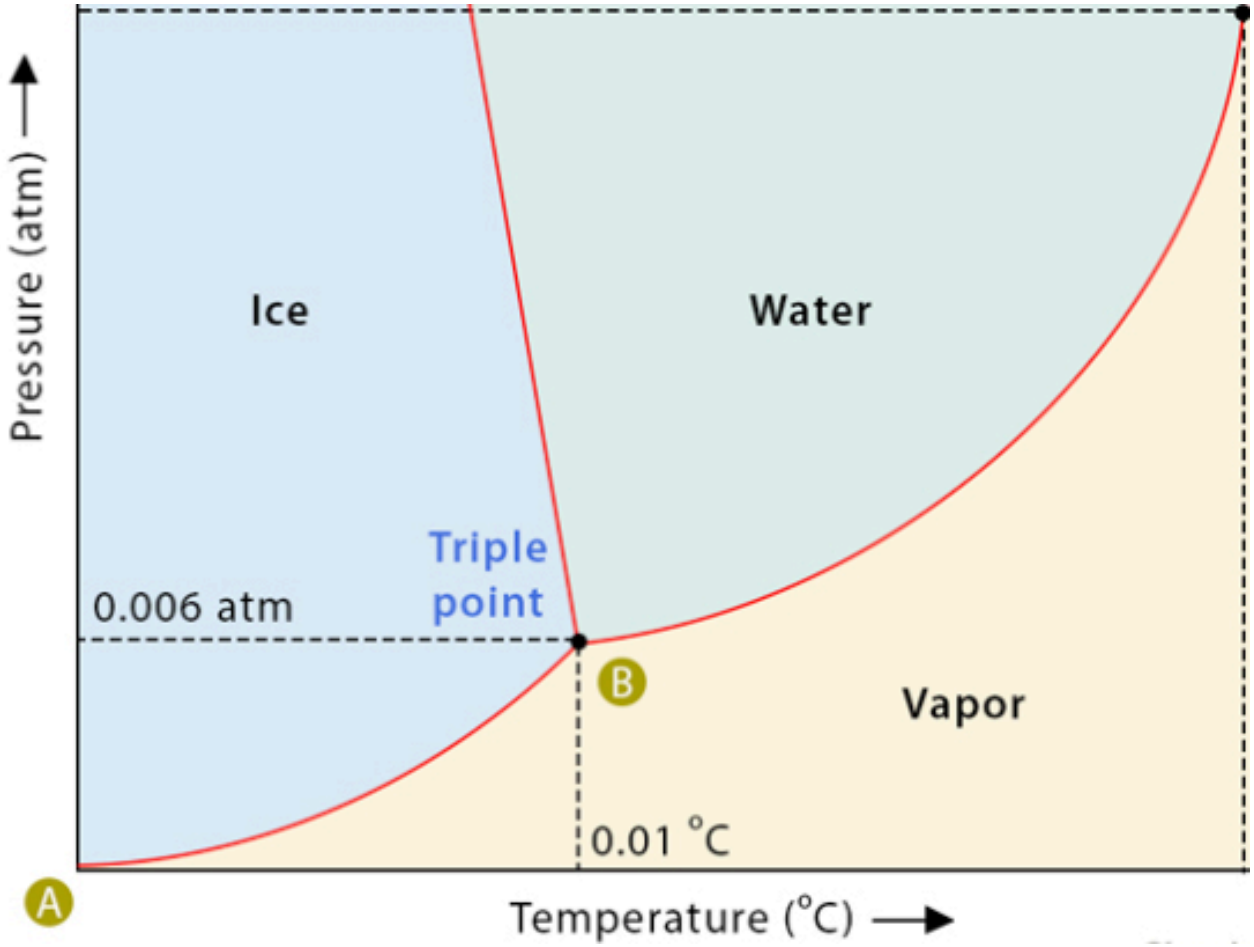
Phase diagram for carbon dioxide:
O (triple point): –56.7 °C & 5.1 atm
Sublimation temperature: –78.5 °C

O – the triple point which defines the unique T and P at which the three states of matter coexists in eqm.

Phase diagram of H_2O







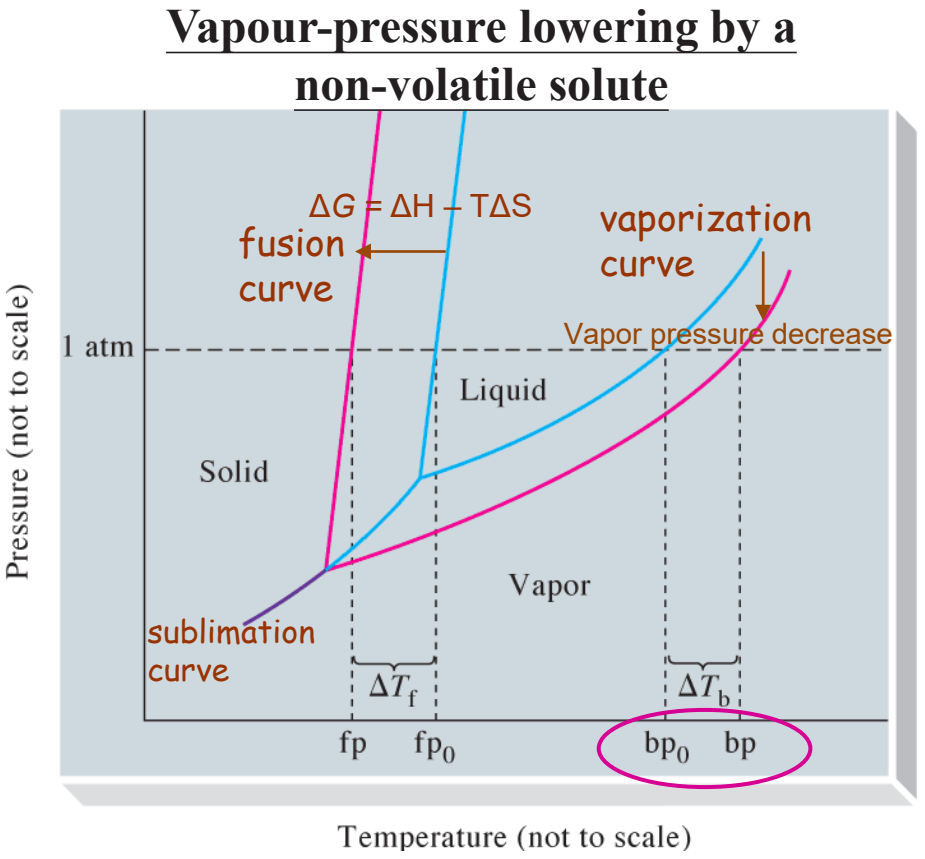
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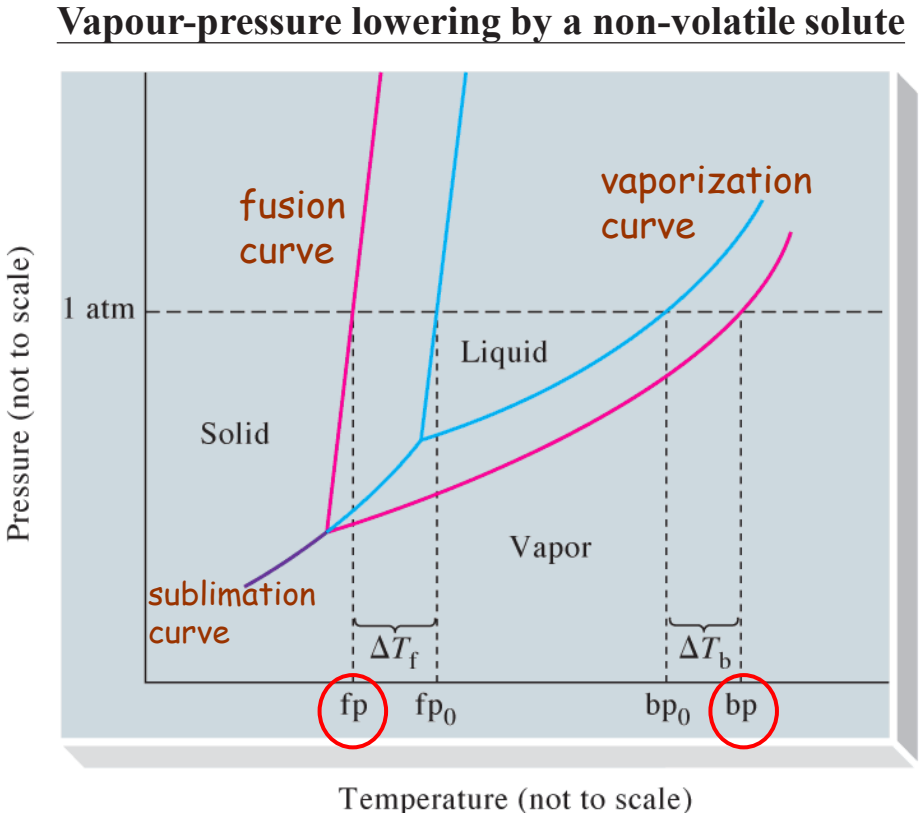
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  (i) the solute is non-volatile.
  (ii) the solid that freezes from a solution is pure solvent.

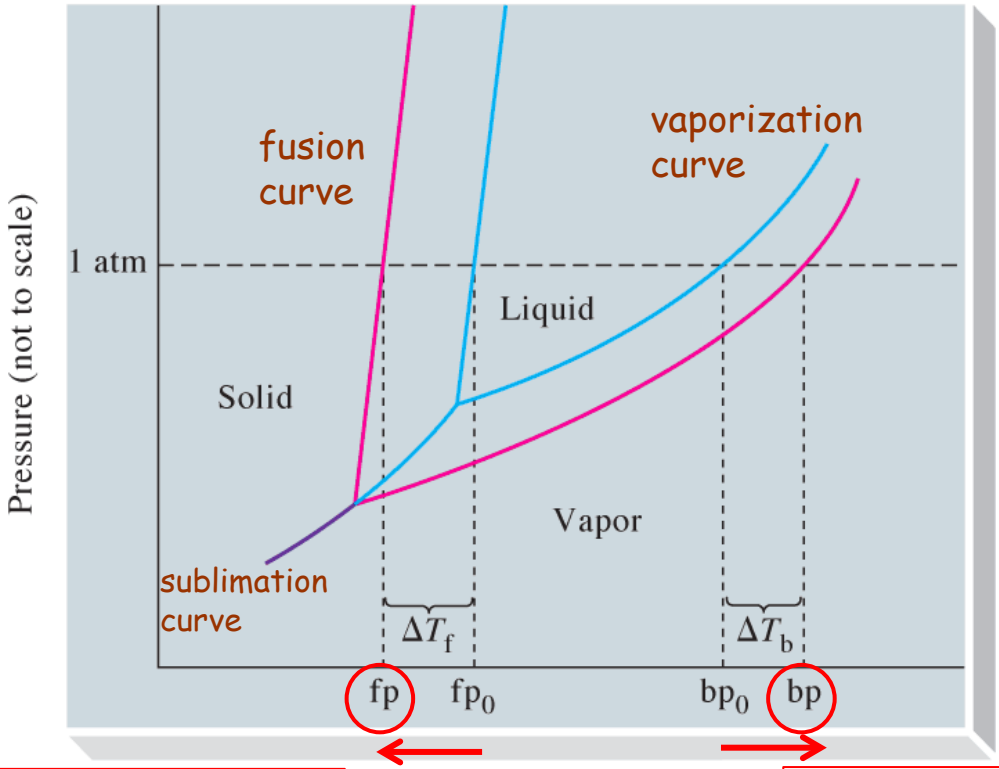
- The vapour pressure curve of the solution (red) intersects the sublimation curve at a lower T than in the case for the pure solvent.

- The normal freezing point and normal boiling point of the pure solvent are f_{p0} and b_{p0}, respectively. The corresponding points for the solution are f_p and b_p.



- The freezing-point depression, ΔT_f, and the boiling-point elevation, ΔT_b, are indicated. Because the solute is assumed to be ***insoluble*** in the solid solvent, the sublimation curve of the solvent is unaffected by the presence of solute in the liquid solution phase. That is, the sublimation curve is the **same** for the two phase diagrams.

Vapour-pressure lowering by a non-volatile solute



f.p. of the solvent in solution ↓

b.p. of the solvent in solution ↑



The freezing point of the solvent in solution is depressed, and the boiling point is elevated.

- The extent to which the freezing point is lowered or the boiling point raised is proportional to the mole fraction of solute (just as is vapour pressure lowering).

∵ they are colligative properties!

- In dilute solutions of non-electrolytes (< 1 m), the solute mole fraction is proportional to its molality:

$T_f - T_{f0}$

$T_b - T_{b0}$

$$\Delta T_f = -K_f \times m$$
$$\Delta T_b = K_b \times m$$

where ΔT_f = the freezing-point depression;
ΔT_b = boiling-point elevation;
m = solute molality; and
K_f and K_b are proportionality constants.

- The value of K_f depends on the melting point, enthalpy of fusion and molar mass of the solvent.
- The value of K_b depends on the boiling point, enthalpy of vaporization and molar mass of the solvent.
- The units of K_f and K_b are $°C\ m^{-1}$.
- Their values represent the freezing-point depression and boiling-point elevation for a 1 m solution

TABLE 14.3 Freezing-Point Depression and Boiling-Point Elevation Constants[a]

Solvent	Normal Freezing Point, °C	K_f, °C mol^{-1} kg	Normal Boiling Point, °C	K_b, °C mol^{-1} kg
Acetic acid	16.6	3.90	118	3.07
Benzene	5.53	5.12	80.10	2.53
Nitrobenzene	5.7	8.1	210.8	5.24
Phenol	41	7.27	182	3.56
Water	0.00	1.86	100.0	0.512

[a]Values correspond to freezing-point depressions and boiling-point elevations, in degrees Celsius, caused by 1 mol of solute particles dissolved in 1 kg of solvent in an ideal solution.



Class Practice 13

EXAMPLE 14-10 Establishing a Molecular Formula with Freezing-Point Data

Nicotine, extracted from tobacco leaves, is a liquid completely miscible with water at temperatures below 60 °C. **(a)** What is the *molality* of nicotine in an aqueous solution that starts to freeze at −0.450 °C? **(b)** If this solution is obtained by dissolving 1.921 g of nicotine in 48.92 g H_2O, what must be the molar mass of nicotine? **(c)** Combustion analysis shows nicotine to consist of 74.03% C, 8.70% H, and 17.27% N, by mass. What is the molecular formula of nicotine?

$$\Delta T_f = -K_f \times m$$

Analyze

(a) We can establish the molality of the nicotine by using equation (14.5) with the value of K_f for water listed in Table 14.3. **(b)** Once we know the molality, we can use the definition of molality, but with a known molality (from part a) and an unknown molar mass of solute (M). **(c)** To establish the empirical formula of nicotine, we need to use the method of Example 3-5.



Class Practice 13 (Con't)

Solution:

(a) Note that $T_f = -0.450\ °C$, and that $\Delta T_f = -0.450\ °C - 0.000\ °C = -0.450\ °C$.

$$\text{molality} = \frac{\Delta T_f}{-K_f} = \frac{-0.450\ °C}{-1.86\ °C\ \text{mol}^{-1}\ \text{kg}} = 0.242\ \frac{\text{mol}}{\text{kg}} = 0.242\ \text{mol kg}^{-1}$$

(b) Let's represent the molar mass as x g/mol. The amount, in moles, contained in a 1.921 g sample is $1.921\ \text{g} \times (1\ \text{mol}/x\ \text{g}) = (1.921/x)\ \text{mol}$. Thus,

$$\text{molality} = \frac{1.921\ x^{-1}}{0.04892\ \text{kg water}} = 0.242\ \text{mol (kg water)}^{-1}$$

$$x = 162$$

The molar mass is 162 g/mol.

(c) This calculation is left as an exercise for you to do. The result you should obtain is C_5H_7N. The formula mass based on this empirical formula is 81 u. The molecular mass obtained from the molar mass in part (b) is exactly twice this value—162 u. The molecular formula is twice C_5H_7N, or $C_{10}H_{14}N_2$.

Assess

Using the freezing-point data is another experimental technique that can be used to obtain the chemical properties of a substance. In this example we were able to determine the molecular formula from the freezing-point depression and a known amount of substance dissolved in a solvent. Note that water was used as the solvent here; however, other solvents can be used in freezing-point experiments.

Practical applications of freezing-point depression and boiling-point elevation

- Common automobile antifreeze is ethylene glycol, $HOCH_2CH_2OH$. In summer it helps by raising the boiling point of water and preventing the cooling system from boiling over.
- The juice in the fruit has enough dissolved solutes to lower the freezing point by a degree or two below 0 °C. The growers must protect lemons sooner than oranges as they have lower concentration of dissolved solutes (sugars).
- Salts can be used to prepare a slush bath to cool or freeze something. E.g. the mixture of ice and NaCl(s) used to freeze ice-cream.
- NaCl is also useful for de-icing roads at temperatures as low as –21 °C.







Solutions of Electrolytes

An electrolyte is a substance that dissociates in water into charged particles called ions.

Ex. NaCl an HCl

Colligative Properties for electrolytes

- Certain solutes produce a *greater* effect on colligative properties than expected.

molality

E.g.) Consider a 0.0100 m aqueous solution, the predicted freezing-point depression of the solution is

$$\Delta T_f = -K_f \times m = -1.86℃m^{-1} \times 0.0100m = -0.0186℃$$

- If the solution is 0.0100 m urea, the measured freezing point is just about –0.0186 ℃.
- However, if the solution is 0.0100 m NaCl, the measured freezing point is about – 0.0361 ℃。

- Van't Hoff factor i is the ratio of the measured value of a colligative property to the expected value if the solute is an electrolyte

0.0100 *m* NaCl solution ⟶ 0.0200 *m* ions in solution

- Electrolytic dissociation allows us to explain different values of the van't Hoff factor i for different solutes.

- For **non-electrolytes** (e.g. urea, glycerol and sucrose), $i = 1$.

- For **strong electrolytes** that produce two moles of ions in solution per mole of solute dissolved (e.g. NaCl), the freezing-point depression is expected to be twice as great as for a non-electrolyte, $\therefore\ i = 2$. Similarly for $MgCl_2$, $i = 3$.

- For weak acid that only slightly ionized in aqueous solutions (e.g. CH_3COOH), $1 < i < 2$.

- The equations discussed before should be rewritten as follows:

$$\pi = \frac{n}{V}RT = i \times M \times RT$$

$$\Delta T_f = -i \times K_f \times m$$

$$\Delta T_b = i \times K_b \times m$$

if these equations are used for non-electrolytes, simply substitute $i = 1$.

Class Practice 14

EXAMPLE 14-11 Predicting Colligative Properties for Electrolyte Solutions

Predict the freezing point of aqueous 0.00145 mol kg^{-1} $MgCl_2$.

Analyze

We will use a modified freezing-point depression equation in which the van't Hoff factor i is included. We first note that $MgCl_2$ is a salt that completely dissociates when it is dissolved in water. So we determine the value of i for $MgCl_2$. We can do this by writing an equation to represent the dissociation of $MgCl_2(s)$. Then we use the appropriate freezing-point depression expression.

Solution:

$$MgCl_2(s) \xrightarrow{H_2O} Mg^{2+}(aq) + 2\,Cl^-(aq)$$

Because *three* moles of ions are obtained per mole of formula units dissolved, we expect the value $i = 3$. Now use the expression

$$\begin{aligned}\Delta T_f &= -i \times K_f \times m \\ &= -3 \times 1.86\ ^\circ C\ mol^{-1}\ kg \times 0.00145\ mol^{-1}\ kg \\ &= -0.0081\ ^\circ C\end{aligned}$$

The predicted freezing point is $-0.0081\ ^\circ C$.

Assess

Because the value of i for $MgCl_2$ is not exactly 3, we are not justified in carrying more than one or two significant figures in our answer. If we had ignored the fact that $MgCl_2$ is a strong electrolyte, then our calculated freezing-point depression would have been three times as small as the experimental value. Always remember to include the van't Hoff factor when ionic compounds are given as part of the problem.

Anomalous Colligative Properties: Interionic Attractions

- The electrical conductivities of 'concentrated' solutions of strong electrolytes are not as great as expected, and the values of the van't Hoff factor i depend on the solution concentrations *(see table below).*

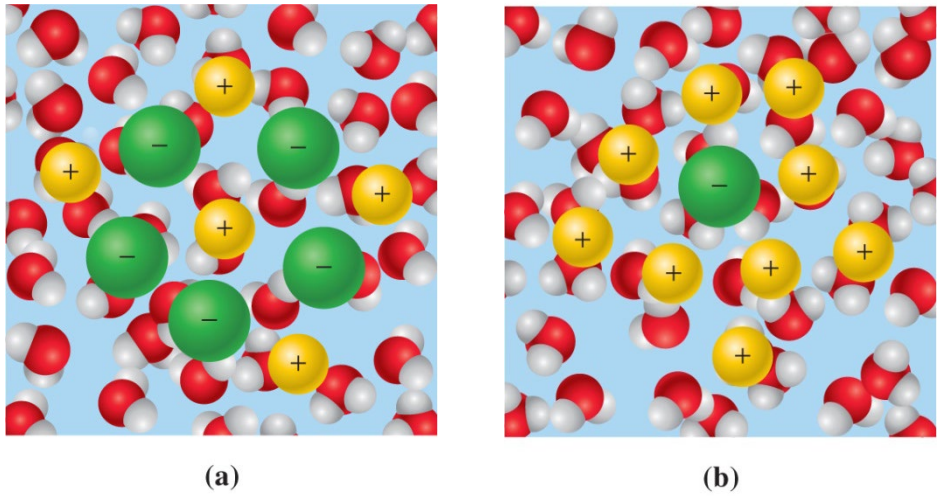
- For strong electrolytes that exist completely in ionic form in aqueous solutions, we would expect i = 2 for NaCl and i = 3 for $MgCl_2$, and so on, regardless of the concentration. However,

TABLE 14.4 Variation of the van't Hoff Factor, *i*, with Solution Molality

	Molality, mol kg^{-1}					
Solute	1.0	0.10	0.010	0.0010	· · ·	Inf. dil.[a]
NaCl	1.81	1.87	1.94	1.97	· · ·	2
$MgSO_4$	1.09	1.21	1.53	1.82	· · ·	2
$Pb(NO_3)_2$	1.31	2.13	2.63	2.89	· · ·	3

[a]The limiting values: i = 2, 2, and 3 are reached when the solution is infinitely dilute. Note that a solute whose ions are singly charged (for example, NaCl) approaches its limiting value more quickly than does a solute whose ions carry higher charges. Interionic attractions are greater in solutes with more highly charged ions.

- By a theory of electrolyte solutions proposed by Peter Debye and Erich Hückel, ions in solution do not behave independently of one another.
- Each ion is enveloped by an ***ionic atmosphere*** with a net charge opposite its own due to interionic interactions. (In an electric field, the mobility of each ion is reduced because of the attraction or drag exerted by its ionic atmosphere)
- Similarly, the magnitudes of colligative properties are ***reduced***. This explains why, for example, $i = 1.94$ but not = 2 for 0.010 m NaCl.



(a) (b)

Interionic attractions in aqueous solution

(a) A positive ion in aqueous solution is surrounded by a shell of negative ions.

(b) A negative ion attracts positive ions to its immediate surroundings

- Each type of ion in an aqueous solution has two "concentrations":
  - *Stoichiometric concentration* is based on the amount of solute dissolved.
  - "Effective" concentration, called the ***activity***, takes into account interionic attractions
- The two are related through a factor called *activity coefficient*.
- In sufficiently dilute solutions, activity coefficient ≈ 1.

Refer to the next few slides for the supplementary notes on Activity

Activity:

- For a dilute gas, the ratio of the partial pressure to a reference pressure of 1 bar. $a_x = \frac{\gamma P_x}{P^0} = \gamma P_x$
- For a dilute solute species, the ratio of the concentration to a reference concentration of 1M. $a_x = \frac{\gamma[X]}{c^0} = \gamma[X]$

When concentrations are used to express the equilibrium constant of a solution system, it is assumed that the observed solubility of a salt and the concentrations of the component ions are the same in the solution. However, this may not be always true and therefore a correction factor is needed.

Activity coefficient:

- A correction factor that relates the activity of a non-ideal gaseous species to its partial pressure, or the activity of a species in a non-ideal solution to its concentration.

Supplementary information about Activity

- The equilibrium constant expression is written as a ratio of the product concentrations divided by the reactant concentrations.
- G.N. Lewis proposed a more appropriate way to represent the equilibrium constant expression that uses the **activities** of the reactants and products.
- **Activity** is a thermodynamic concept introduced by Lewis. It is a **dimensionless** ratio referred to a chosen reference state.

$$a_x = \frac{\gamma[X]}{c^0} = \gamma[X]$$

$$a_x = \frac{\gamma P_x}{P^0} = \gamma P_x$$

γ is the activity coefficient,
[X] represents a particular concentration,
c^0 corresponds to the concentration in a chosen reference state.

Substance	Reference / Standard state
Solutions	1 mol/L
Gases	1 bar
Solids	Pure solids (assigned activities of 1)
Liquids	Pure liquids (assigned activities of 1)

$$a_x = \frac{\gamma[X]}{c^0} = \gamma[X] \qquad a_x = \frac{\gamma P_x}{P^0} = \gamma P_x$$

- Under conditions where gases do not obey the ideal gas law or solutions depart from ideal behavior, equilibrium constant values may vary with total concentration or pressure. This problem is eliminated when activities are used because activities are effective or active concentrations.



In our discussion, we will assume that the systems are **ideal**, and that activities can be replaced by the numerical values of concentrations or partial pressures.

$$2\,Cu^{2+}(aq) + Sn^{2+}(aq) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2\,Cu^{+}(aq) + Sn^{4+}(aq)$$

$$K = \frac{[Cu^+]^2_{eq}[Sn^{4+}]_{eq}}{[Cu^{2+}]^2_{eq}[Sn^{2+}]_{eq}} = 1.48$$

is now replaced by $K = \dfrac{a^2_{Cu^+}a_{Sn^{4+}}}{a^2_{Cu^{2+}}a_{Sn^{2+}}} = 1.48$

- The relationship between an eqm constant expressed in activities and the corresponding one expressed in concentration is shown as follows:

The activity of each species is

$$a_{Cu^+} = \frac{\gamma_{Cu^+}[Cu^+]}{c^\circ}; \qquad a_{Sn^{4+}} = \frac{\gamma_{Sn^{4+}}[Sn^{4+}]}{c^\circ}$$

$$a_{Cu^{2+}} = \frac{\gamma_{Cu^{2+}}[Cu^{2+}]}{c^\circ}; \qquad a_{Sn^{2+}} = \frac{\gamma_{Sn^{2+}}[Sn^{2+}]}{c^\circ}$$

The reference state (c°) is 1 mol/L
→ The values of γ = 1.

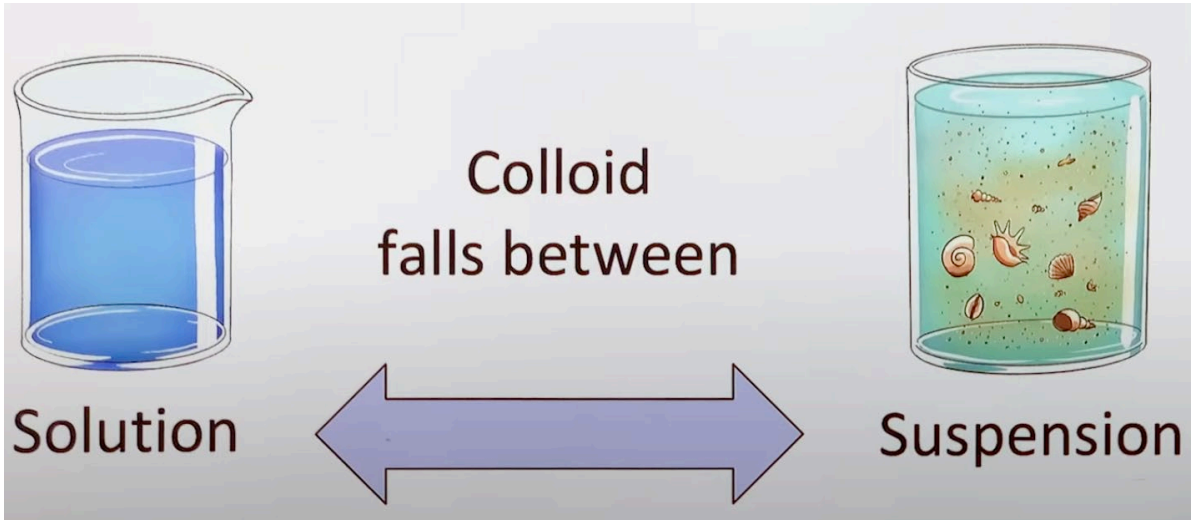
$$K = \frac{\left(\frac{\gamma_{Cu^+}[Cu^+]_{eq}}{c^\circ}\right)^2 \frac{\gamma_{Sn^{4+}}[Sn^{4+}]_{eq}}{c^\circ}}{\left(\frac{\gamma_{Cu^{2+}}[Cu^{2+}]_{eq}}{c^\circ}\right)^2 \frac{\gamma_{Sn^{2+}}[Sn^{2+}]_{eq}}{c^\circ}} \approx \frac{[Cu^+]^2_{eq}[Sn^{4+}]_{eq}}{[Cu^{2+}]^2_{eq}[Sn^{2+}]_{eq}} = 1.48$$

Colloidal Mixtures

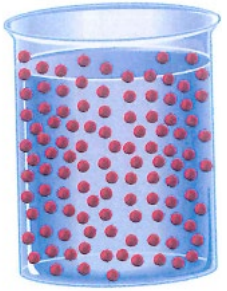
E.g. aerosol sprays, smoke, milk, mayonnaise, and paint

- A **colloid** is a solution that contains particles with one or more of its dimensions (length, width or thickness) in the approximate range of 1-1000 nm. (10^{-9} to 10^{-6} m)

- E.g. In a mixture of water and 40% sand (silica, SiO_2) by mass, the sand, being not present as ions or molecules, may be dispersed in the medium for many years.







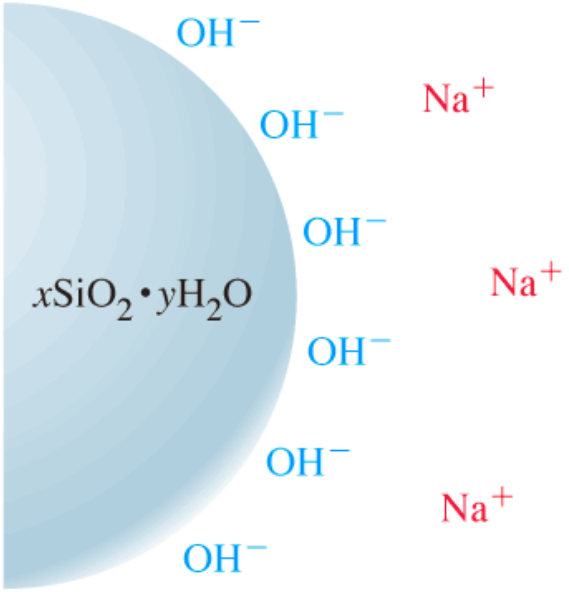



itech

- Different colloidal particles may have different shapes: spherical, rod-like, disc-like, in the form of thin films or randomly coiled filaments.

- In colloidal silica, ***what keeps the SiO_2 particles suspended?***

Electrostatic repulsion



- The surfaces of the particles adsorb ions from the solution and they preferentially adsorb one type of ion over others.
- In the case of SiO_2, the preferred ions to adsorb are OH^- ions. Hence the particles acquire a net negative charge and they repel one another.
- These mutual repulsions overcome the force of gravity. The particles remain suspended indefinitely.

Tyndall effect:

- When light passes through a solution, observe from a direction perpendicular to the light beam.
  - True solution (left): no light is observed.
  - Colloidal dispersion (right): light is scattered in many directions.



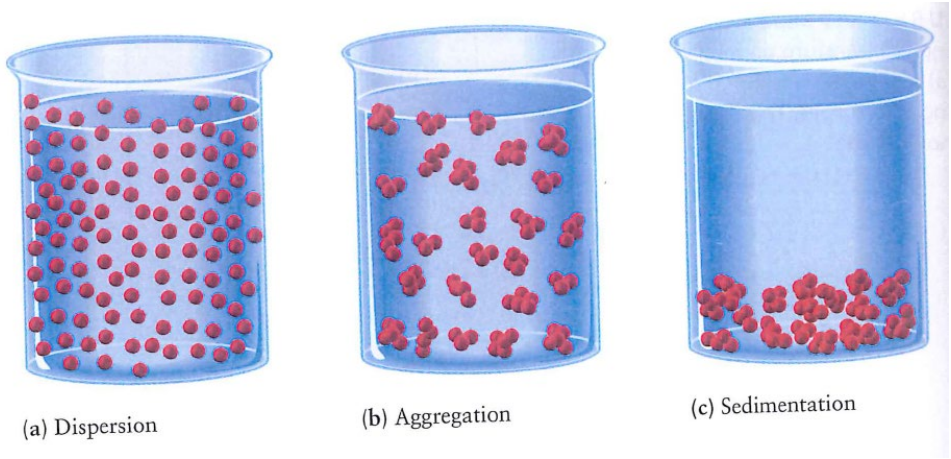
Colloid shows the light path

- A **high** concentration of ions can bring about the ***coagulation***, or precipitation, of a colloid. This is caused by particles carrying a charge opposite to that on the colloidal particles.



Coagulation of colloidal iron oxide

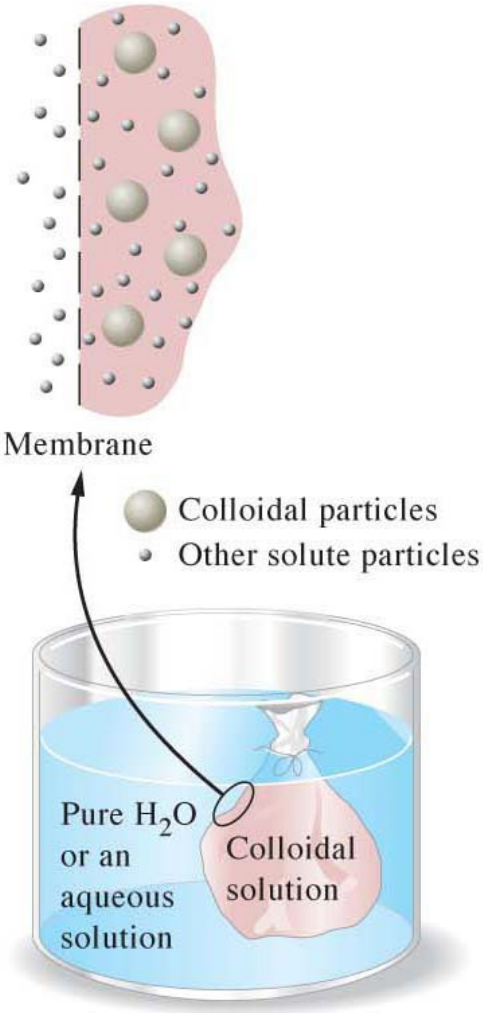
(Left) Red colloidal hydrous Fe_2O_3, obtained by adding $FeCl_3(aq)$ to boiling water. When a few drops of $Al_2(SO_4)_3$ (aq) are added, (right) the suspended particles rapidly coagulate into a precipitate of $Fe_2O_3(s)$.



- ***Dialysis*** (a process similar to osmosis) can be used to remove excess ions from a colloidal mixture. (See figure) Molecules of solvent and molecules/ions of solute pass through semi-permeable membrane, but the larger colloidal particles do not.

- In ***electrodialysis***, ions are attracted out of a colloidal mixture by an electrode carrying the opposite charge.

- E.g.) A human kidney dialyzes blood (a colloid) to remove excess electrolytes produced by metabolic processes.



▲ FIGURE 13-24
The principles of dialysis

Water molecules, other solute molecules, and dissolved ions are all free to pass through the pores of the membrane (e.g. a film of cellophane). The direction of net flow of these species depends on their relative concentrations on either side of the membrane. But colloidal particles cannot pass through the pores of the membrane.

TABLE 14.5 Some Common Types of Colloids

Dispersed Phase	Dispersion Medium	Type	Examples
Solid	Liquid	Sol	Clay sols,[a] colloidal silica, colloidal gold
Liquid	Liquid	Emulsion	Oil in water, milk, mayonnaise
Gas	Liquid	Foam	Soap and detergent suds, whipped cream, meringues
Solid	Gas	Aerosol[b]	Smoke, dust-laden air[c]
Liquid	Gas	Aerosol[b]	Fog, mist (as in aerosol products)
Solid	Solid	Solid sol	Ruby glass, certain natural and synthetic gems, blue rock salt, black diamond
Liquid	Solid	Solid emulsion	Opal, pearl
Gas	Solid	Solid foam	Pumice, lava, volcanic ash

[a]In water purification, it is sometimes necessary to precipitate clay particles or other suspended colloidal materials. This is often done by treating the water with an appropriate electrolyte. Clay sols are suspected of adsorbing organic substances, such as pesticides, and distributing them in the environment.

[b]Smogs are complex materials that are at least partly colloidal. The suspended particles are both solid and liquid. Other constituents of smog are molecular, for example, sulfur oxides, nitrogen oxides, and ozone.

[c]The bluish haze of tobacco smoke and the brilliant sunsets in desert regions are both attributable to the scattering of light by colloidal particles suspended in air.



Class Practice 15

State two main differences between a colloid and a solution.

Solution:

Colloid versus solution

- colloidal particles are much larger than solute molecules.
- colloidal suspension is not as homogeneous as a solution.